Acceptance of the Flower implementation and timing (MNIST-LT and CIFAR-10-LT)

Report generated 2026-10-24 12:38 UTC. 7 runs completed (2 of them on a second attempt), 0 failed. Framework: Flower simulation (one server, one client process per simulation); every client trains on its own data only. Methods marked LongTailLib run in the authors' own code (LongTailLib, FedLTLib benchmark) on the same client split.

TABLE I
Software and hardware.

Item	Value
Python	3.13.15
PyTorch	2.11.0+cu128
Flower	1.42.0
GPU	Tesla T4
Code version	see results/code_version.txt

TABLE II
Methods in this experiment, their reference and the origin of their code.

Method	Reference	Code
FedECS-LT	This work	This suite (Flower)
GeFL-F	Kang et al., IEEE TMC 2025	Re-implementation of the authors' Algorithm 3, checked against the earlier experiments
FedTGP	Zhang et al., AAAI 2024	HtFLlib (Zhang et al., KDD 2025), Apache-2.0, ported to Flower
FedAvg (LongTailLib)	McMahan et al., AISTATS 2017	LongTailLib (FedLTLib benchmark), authors' code unchanged, downloaded at run time

Setting: CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Setting: CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Setting: MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

TABLE III
Acceptance of the Flower implementation against the earlier experiments (seed 3): overall PASS. Balanced accuracy new / old (tolerance), as trained and with the exact readout; class separation F new / old.

Check	Verdict	Same splits	Bal. acc.	Bal. acc. (ER)	F
mnist GeFL-F	PASS	yes	74.2 / 75.0 (±8.3)	90.5 / 90.4 (±2.0)	-
mnist FedECS-LT	PASS	yes	94.3 / 94.5 (±2.0)	93.5 / 93.8 (±2.0)	0.325 / 0.314
cifar10 GeFL-F	PASS	yes	32.3 / 33.0 (±2.4)	49.3 / 50.3 (±2.6)	-
cifar10 FedECS-LT	PASS	yes	40.7 / 42.3 (±3.9)	51.0 / 52.3 (±3.0)	0.082 / 0.076

1. Results

CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

TABLE IV

Main metrics (%, test set, heads as trained; mean ± standard deviation over seeds; shaded = best). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	40.7 ± 0.0	38.5 ± 0.0	45.6 ± 0.0	35.1 ± 0.0	34.9 ± 0.0	29.0 ± 0.0	17.7 ± 0.0	42.7 ± 0.0
GeFL-F	32.3 ± 0.0	25.6 ± 0.0	35.9 ± 0.0	26.7 ± 0.0	14.7 ± 0.0	13.1 ± 0.0	3.0 ± 0.0	32.3 ± 0.0
FedTGP	23.3 ± 0.0	16.1 ± 0.0	25.0 ± 0.0	17.4 ± 0.0	7.8 ± 0.0	7.7 ± 0.0	1.1 ± 0.0	26.7 ± 0.0

TABLE V

The same heads with the exact class-balanced readout (last layer recomputed from secure sums of every client's embeddings). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Bal. acc.	Macro-F1	Tail rec.	Worst rec.	MCC	G-mean
FedECS-LT	51.0 ± 0.0	50.7 ± 0.0	48.4 ± 0.0	36.6 ± 0.0	45.7 ± 0.0	49.7 ± 0.0
GeFL-F	49.3 ± 0.0	49.0 ± 0.0	46.6 ± 0.0	35.5 ± 0.0	43.7 ± 0.0	48.0 ± 0.0

TABLE VI

Further metrics (%, test set, heads as trained). Own-mix accuracy: accuracy re-weighted to each architecture's own clients' class mix. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Accuracy	Kappa	Top-5	AUC-ROC	Head rec.	Med. rec.	Recall SD	Worst arch.	Own-mix acc.
FedECS-LT	40.7 ± 0.0	34.1 ± 0.0	89.5 ± 0.0	83.8 ± 0.0	62.4 ± 0.0	34.6 ± 0.0	17.0 ± 0.0	35.7 ± 0.0	68.6 ± 0.0
GeFL-F	32.3 ± 0.0	24.8 ± 0.0	82.0 ± 0.0	74.2 ± 0.0	57.6 ± 0.0	32.6 ± 0.0	20.2 ± 0.0	24.6 ± 0.0	73.3 ± 0.0
FedTGP	23.3 ± 0.0	14.8 ± 0.0	74.6 ± 0.0	74.3 ± 0.0	42.4 ± 0.0	25.0 ± 0.0	17.0 ± 0.0	15.6 ± 0.0	-

TABLE VII

Calibration (ECE and class-wise ECE in %, NLL in nats, Brier score). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	ECE	Class-wise ECE	NLL	Brier
FedECS-LT	39.0 ± 0.0	8.3 ± 0.0	2.9	0.937
GeFL-F	50.3 ± 0.0	10.8 ± 0.0	4.32	1.13
FedTGP	8.9 ± 0.0	2.3 ± 0.0	2.18	0.872

TABLE VIII

Balanced accuracy on the validation set (images no client holds), used to select settings without the test set. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Val. bal. acc.	Val. bal. acc. (ER)
FedECS-LT	41.5 ± 0.0	51.7 ± 0.0
GeFL-F	32.8 ± 0.0	50.0 ± 0.0
FedTGP	22.8 ± 0.0	-

TABLE IX

Cost: communication per client over the whole training, wall-clock time and peak GPU memory on one GPU. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Comm./client (MB)	Time (s)	Stage I (s)	Stage II (s)	Peak GPU (MB)
FedECS-LT	11,268	6,546	433	4,288	5,826
GeFL-F	8,659	6,713	433	6,161	8,679
FedTGP	18.8	7,565	433	-	889

TABLE X

Every run (one row per method and seed): balanced accuracy as trained and with the exact readout, tail recall, macro-F1 (%), time (s). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Seed	Bal.	Bal. (ER)	Tail	F1	Time
FedECS-LT	3	40.7 ± 0.0	51.0 ± 0.0	29.0 ± 0.0	38.5 ± 0.0	6546
GeFL-F	3	32.3 ± 0.0	49.3 ± 0.0	13.1 ± 0.0	25.6 ± 0.0	6713
FedTGP	3	23.3 ± 0.0	-	7.7 ± 0.0	16.1 ± 0.0	7565

TABLE XI

Per-class recall (%, test set, mean over seeds); class 0 is the most frequent. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	0	1	2	3	4	5	6	7	8	9
FedECS-LT	68	75	44	36	37	31	39	34	25	18
GeFL-F	68	63	43	31	37	30	26	15	8	3
FedTGP	50	43	35	12	20	44	17	8	5	1

CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

TABLE XII

Main metrics (%, test set, heads as trained; mean ± standard deviation over seeds; shaded = best). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	59.6 ± 0.0	58.8 ± 0.0	62.7 ± 0.0	55.6 ± 0.0	57.3 ± 0.0	48.0 ± 0.0	33.7 ± 0.0	67.3 ± 0.0
FedAvg (LongTailLib)	61.1 ± 0.0	59.2 ± 0.0	70.6 ± 0.0	57.9 ± 0.0	55.6 ± 0.0	44.2 ± 0.0	19.4 ± 0.0	77.0 ± 0.0

TABLE XIII

The same heads with the exact class-balanced readout (last layer recomputed from secure sums of every client's embeddings). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Bal. acc.	Macro-F1	Tail rec.	Worst rec.	MCC	G-mean
FedECS-LT	56.9 ± 0.0	55.5 ± 0.0	36.3 ± 0.0	23.1 ± 0.0	53.2 ± 0.0	52.2 ± 0.0

TABLE XIV

Further metrics (%, test set, heads as trained). Own-mix accuracy: accuracy re-weighted to each architecture's own clients' class mix. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Accuracy	Kappa	Top-5	AUC-ROC	Head rec.	Med. rec.	Recall SD	Worst arch.	Own-mix acc.
FedECS-LT	59.6 ± 0.0	55.1 ± 0.0	94.8 ± 0.0	92.5 ± 0.0	78.9 ± 0.0	55.8 ± 0.0	17.0 ± 0.0	59.6 ± 0.0	77.4 ± 0.0
FedAvg (LongTailLib)	61.1 ± 0.0	56.8 ± 0.0	96.4 ± 0.0	95.3 ± 0.0	90.9 ± 0.0	53.9 ± 0.0	23.8 ± 0.0	61.1 ± 0.0	-

TABLE XV

Calibration (ECE and class-wise ECE in %, NLL in nats, Brier score). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	ECE	Class-wise ECE	NLL	Brier
FedECS-LT	18.5 ± 0.0	4.3 ± 0.0	1.42	0.588
FedAvg (LongTailLib)	28.6 ± 0.0	6.4 ± 0.0	2.1	0.644

TABLE XVI

Balanced accuracy on the validation set (images no client holds), used to select settings without the test set. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Val. bal. acc.	Val. bal. acc. (ER)
FedECS-LT	58.7 ± 0.0	57.5 ± 0.0

TABLE XVII

Cost: communication per client over the whole training, wall-clock time and peak GPU memory on one GPU. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Comm./client (MB)	Time (s)	Stage I (s)	Stage II (s)	Peak GPU (MB)
FedECS-LT	2,984	7,409	3,101	1,070	5,826
FedAvg (LongTailLib)	-	12,018	-	-	-

TABLE XVIII

Every run (one row per method and seed): balanced accuracy as trained and with the exact readout, tail recall, macro-F1 (%), time (s). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	Seed	Bal.	Bal. (ER)	Tail	F1	Time
FedECS-LT	3	59.6 ± 0.0	56.9 ± 0.0	48.0 ± 0.0	58.8 ± 0.0	7409

Method	Seed	Bal.	Bal. (ER)	Tail	F1	Time
FedAvg (LongTailLib)	3	61.1 ± 0.0	-	44.2 ± 0.0	59.2 ± 0.0	12018

TABLE XIX

Per-class recall (%, test set, mean over seeds); class 0 is the most frequent. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

Method	0	1	2	3	4	5	6	7	8	9
FedECS-LT	86	92	60	63	60	44	60	52	46	34
FedAvg (LongTailLib)	92	97	84	60	44	58	67	57	33	19

MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

TABLE XX

Main metrics (%, test set, heads as trained; mean ± standard deviation over seeds; shaded = best). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	94.3 ± 0.0	94.3 ± 0.0	94.5 ± 0.0	93.8 ± 0.0	94.2 ± 0.0	90.0 ± 0.0	83.6 ± 0.0	98.6 ± 0.0
GeFL-F	74.2 ± 0.0	70.2 ± 0.0	75.3 ± 0.0	72.6 ± 0.0	56.9 ± 0.0	52.0 ± 0.0	27.0 ± 0.0	84.9 ± 0.0

TABLE XXI

The same heads with the exact class-balanced readout (last layer recomputed from secure sums of every client's embeddings). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Bal. acc.	Macro-F1	Tail rec.	Worst rec.	MCC	G-mean
FedECS-LT	93.5 ± 0.0	93.5 ± 0.0	88.3 ± 0.0	85.7 ± 0.0	92.9 ± 0.0	93.3 ± 0.0
GeFL-F	90.5 ± 0.0	90.5 ± 0.0	83.5 ± 0.0	80.6 ± 0.0	89.6 ± 0.0	90.1 ± 0.0

TABLE XXII

Further metrics (%, test set, heads as trained). Own-mix accuracy: accuracy re-weighted to each architecture's own clients' class mix. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Accuracy	Kappa	Top-5	AUC-ROC	Head rec.	Med. rec.	Recall SD	Worst arch.	Own-mix acc.
FedECS-LT	94.4 ± 0.0	93.8 ± 0.0	99.8 ± 0.0	99.8 ± 0.0	98.5 ± 0.0	96.0 ± 0.0	4.6 ± 0.0	93.3 ± 0.0	98.3 ± 0.0
GeFL-F	74.5 ± 0.0	71.6 ± 0.0	97.9 ± 0.0	96.3 ± 0.0	96.6 ± 0.0	81.4 ± 0.0	25.5 ± 0.0	69.7 ± 0.0	97.6 ± 0.0

TABLE XXIII

Calibration (ECE and class-wise ECE in %, NLL in nats, Brier score). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	ECE	Class-wise ECE	NLL	Brier
FedECS-LT	2.0 ± 0.0	0.7 ± 0.0	0.203	0.0861
GeFL-F	7.9 ± 0.0	3.3 ± 0.0	0.841	0.358

TABLE XXIV

Balanced accuracy on the validation set (images no client holds), used to select settings without the test set. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Val. bal. acc.	Val. bal. acc. (ER)
FedECS-LT	93.4 ± 0.0	92.4 ± 0.0
GeFL-F	73.4 ± 0.0	89.0 ± 0.0

TABLE XXV

Cost: communication per client over the whole training, wall-clock time and peak GPU memory on one GPU. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Comm./client (MB)	Time (s)	Stage I (s)	Stage II (s)	Peak GPU (MB)
FedECS-LT	4,739	2,066	99	1,161	8,674
GeFL-F	4,296	1,651	99	1,267	4,203

TABLE XXVI

Every run (one row per method and seed): balanced accuracy as trained and with the exact readout, tail recall, macro-F1 (%), time (s). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	Seed	Bal.	Bal. (ER)	Tail	F1	Time
FedECS-LT	3	94.3 ± 0.0	93.5 ± 0.0	90.0 ± 0.0	94.3 ± 0.0	2066

Method	Seed	Bal.	Bal. (ER)	Tail	F1	Time
GeFL-F	3	74.2 ± 0.0	90.5 ± 0.0	52.0 ± 0.0	70.2 ± 0.0	1651

TABLE XXVII

Per-class recall (%, test set, mean over seeds); class 0 is the most frequent. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Method	0	1	2	3	4	5	6	7	8	9
FedECS-LT	99	99	97	96	97	95	96	91	84	90
GeFL-F	98	99	93	75	94	75	89	63	27	29

2. Figures

Every figure is drawn at its printed IEEE size (one column 3.5 in, two columns 7.16 in, Times 8 pt); shown here as previews. The vector PDF and the plotted data (CSV) of each are in results.zip, folder figures/.

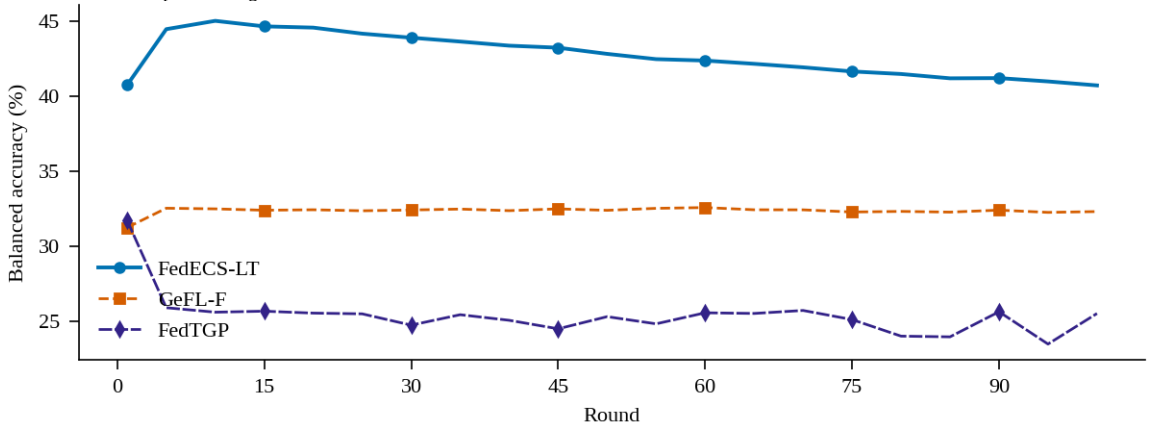


Fig. 1. Test-set balanced accuracy of the aggregated heads during head training (Stage III), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

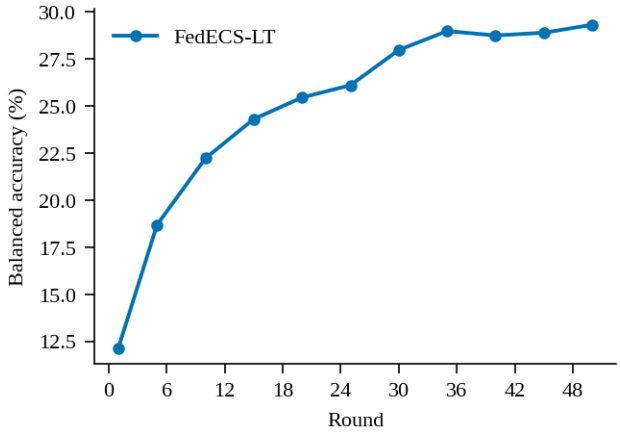


Fig. 2. Test-set balanced accuracy of the shared feature extractor with the architecture heads during Stage I, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

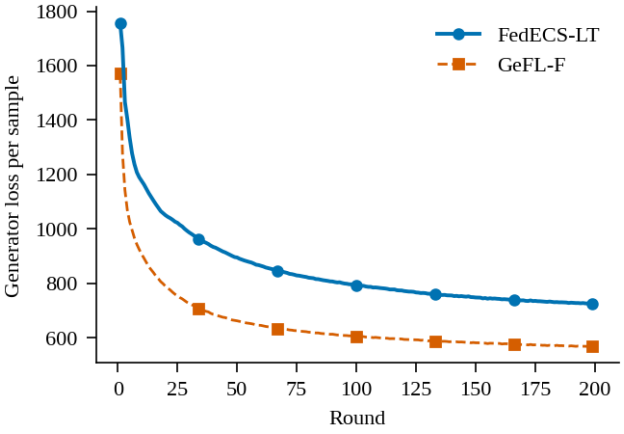


Fig. 3. Mean training loss of the clients' generator copies (reconstruction + KL) during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

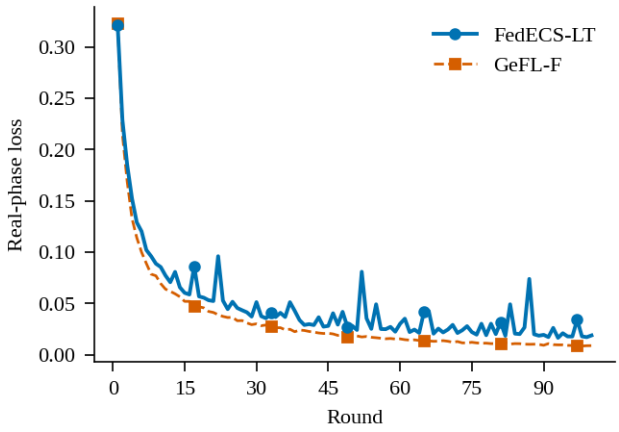


Fig. 4. Mean real-feature training loss of the clients' heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

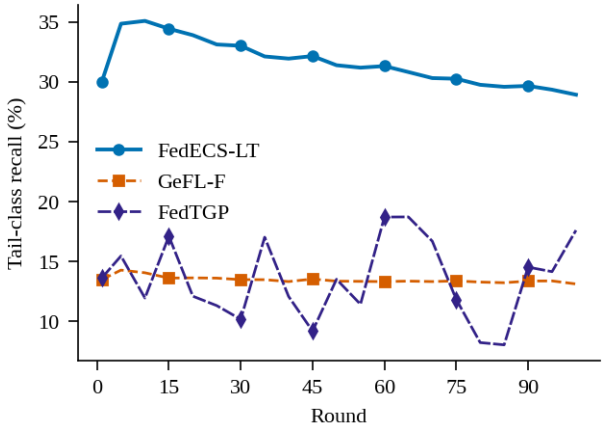


Fig. 5. Test-set mean recall of the tail classes (the rarest 40%) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

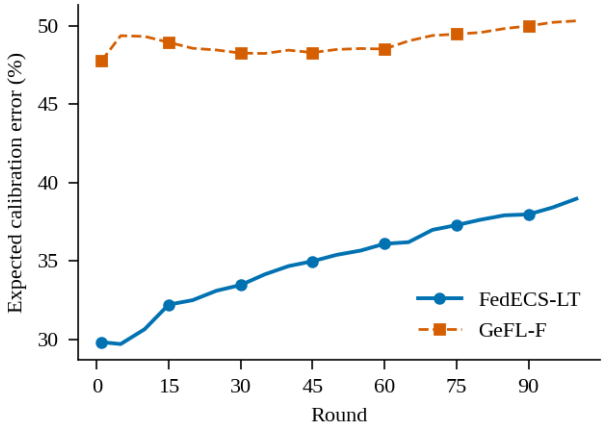


Fig. 6. Test-set expected calibration error (15 bins) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

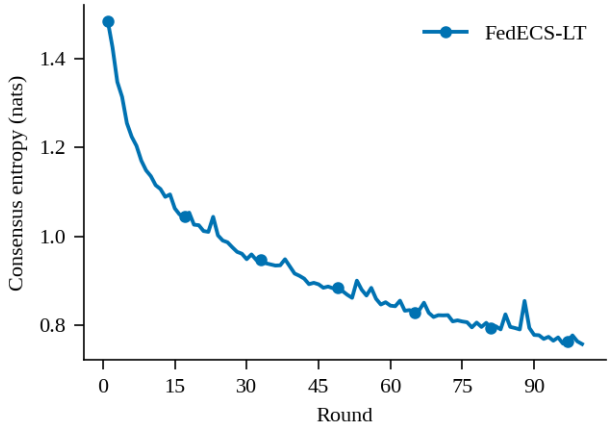


Fig. 7. Mean entropy of the consensus soft labels on the round's generated features, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

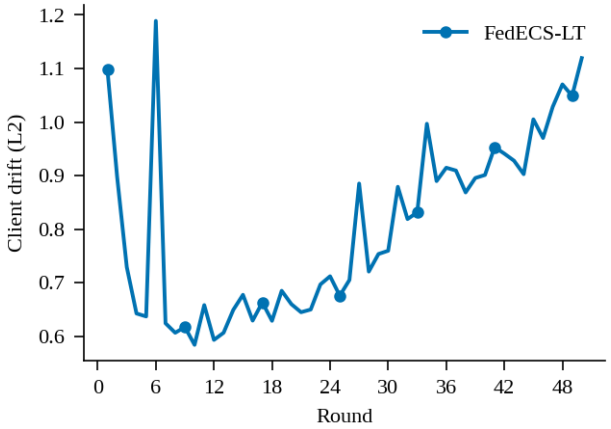


Fig. 8. Mean L2 distance of the clients' feature extractors from their average after each Stage I round (client drift), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

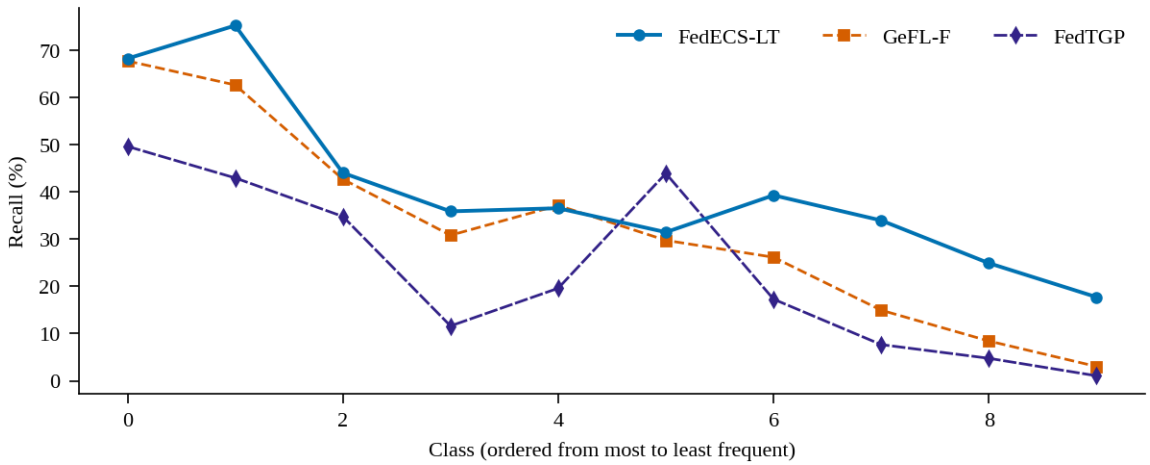


Fig. 9. Per-class recall on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

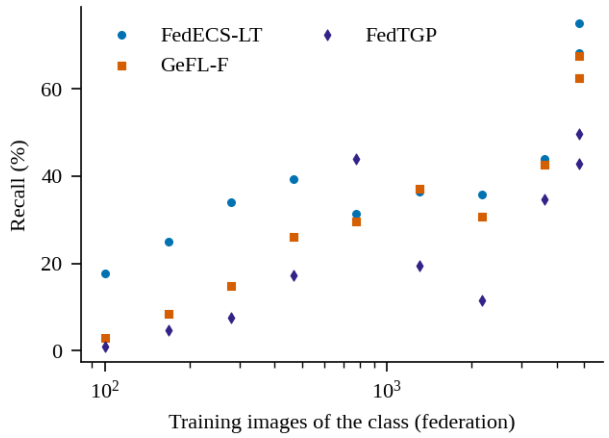


Fig. 10. Per-class recall against the number of training images of the class in the federation (logarithmic axis); each point is one class, mean over seeds. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

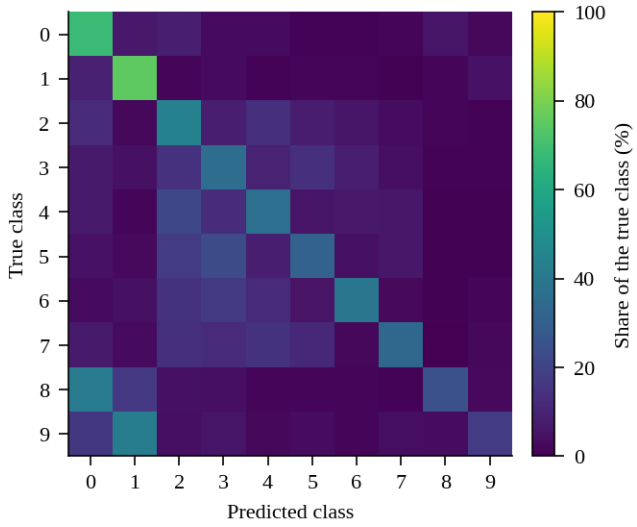


Fig. 11. Row-normalised confusion matrix of FedECS-LT on the test set (all client models, summed over seeds); classes ordered from most to least frequent. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

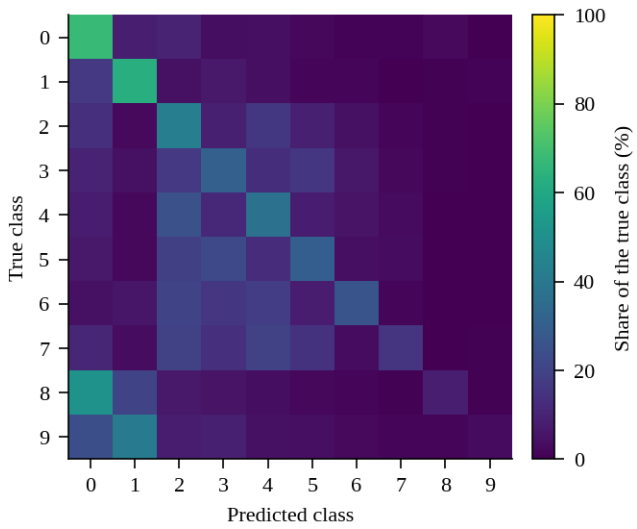


Fig. 12. Row-normalised confusion matrix of GeFL-F [Kang et al., IEEE TMC 2025] on the test set (all client models, summed over seeds); classes ordered from most to least frequent. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

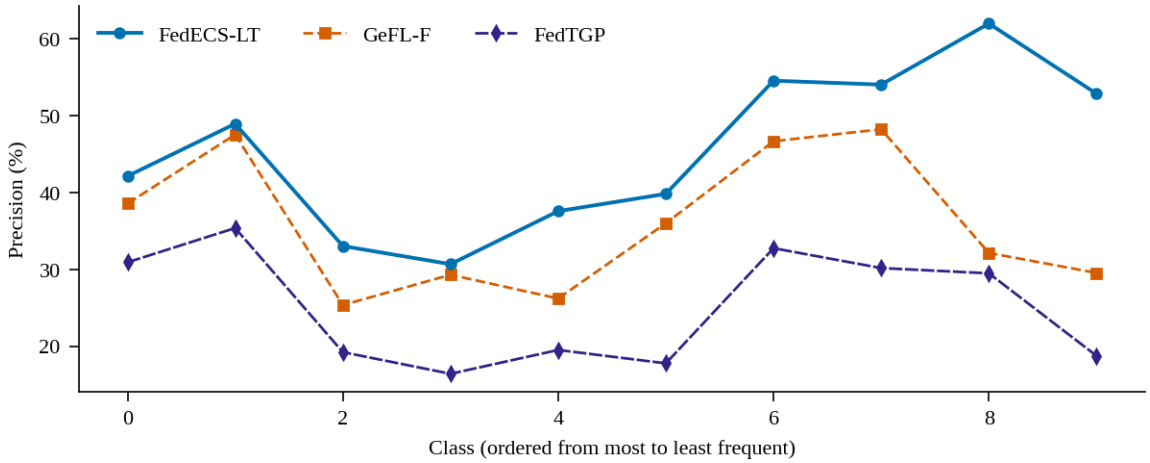


Fig. 13. Per-class precision on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

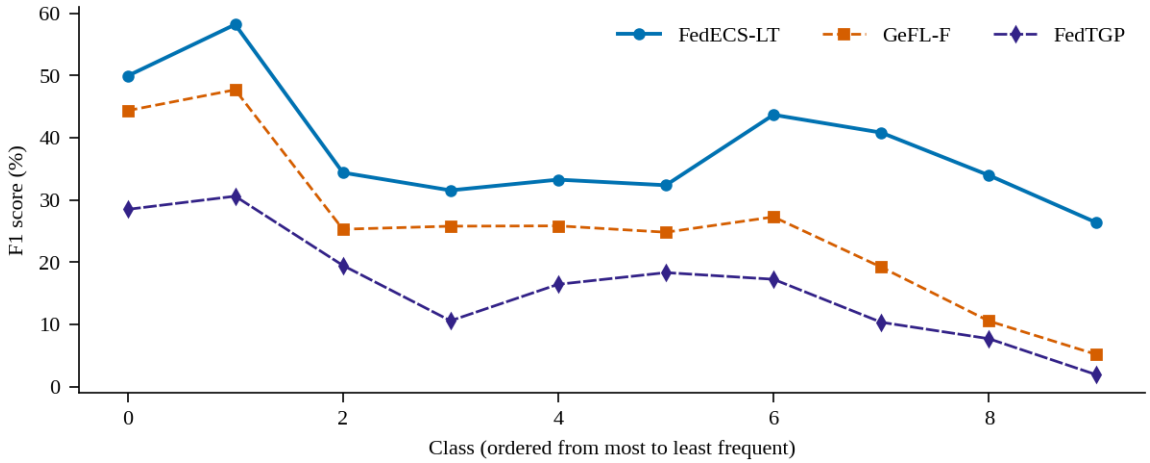


Fig. 14. Per-class F1 score on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

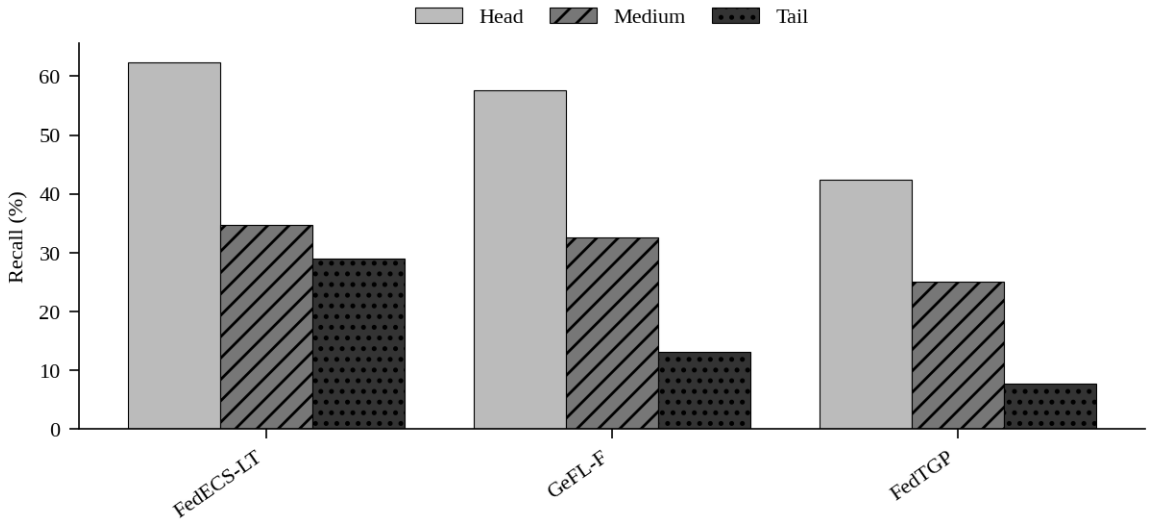


Fig. 15. Mean recall of the head (most frequent 30%), medium (next 30%) and tail (rarest 40%) classes; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

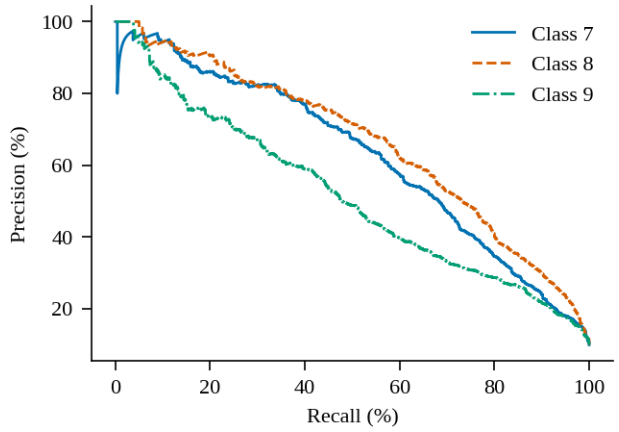


Fig. 16. Precision-recall curves of the three rarest classes for FedECS-LT (averaged softmax of all client models, first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

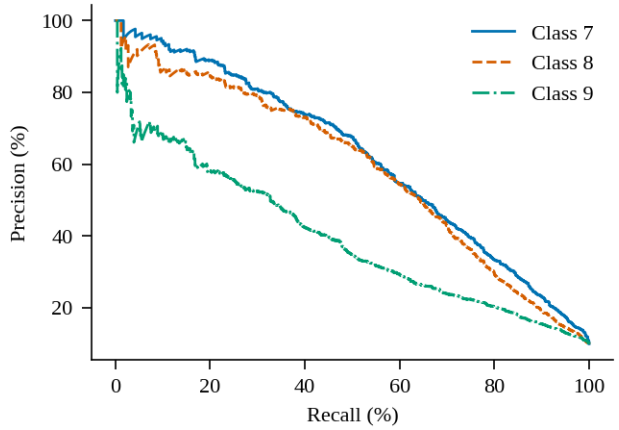


Fig. 17. Precision-recall curves of the three rarest classes for GeFL-F [Kang et al., IEEE TMC 2025] (averaged softmax of all client models, first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

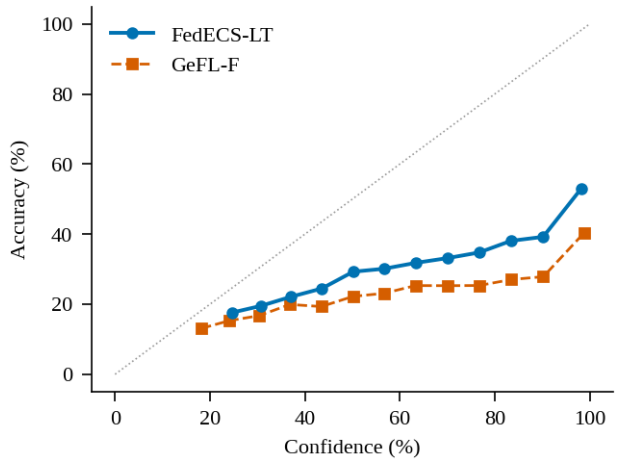


Fig. 18. Reliability diagram: accuracy against confidence in 15 equal-width bins (bins with more than 20 test images); the dotted line is perfect calibration. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

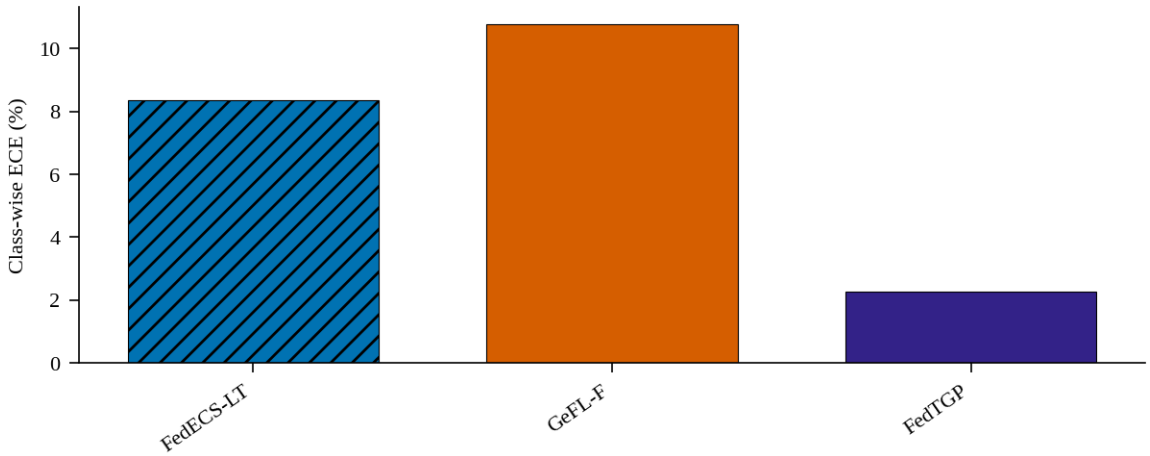


Fig. 19. Class-wise expected calibration error (mean over classes of the calibration error of p(c|x)); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

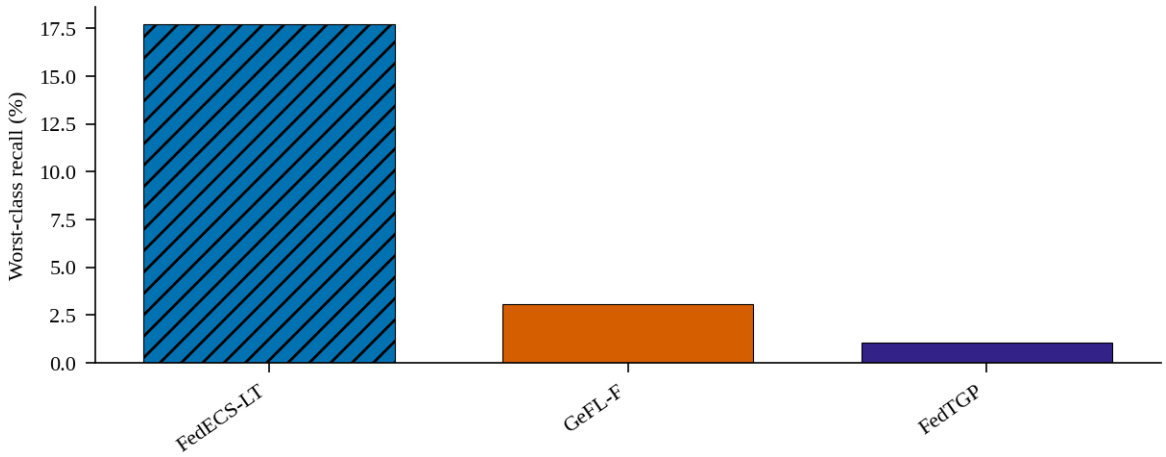


Fig. 20. Recall of the worst class; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

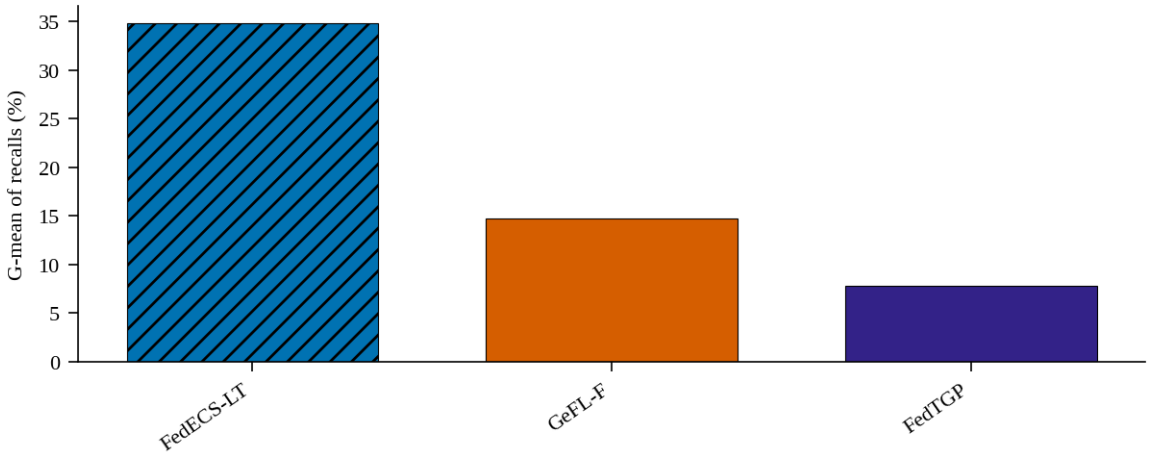


Fig. 21. Geometric mean of the class recalls (each floored at 1%); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

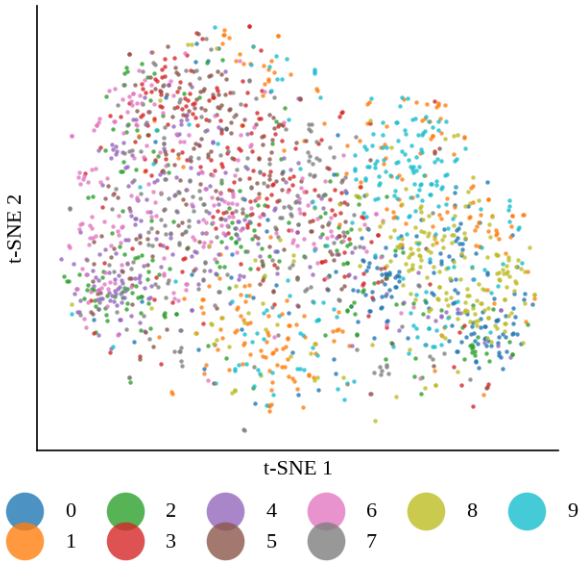


Fig. 22. t-SNE of the test-set embeddings of FedECS-LT's head CNN-1 (2,000 test images, first seed), coloured by true class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

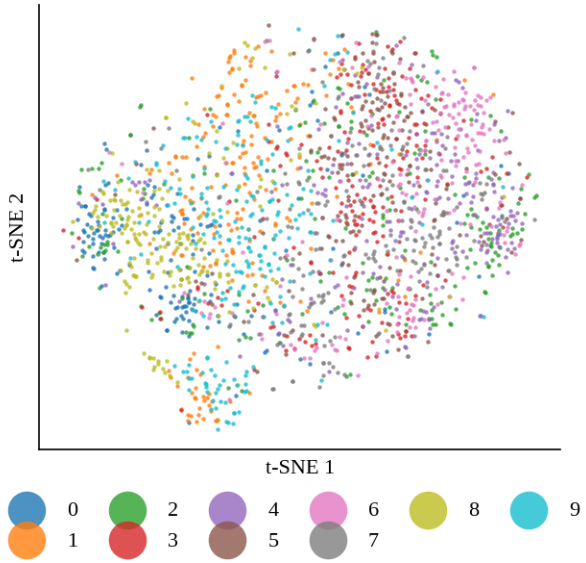
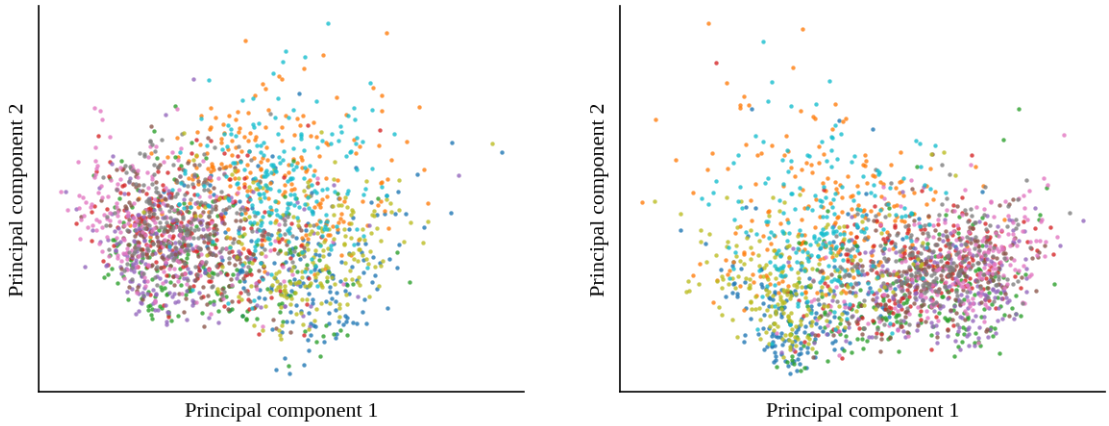


Fig. 23. t-SNE of the test-set embeddings of GeFL-F [Kang et al., IEEE TMC 2025]'s head CNN-1 (2,000 test images, first seed), coloured by true class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.



(a) (b)

Fig. 24. First two principal components of the test-set embeddings: (a) FedECS-LT (b) GeFL-F [Kang et al., IEEE TMC 2025]; coloured by true class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

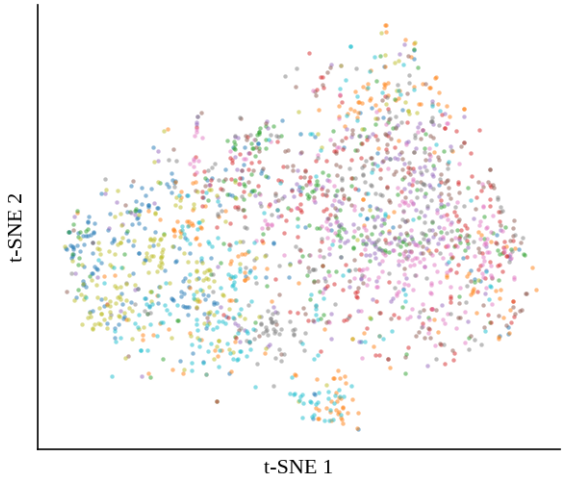


Fig. 25. t-SNE of real validation features (circles) and features generated by FedECS-LT's generator (crosses) in the frozen feature space, coloured by class (first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

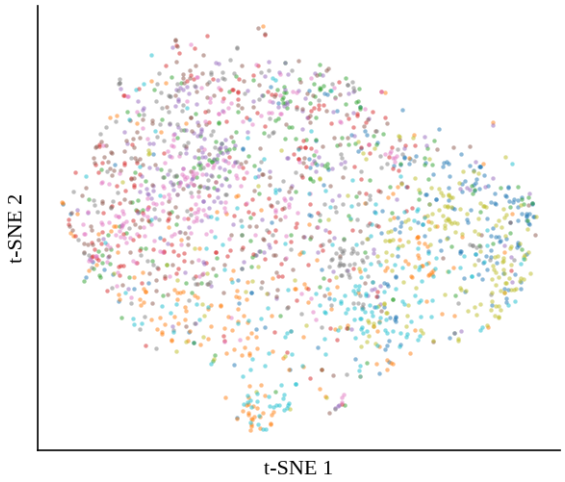


Fig. 26. t-SNE of real validation features (circles) and features generated by GeFL-F [Kang et al., IEEE TMC 2025]'s generator (crosses) in the frozen feature space, coloured by class (first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

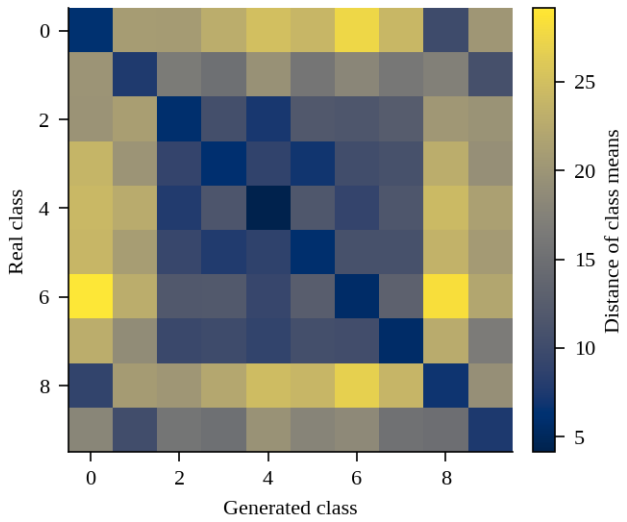


Fig. 27. Euclidean distance between the mean real feature of each class (validation set) and the mean generated feature of each class for FedECS-LT; a dark diagonal means generated classes sit at their real class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

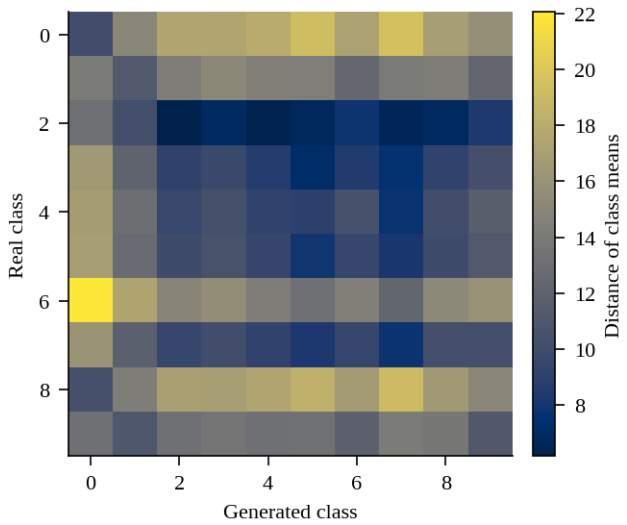


Fig. 28. Euclidean distance between the mean real feature of each class (validation set) and the mean generated feature of each class for GeFL-F [Kang et al., IEEE TMC 2025]; a dark diagonal means generated classes sit at their real class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

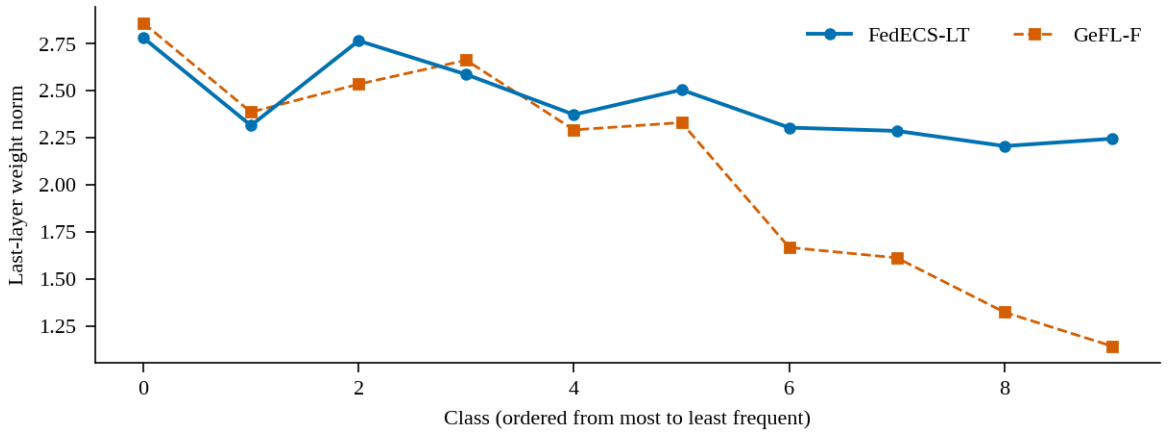


Fig. 29. L2 norm of each class's last-layer weight vector after Stage III, averaged over architectures and seeds; a falling line means the classifier favours frequent classes. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

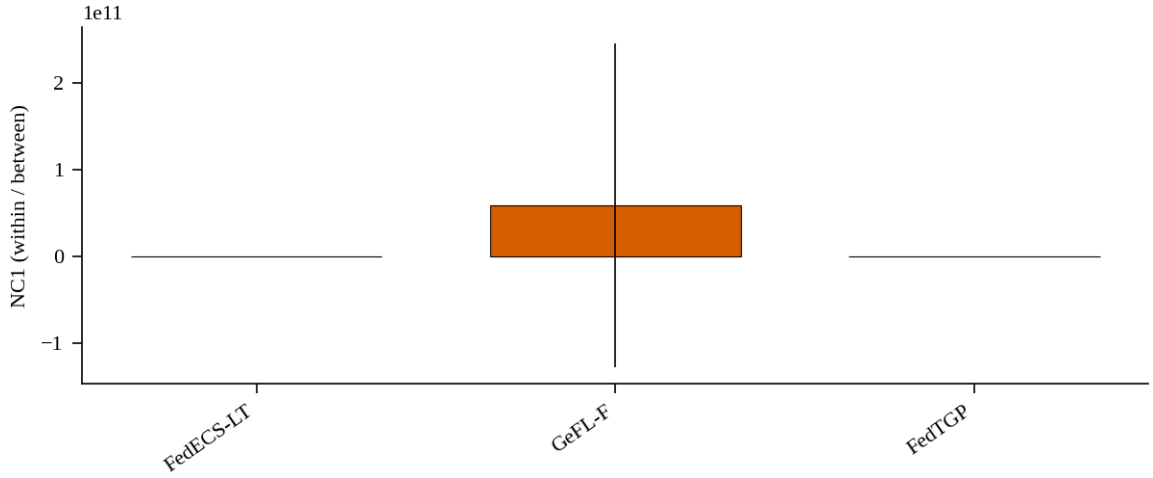


Fig. 30. Neural-collapse statistic NC1 = tr(S_W S_B^+)/C of the test-set embeddings (lower = tighter, better separated classes), mean over architectures and seeds. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

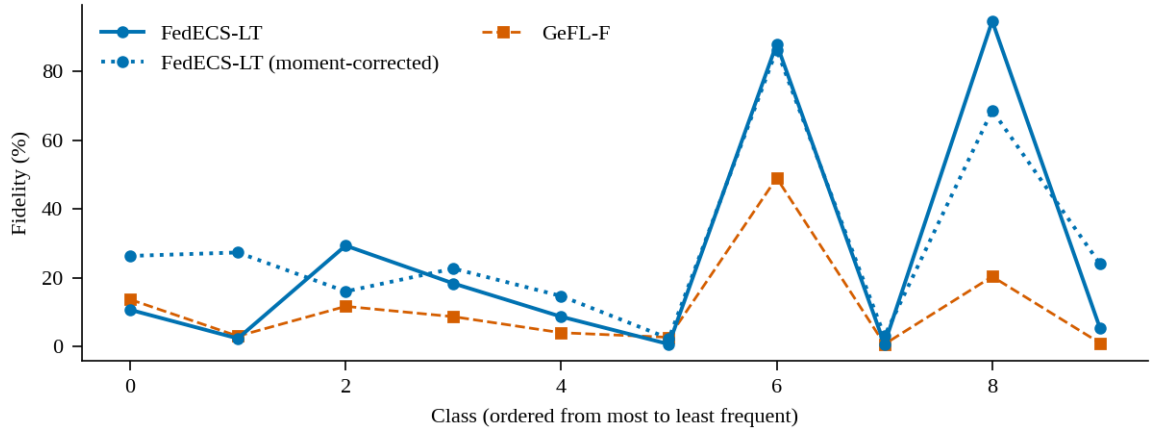


Fig. 31. Per-class fidelity of the trained generator: share of 300 generated class-c features that a referee classifier trained on the server's validation features assigns to class c; mean over seeds. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

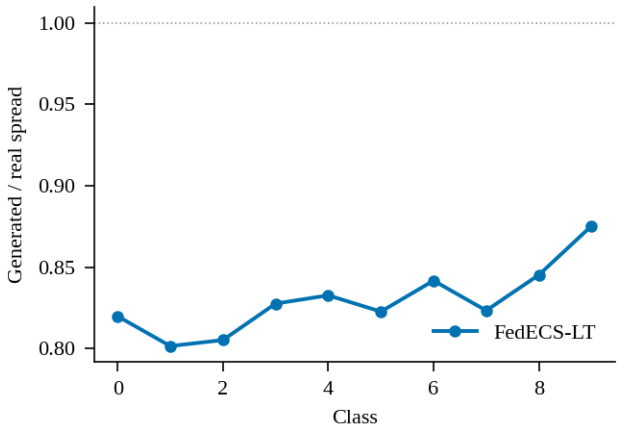


Fig. 32. Within-class spread of the generated features over the exact real spread (1 = realistic diversity, 0 = copies of the class mean), per class. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

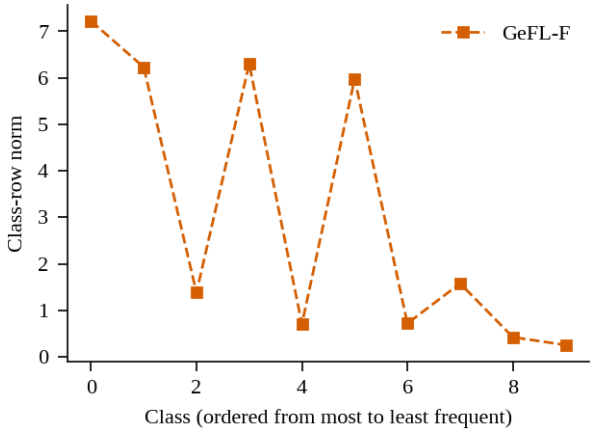


Fig. 33. L2 norm of each class's conditioning row of the generator after Stage II (class-row generators only); mean over seeds. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

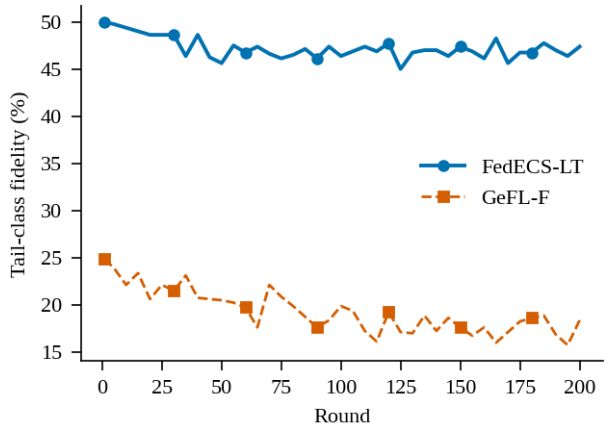


Fig. 34. Fidelity of the aggregated generator on the tail classes during Stage II (share of generated features the server's referee assigns to their class), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

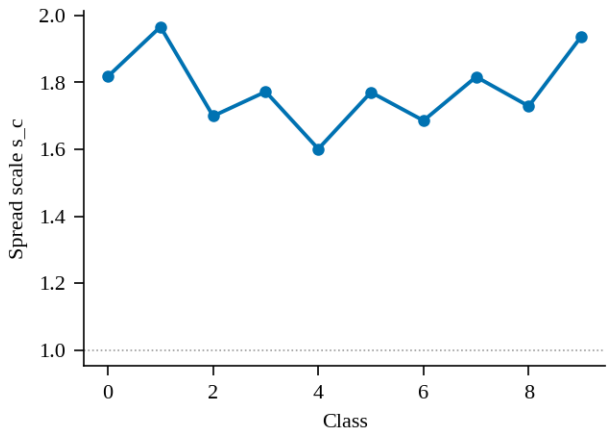


Fig. 35. Moment-correction scale s_c = sqrt(V_c / V~_c) per class: the factor by which the generator's within-class spread is stretched to the exact spread. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

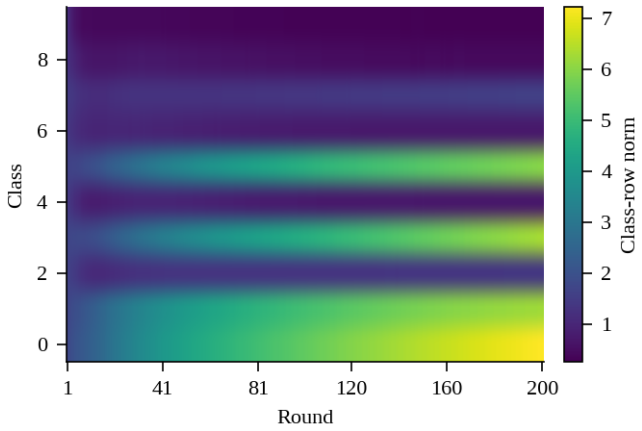


Fig. 36. Norm of each class's conditioning row of the aggregated generator over the rounds of Stage II for GeFL-F [Kang et al., IEEE TMC 2025] (first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

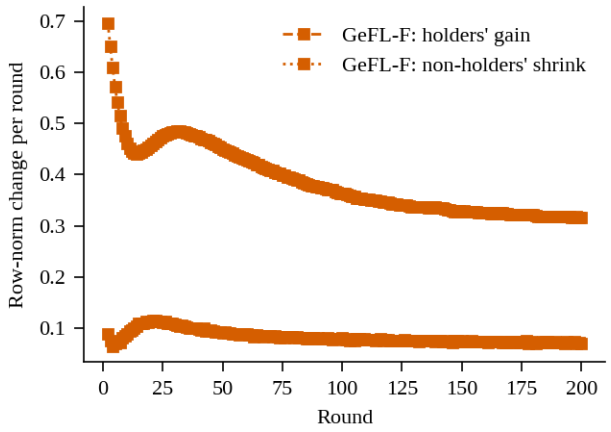


Fig. 37. Mean change of a class row's norm in one round of local training: growth on clients that hold the class against shrinkage on clients that do not (class-row generators, first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

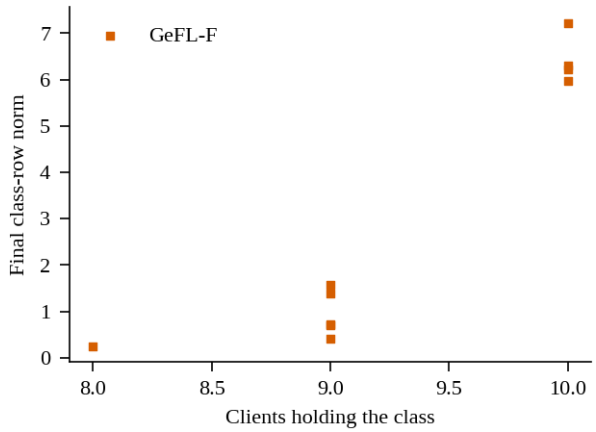


Fig. 38. Final norm of each class's conditioning row against the number of clients that hold the class; one point per class and seed. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

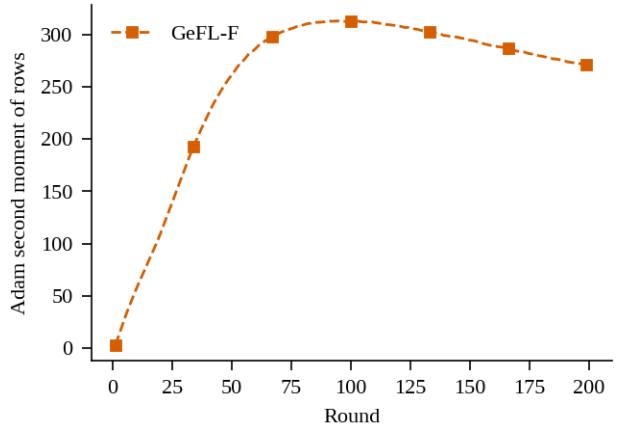


Fig. 39. Mean Adam second-moment estimate of the class rows over clients during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

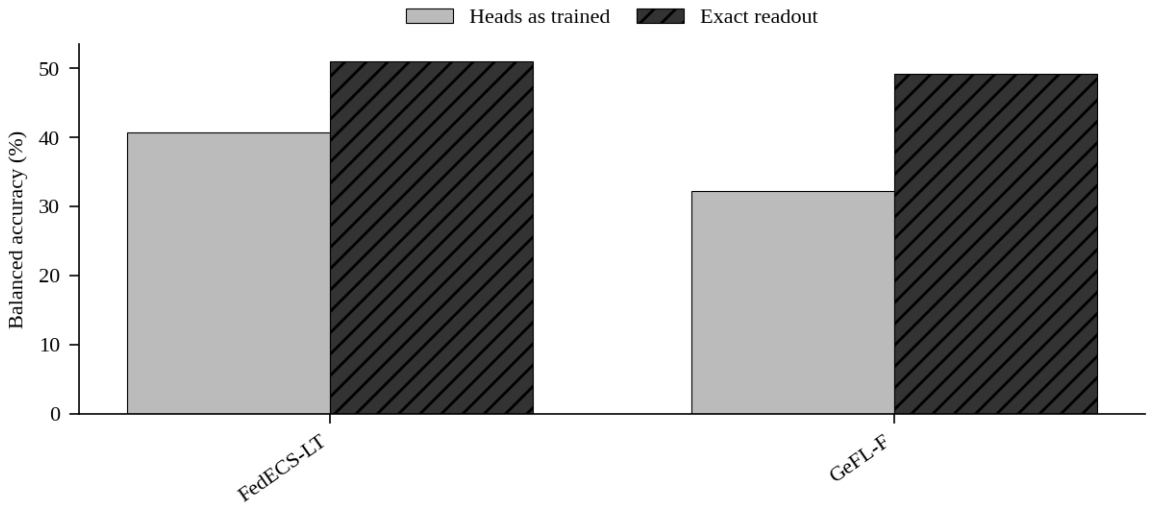


Fig. 40. Balanced accuracy of the heads as trained and after the exact class-balanced readout (last layer recomputed from secure sums of the clients' embeddings); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

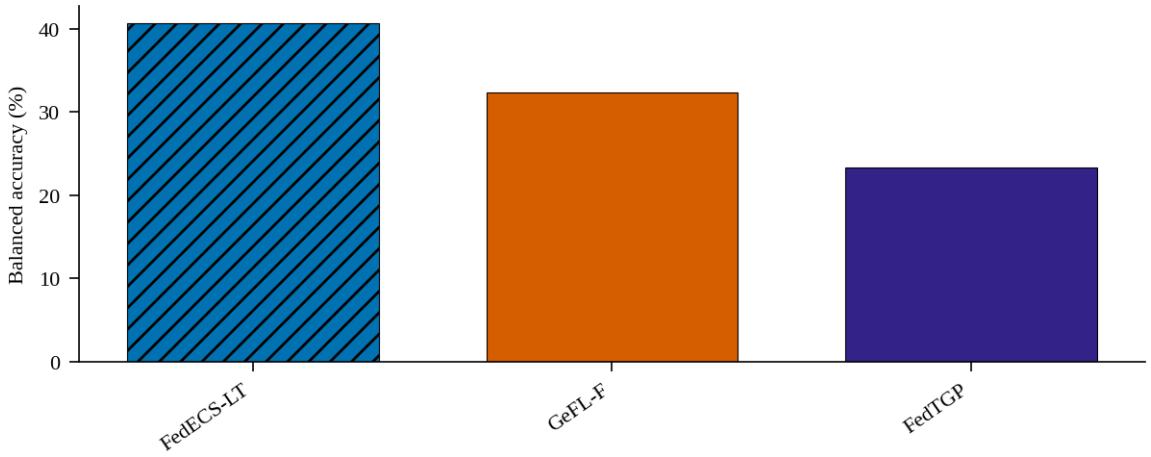


Fig. 41. Balanced accuracy on the test set (mean of the class recalls) of every method; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

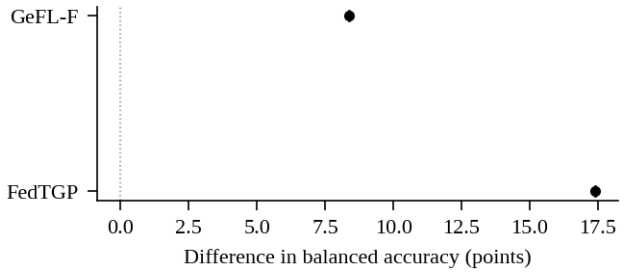


Fig. 42. Paired difference in balanced accuracy, FedECS-LT minus each baseline, over the same seeds and splits; points are means, bars 95% t confidence intervals; right of the dotted line FedECS-LT is better. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

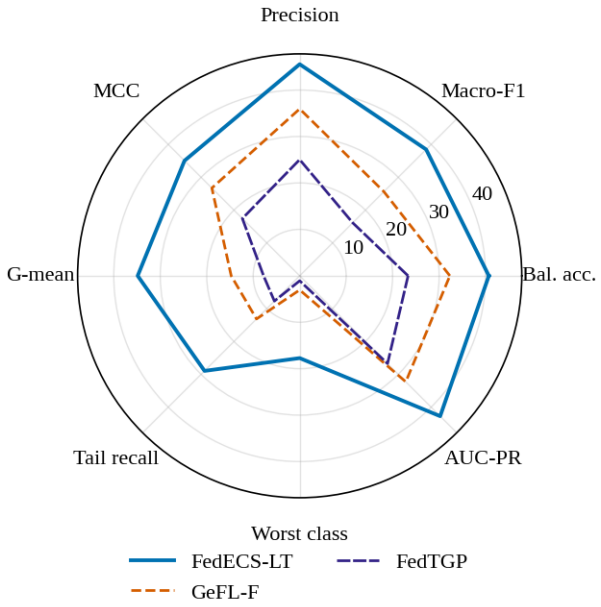


Fig. 43. Main metrics (%, test set, mean over seeds) of FedECS-LT and the five other methods with the highest balanced accuracy; larger is better on every axis. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

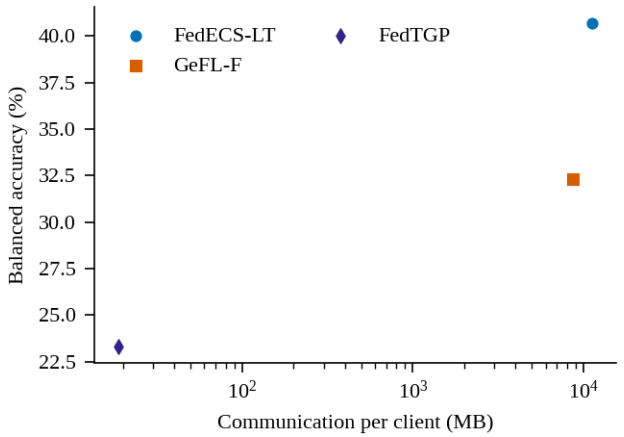


Fig. 44. Balanced accuracy against communication per client over the whole training (logarithmic axis); mean over seeds. Up and left is better. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

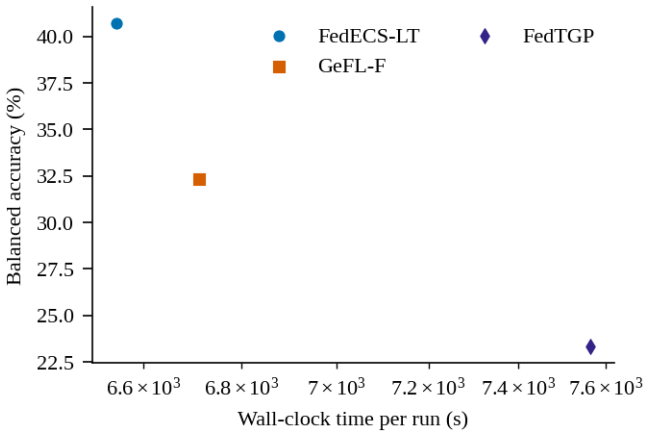


Fig. 45. Balanced accuracy against wall-clock time of one run on one GPU (logarithmic axis); mean over seeds. Up and left is better. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

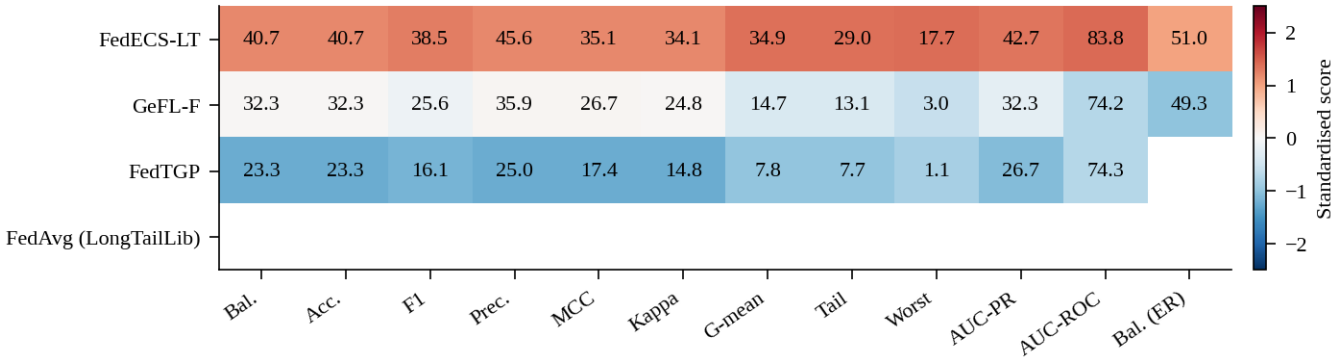


Fig. 46. Every main metric (%, test set, mean over seeds) of every method; colours are standardised per metric (red = better than the column mean). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

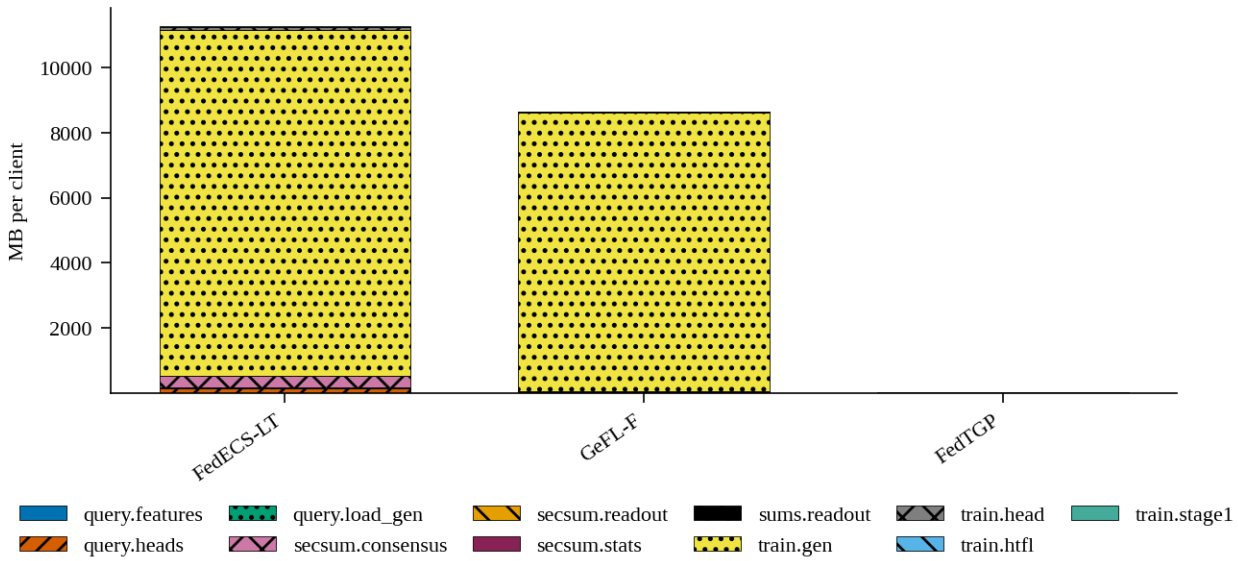


Fig. 47. Communication per client over the whole training, by message type (model weights, statistics, secure sums); first seed. Cached shared stages count once. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

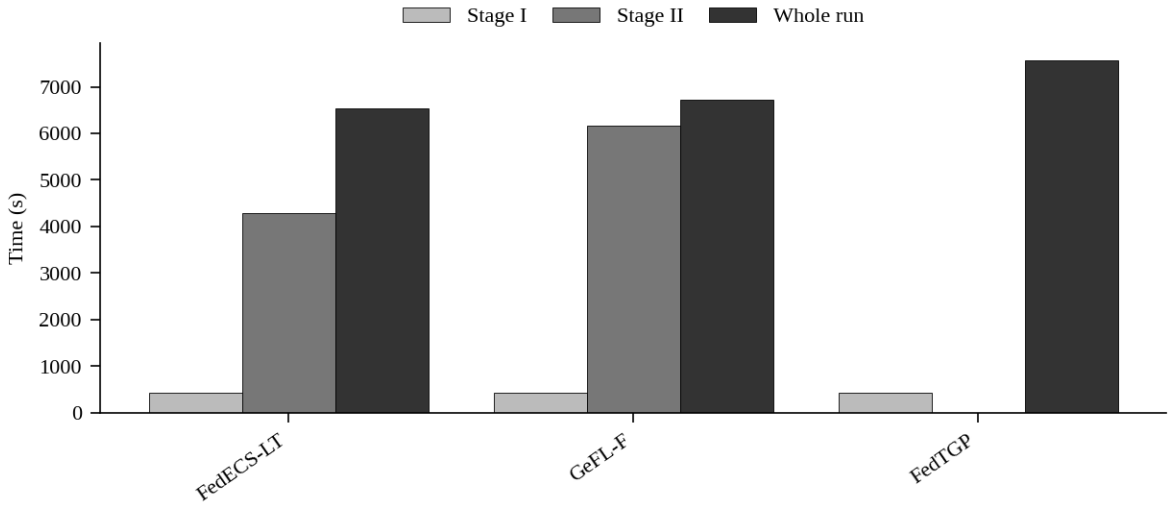


Fig. 48. Wall-clock time per stage on one GPU (simulation of all clients), mean over seeds; shared stages loaded from cache report the time of their first run. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

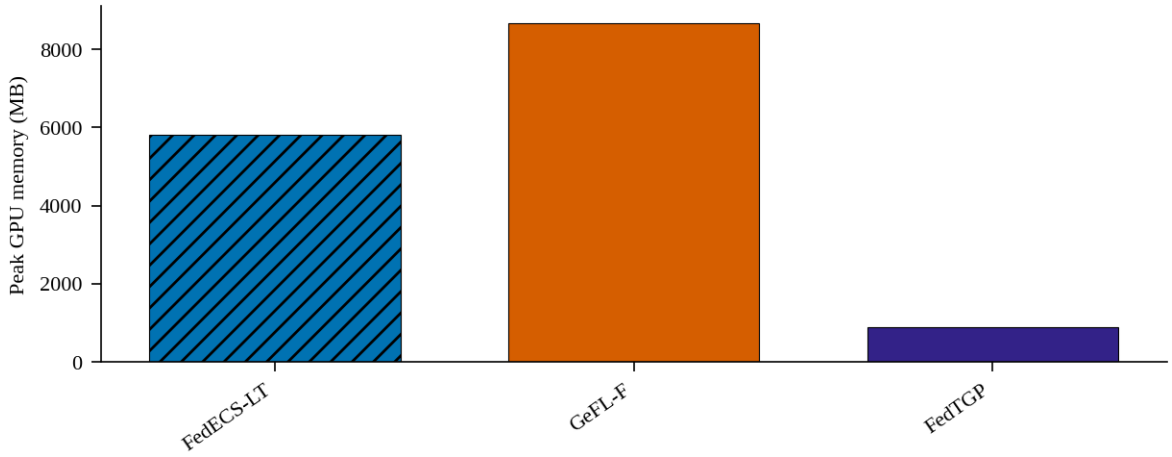


Fig. 49. Peak GPU memory of the server process during one run (mean over seeds). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

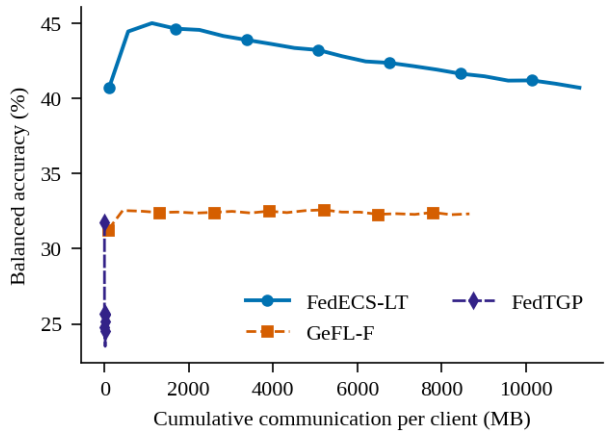


Fig. 50. Balanced accuracy of the aggregated heads against cumulative communication per client (Stage III rounds placed proportionally on the total). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

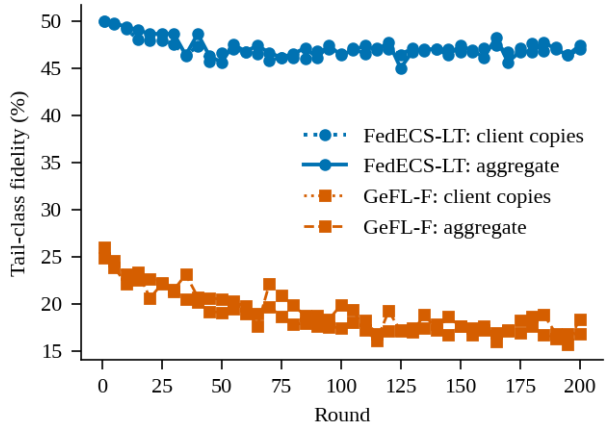


Fig. 51. Tail-class fidelity of the clients' generator copies just before aggregation (mean over clients) and of the aggregated generator, during Stage II. CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

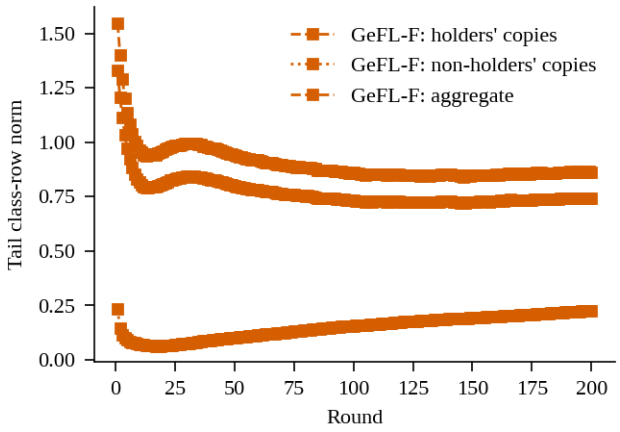


Fig. 52. Norm of the tail classes' conditioning rows in the copies of clients that hold the class, of clients that do not, and in the aggregate, during Stage II (class-row generators, first seed). CIFAR-10-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

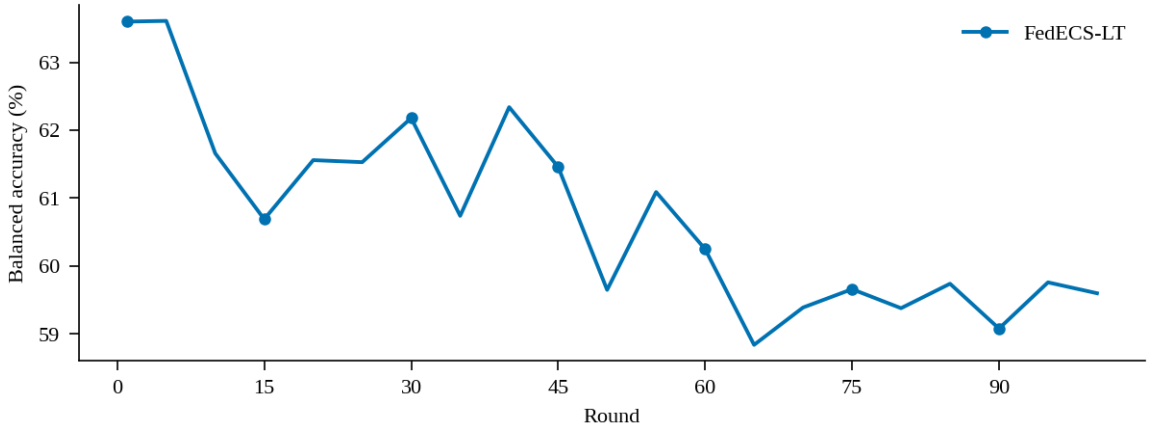


Fig. 53. Test-set balanced accuracy of the aggregated heads during head training (Stage III), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

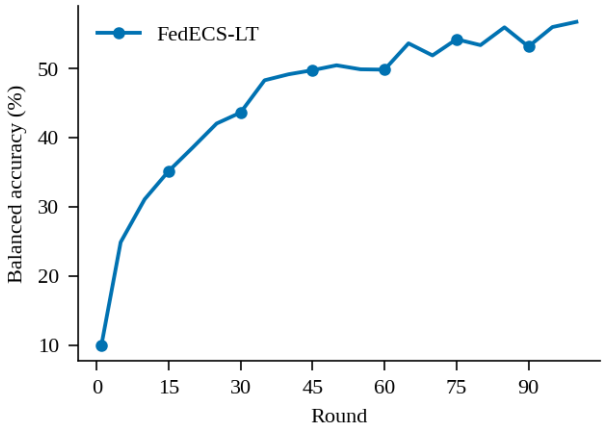


Fig. 54. Test-set balanced accuracy of the shared feature extractor with the architecture heads during Stage I, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

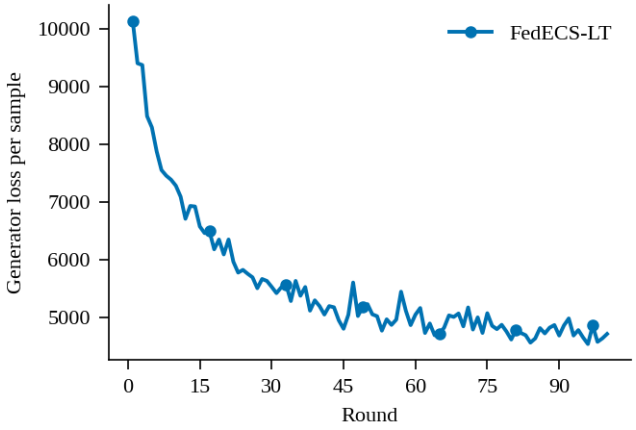


Fig. 55. Mean training loss of the clients' generator copies (reconstruction + KL) during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

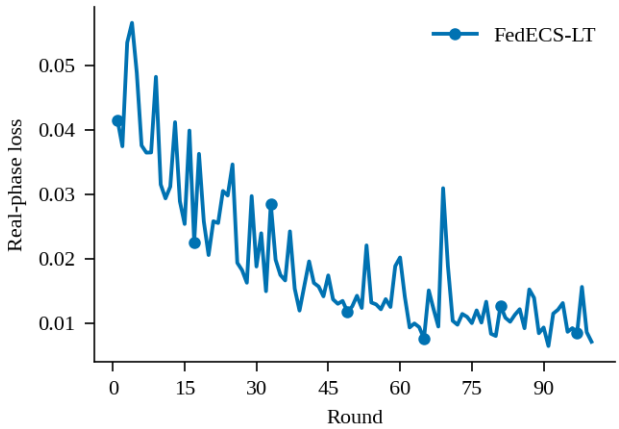


Fig. 56. Mean real-feature training loss of the clients' heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

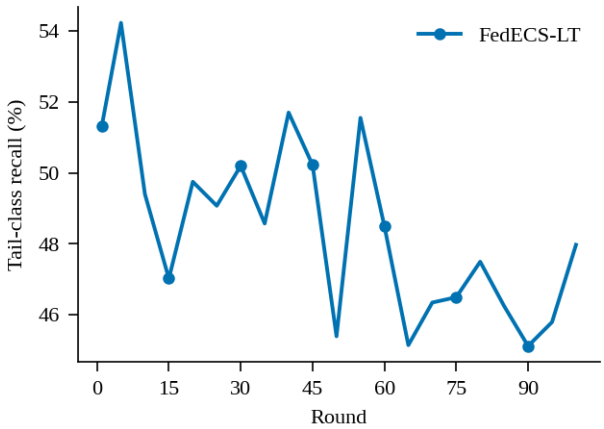


Fig. 57. Test-set mean recall of the tail classes (the rarest 40%) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

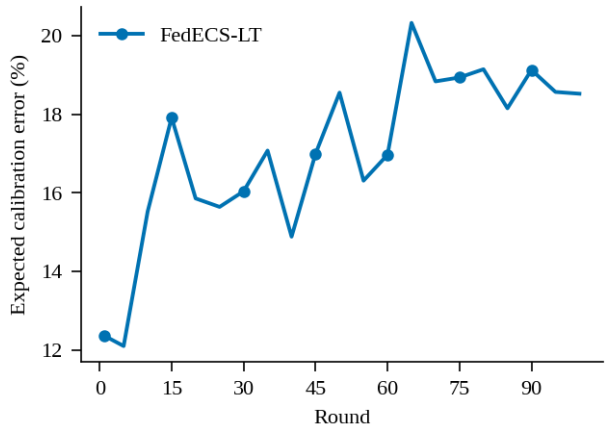


Fig. 58. Test-set expected calibration error (15 bins) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

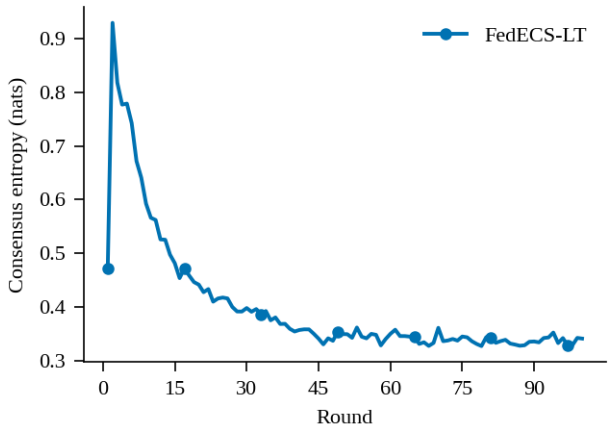


Fig. 59. Mean entropy of the consensus soft labels on the round's generated features, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

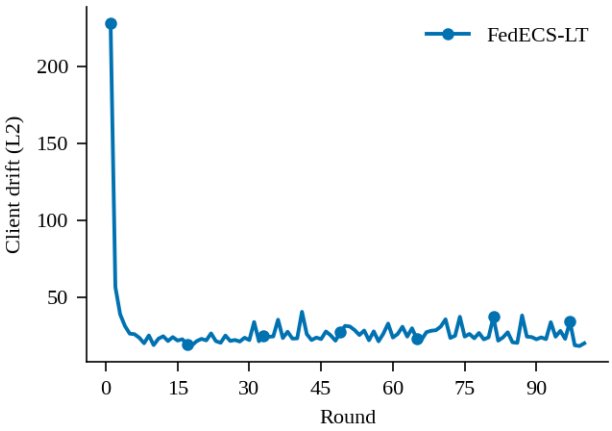


Fig. 60. Mean L2 distance of the clients' feature extractors from their average after each Stage I round (client drift), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

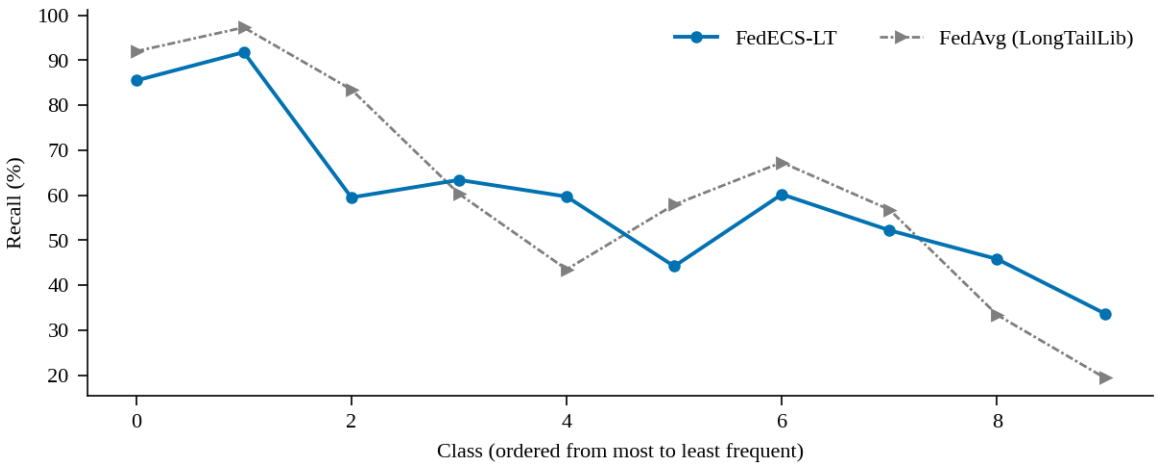


Fig. 61. Per-class recall on the test set after training (the global model, mean over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

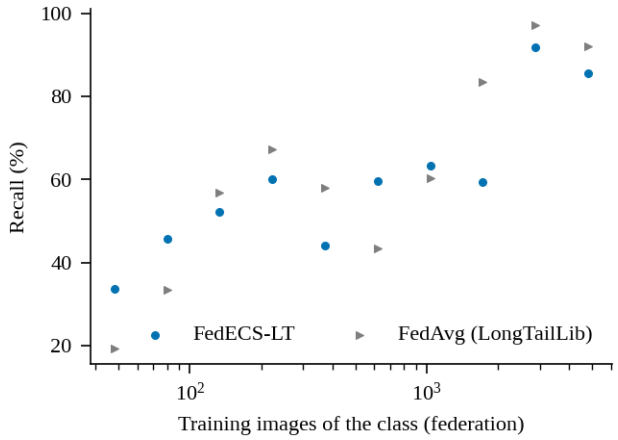


Fig. 62. Per-class recall against the number of training images of the class in the federation (logarithmic axis); each point is one class, mean over seeds. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

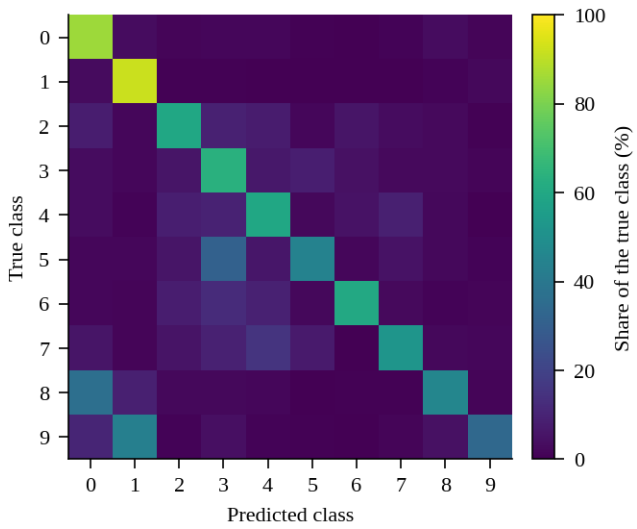


Fig. 63. Row-normalised confusion matrix of FedECS-LT on the test set (all client models, summed over seeds); classes ordered from most to least frequent. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

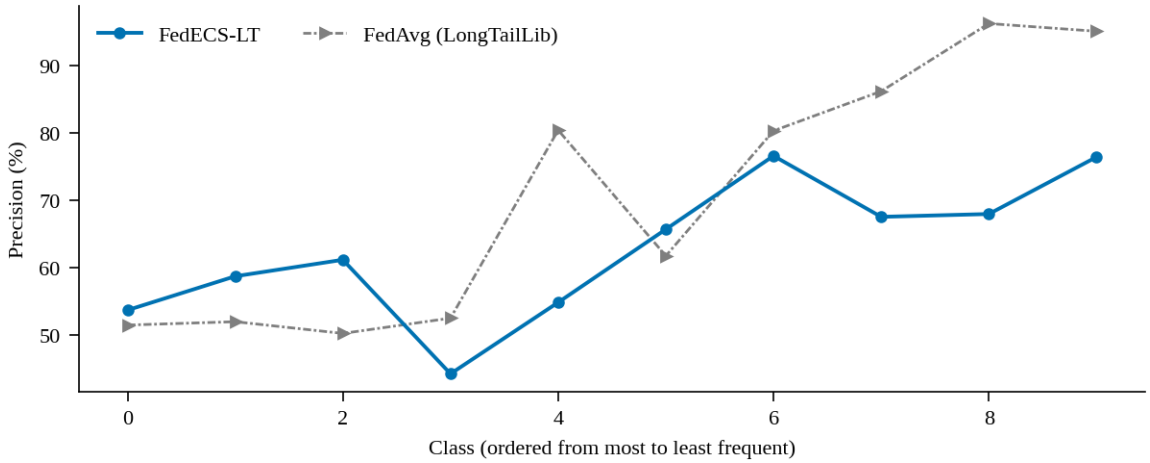


Fig. 64. Per-class precision on the test set after training (the global model, mean over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

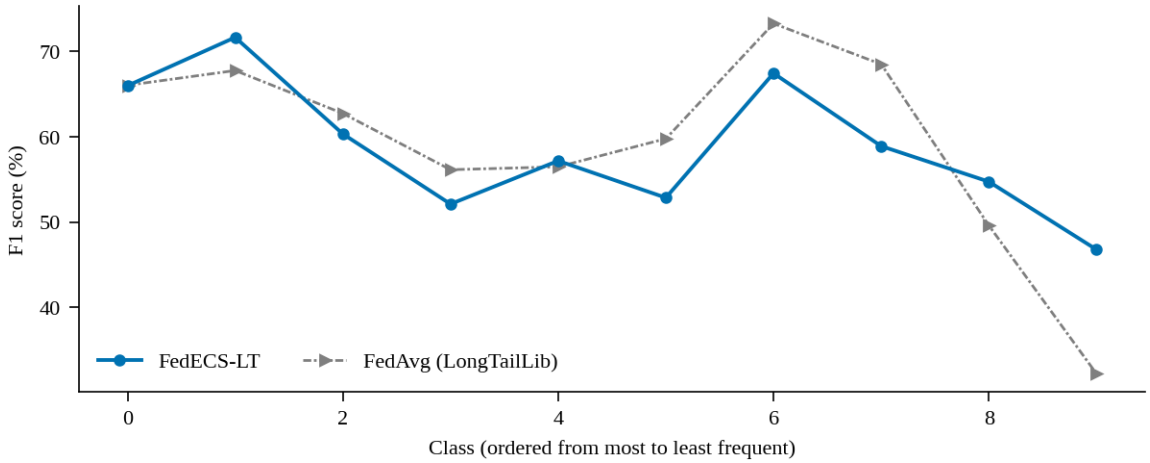


Fig. 65. Per-class F1 score on the test set after training (the global model, mean over seeds); classes are ordered from the most to the least frequent in the federation's training data. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

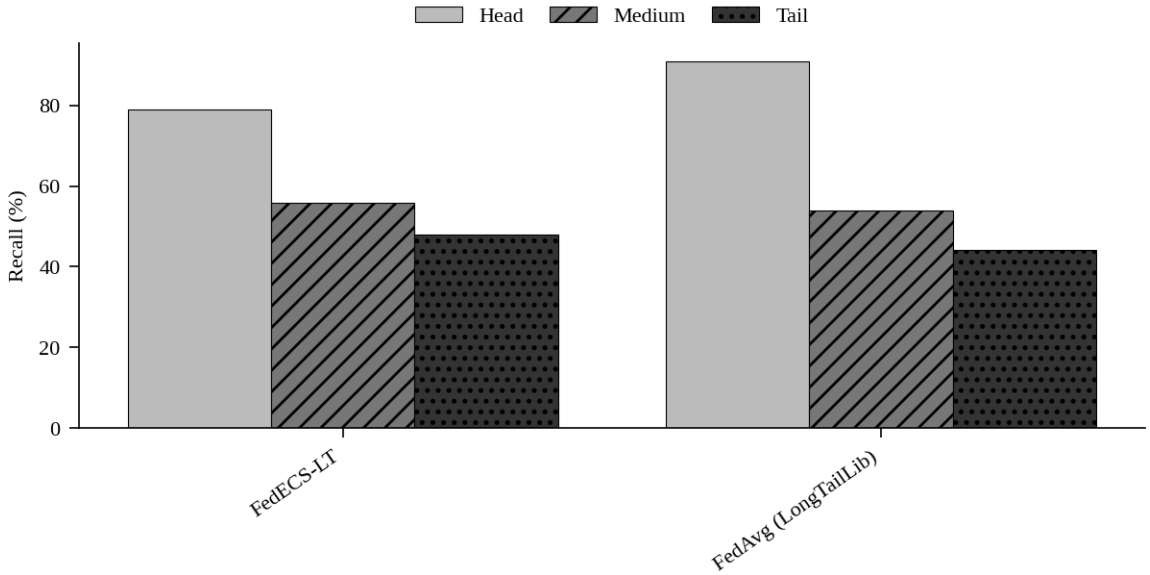


Fig. 66. Mean recall of the head (most frequent 30%), medium (next 30%) and tail (rarest 40%) classes; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

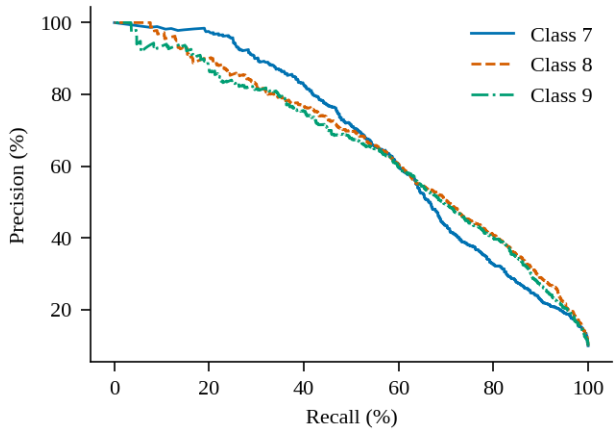


Fig. 67. Precision-recall curves of the three rarest classes for FedECS-LT (averaged softmax of all client models, first seed). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

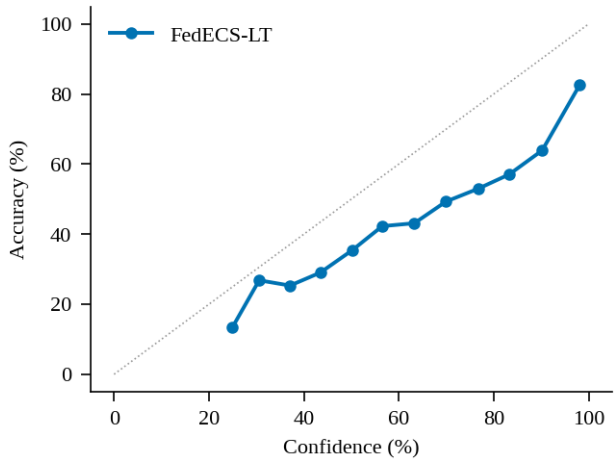


Fig. 68. Reliability diagram: accuracy against confidence in 15 equal-width bins (bins with more than 20 test images); the dotted line is perfect calibration. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

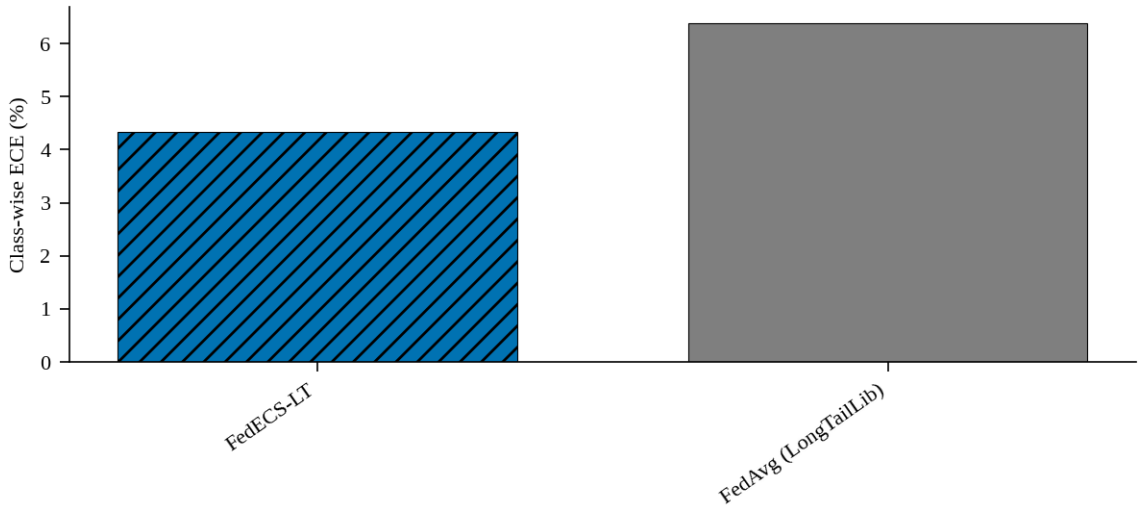


Fig. 69. Class-wise expected calibration error (mean over classes of the calibration error of p(c|x)); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

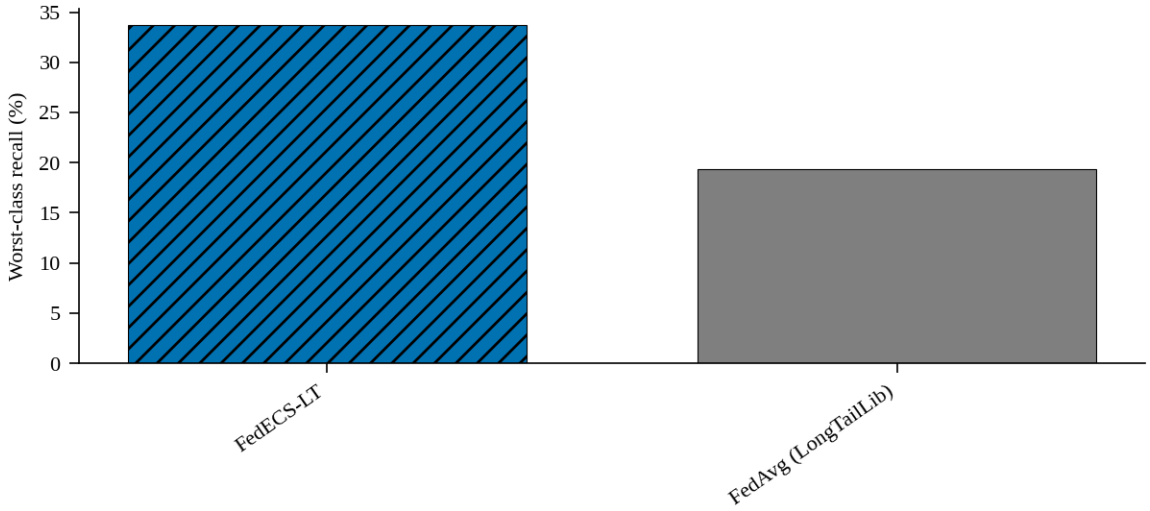


Fig. 70. Recall of the worst class; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

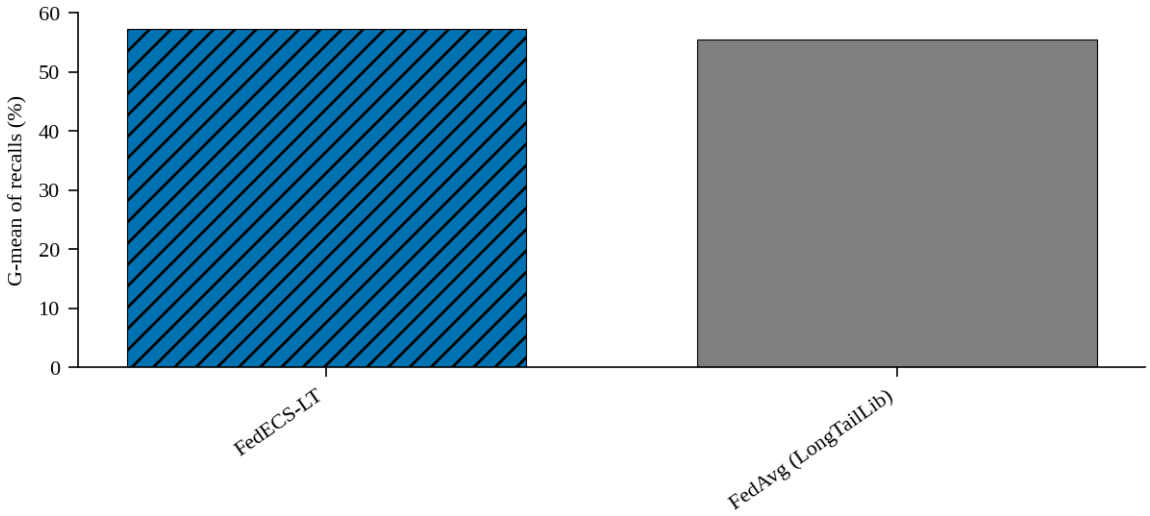


Fig. 71. Geometric mean of the class recalls (each floored at 1%); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

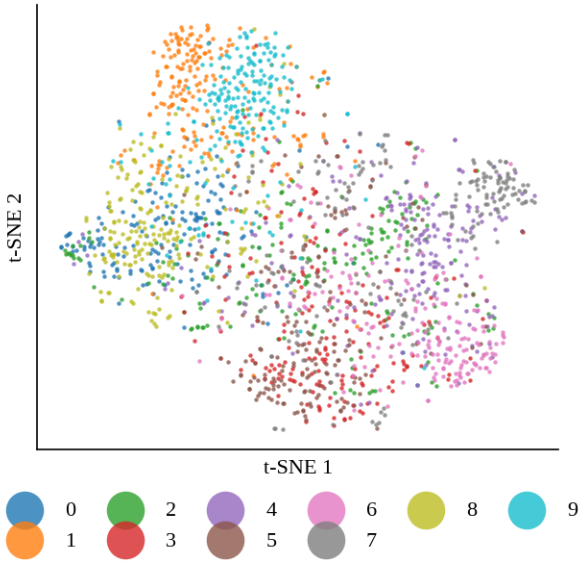


Fig. 72. t-SNE of the test-set embeddings of FedECS-LT's head CNN-1 (2,000 test images, first seed), coloured by true class. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

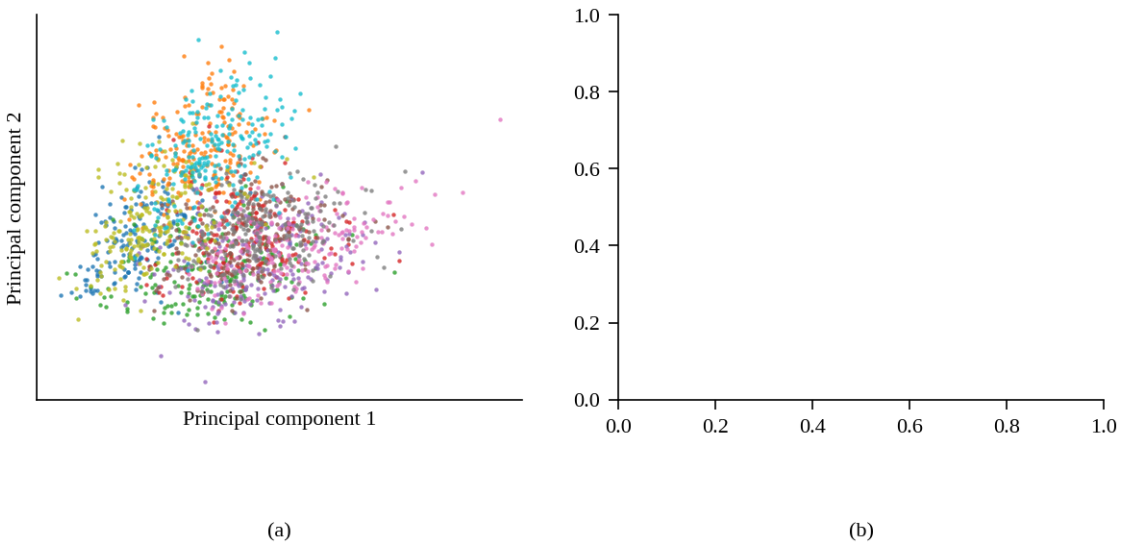


Fig. 73. First two principal components of the test-set embeddings: (a) FedECS-LT (b) GeFL-F [Kang et al., IEEE TMC 2025]; coloured by true class. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

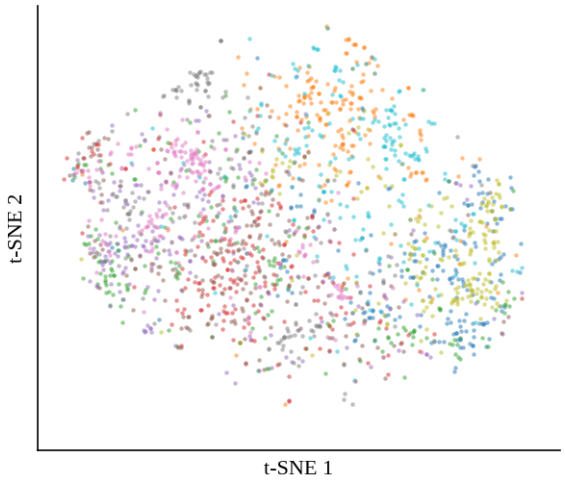


Fig. 74. t-SNE of real validation features (circles) and features generated by FedECS-LT's generator (crosses) in the frozen feature space, coloured by class (first seed). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

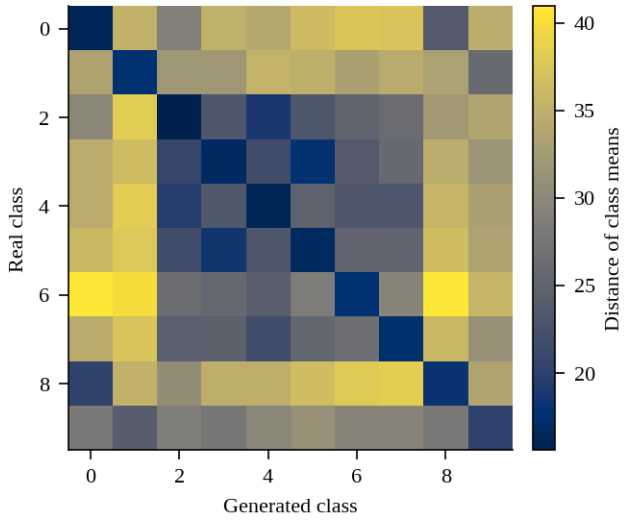


Fig. 75. Euclidean distance between the mean real feature of each class (validation set) and the mean generated feature of each class for FedECS-LT; a dark diagonal means generated classes sit at their real class. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

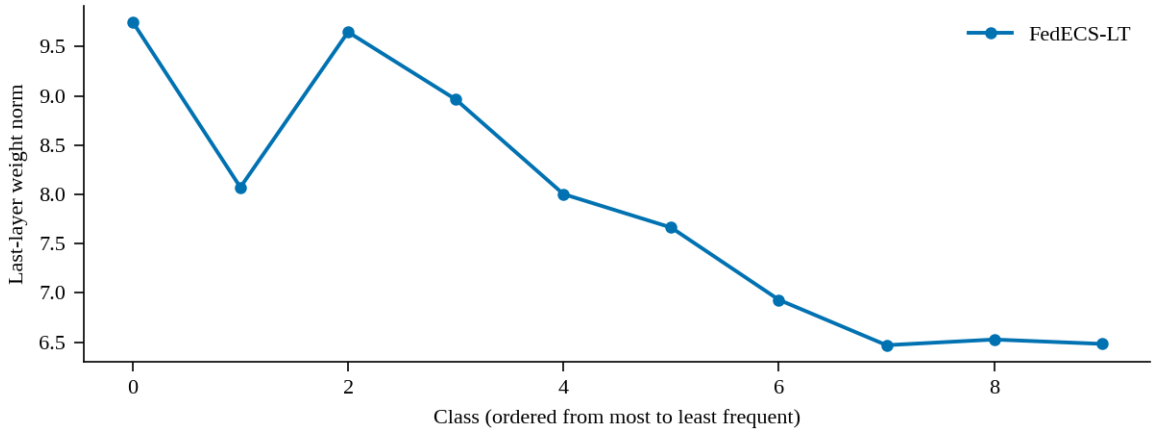


Fig. 76. L2 norm of each class's last-layer weight vector after Stage III, averaged over architectures and seeds; a falling line means the classifier favours frequent classes. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

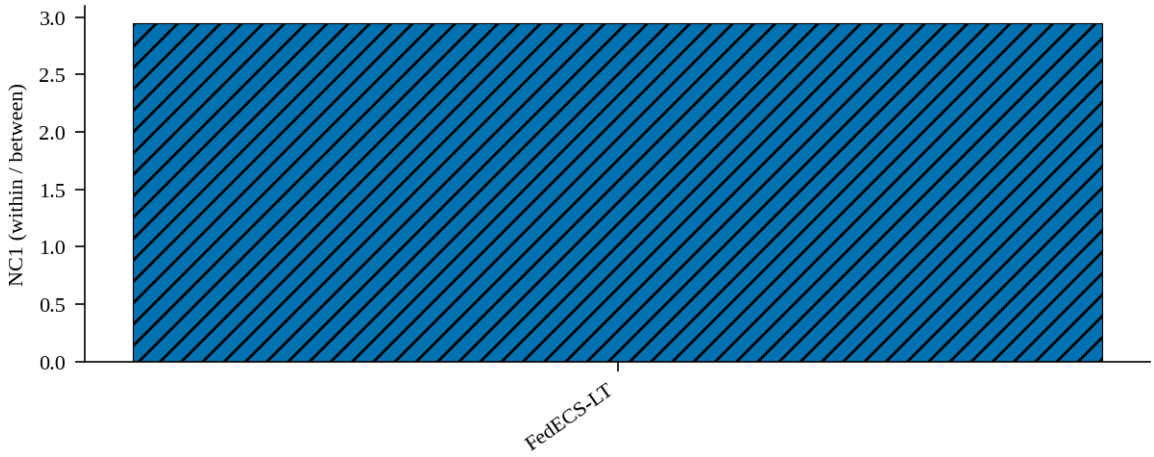


Fig. 77. Neural-collapse statistic NC1 = tr(S_W S_B^+)/C of the test-set embeddings (lower = tighter, better separated classes), mean over architectures and seeds. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

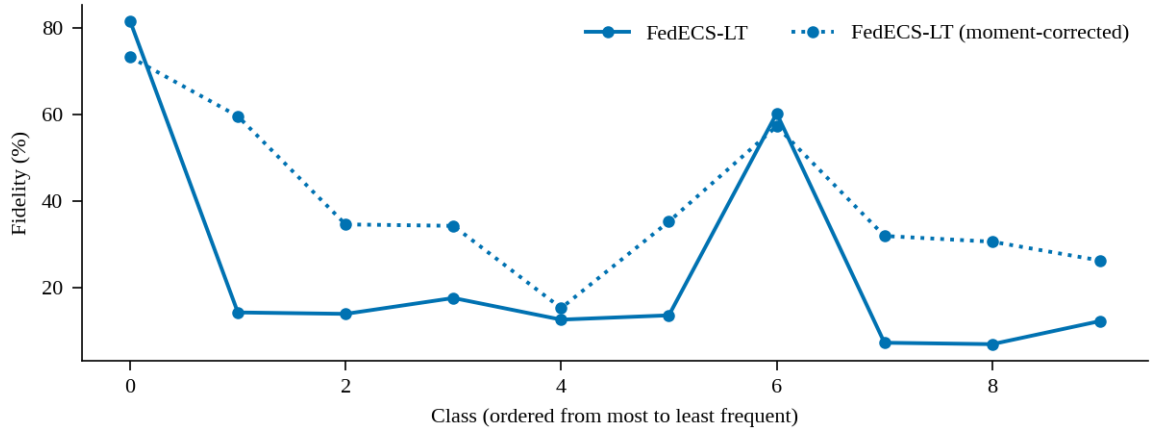


Fig. 78. Per-class fidelity of the trained generator: share of 300 generated class-c features that a referee classifier trained on the server's validation features assigns to class c; mean over seeds. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

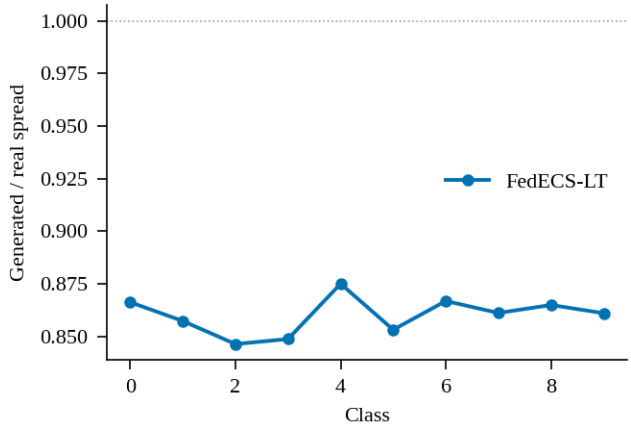


Fig. 79. Within-class spread of the generated features over the exact real spread (1 = realistic diversity, 0 = copies of the class mean), per class. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

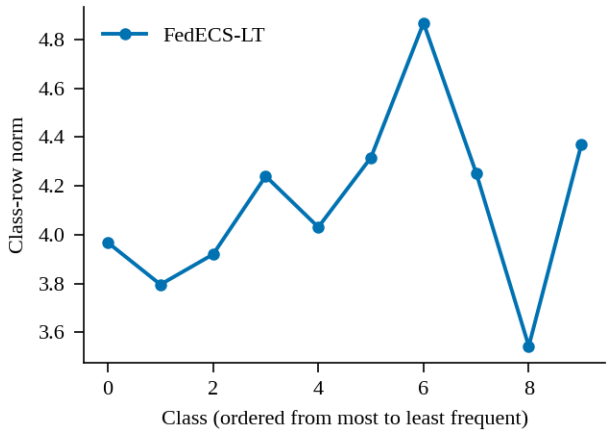


Fig. 80. L2 norm of each class's conditioning row of the generator after Stage II (class-row generators only); mean over seeds. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

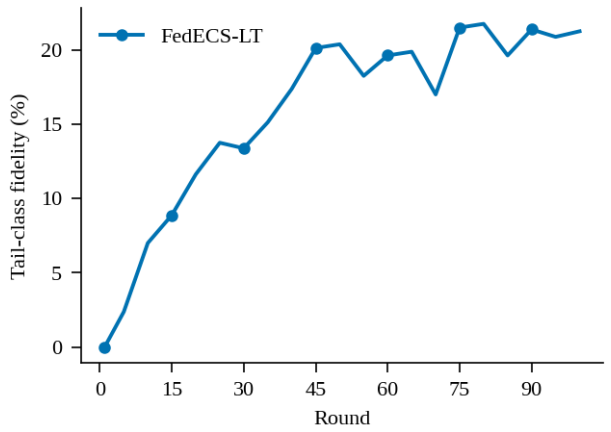


Fig. 81. Fidelity of the aggregated generator on the tail classes during Stage II (share of generated features the server's referee assigns to their class), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

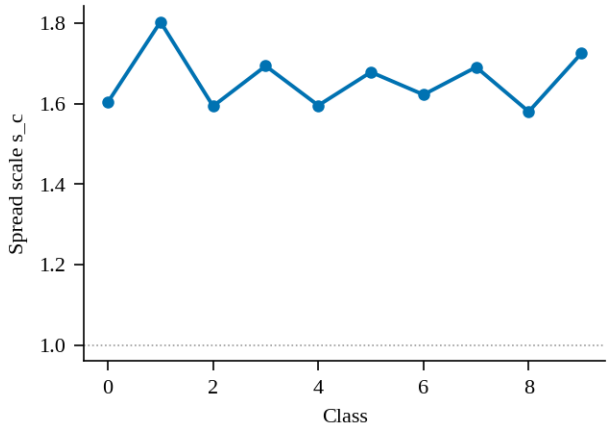


Fig. 82. Moment-correction scale s_c = sqrt(V_c / V~_c) per class: the factor by which the generator's within-class spread is stretched to the exact spread. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

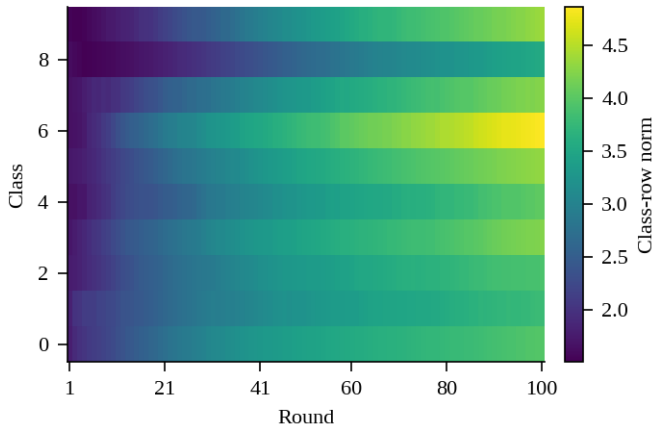


Fig. 83. Norm of each class's conditioning row of the aggregated generator over the rounds of Stage II for FedECS-LT (first seed). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

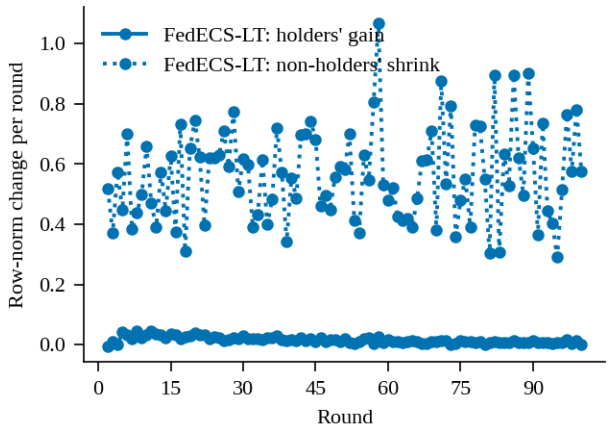


Fig. 84. Mean change of a class row's norm in one round of local training: growth on clients that hold the class against shrinkage on clients that do not (class-row generators, first seed). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

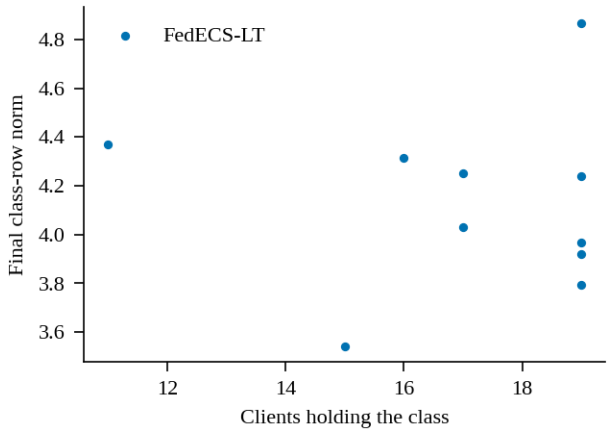


Fig. 85. Final norm of each class's conditioning row against the number of clients that hold the class; one point per class and seed. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

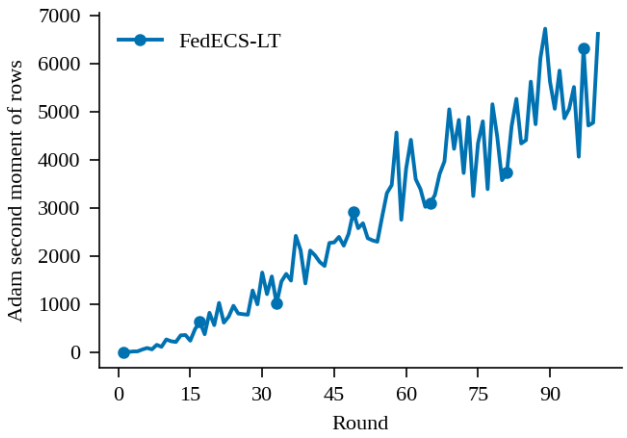


Fig. 86. Mean Adam second-moment estimate of the class rows over clients during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

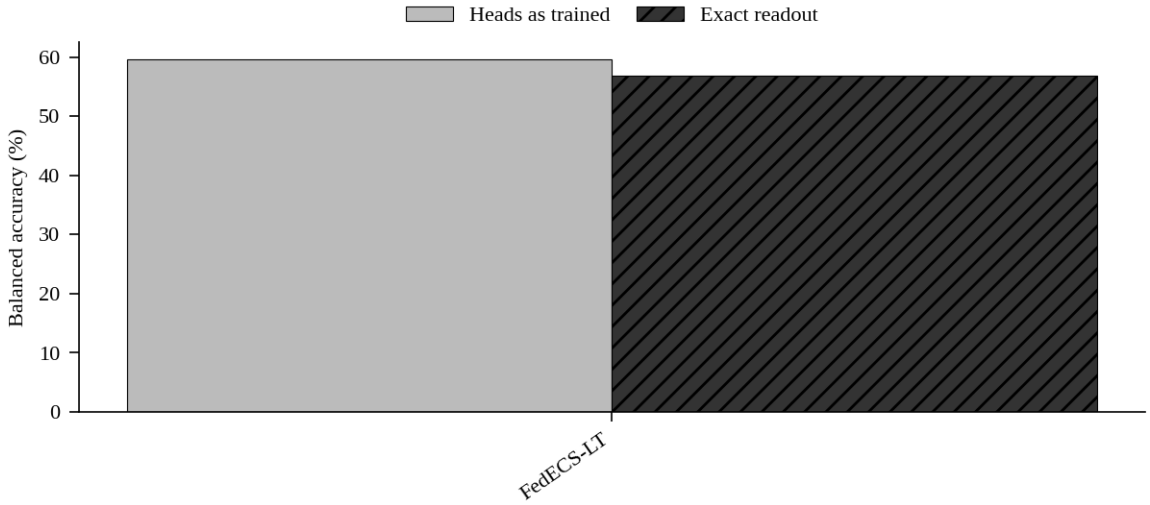


Fig. 87. Balanced accuracy of the heads as trained and after the exact class-balanced readout (last layer recomputed from secure sums of the clients' embeddings); mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

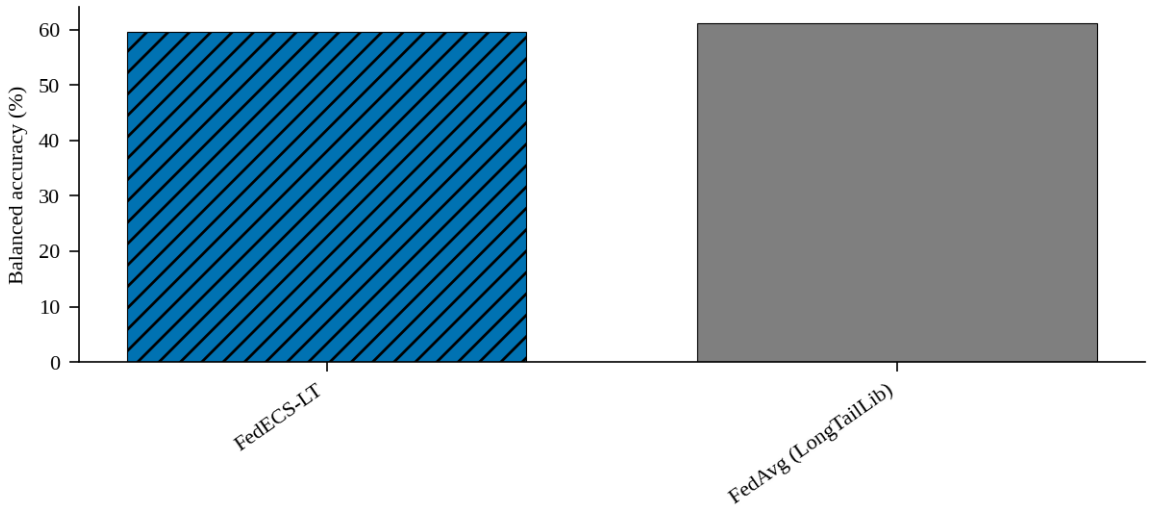


Fig. 88. Balanced accuracy on the test set (mean of the class recalls) of every method; mean over seeds, error bars one standard deviation. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

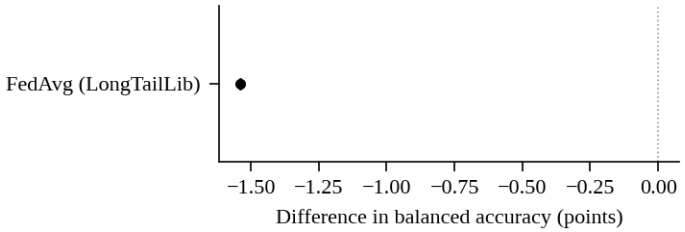


Fig. 89. Paired difference in balanced accuracy, FedECS-LT minus each baseline, over the same seeds and splits; points are means, bars 95% t confidence intervals; right of the dotted line FedECS-LT is better. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

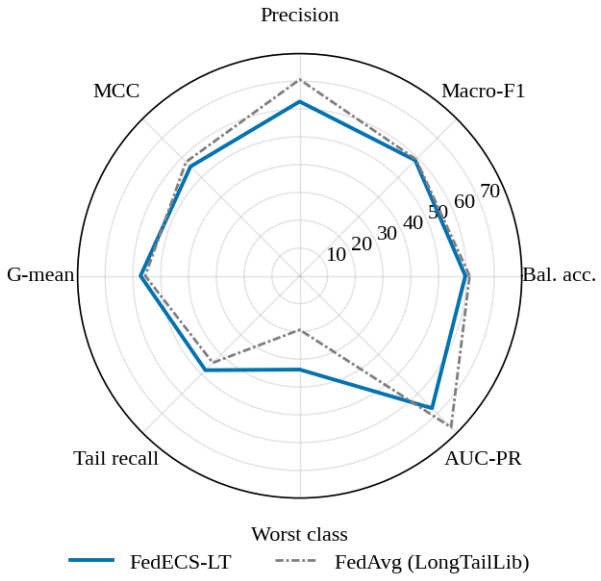


Fig. 90. Main metrics (%, test set, mean over seeds) of FedECS-LT and the five other methods with the highest balanced accuracy; larger is better on every axis. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

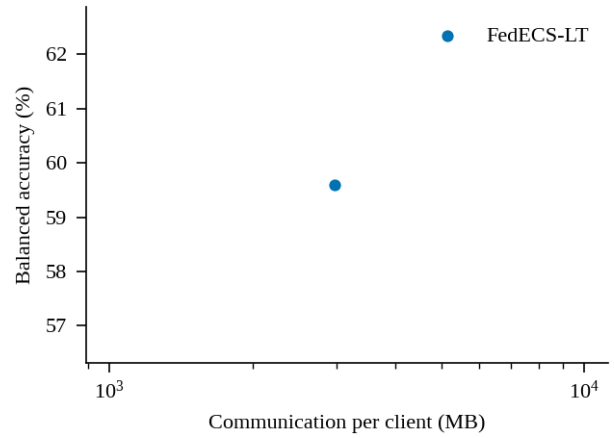


Fig. 91. Balanced accuracy against communication per client over the whole training (logarithmic axis); mean over seeds. Up and left is better. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

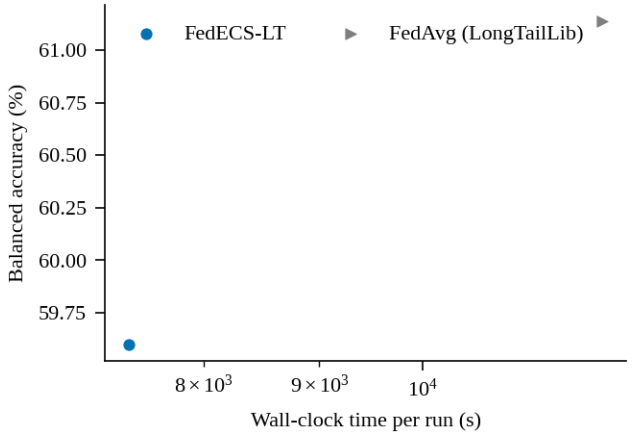


Fig. 92. Balanced accuracy against wall-clock time of one run on one GPU (logarithmic axis); mean over seeds. Up and left is better. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

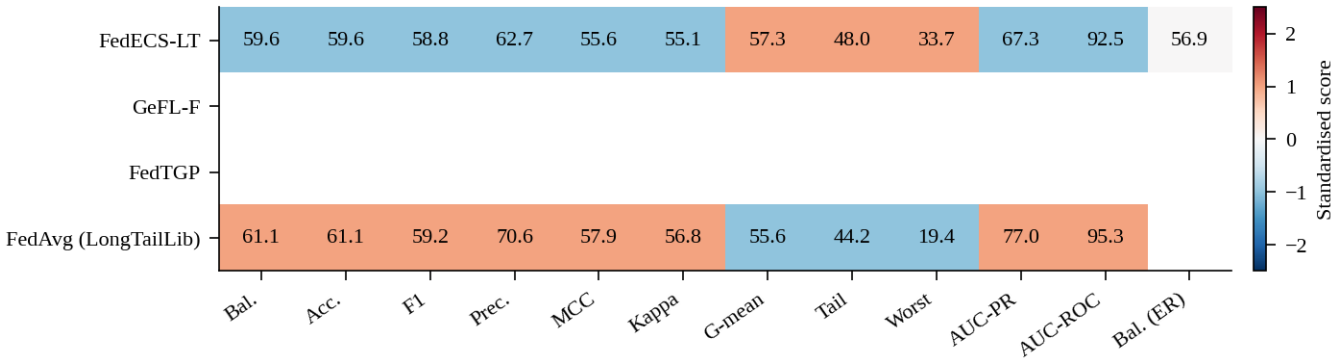


Fig. 93. Every main metric (%, test set, mean over seeds) of every method; colours are standardised per metric (red = better than the column mean). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

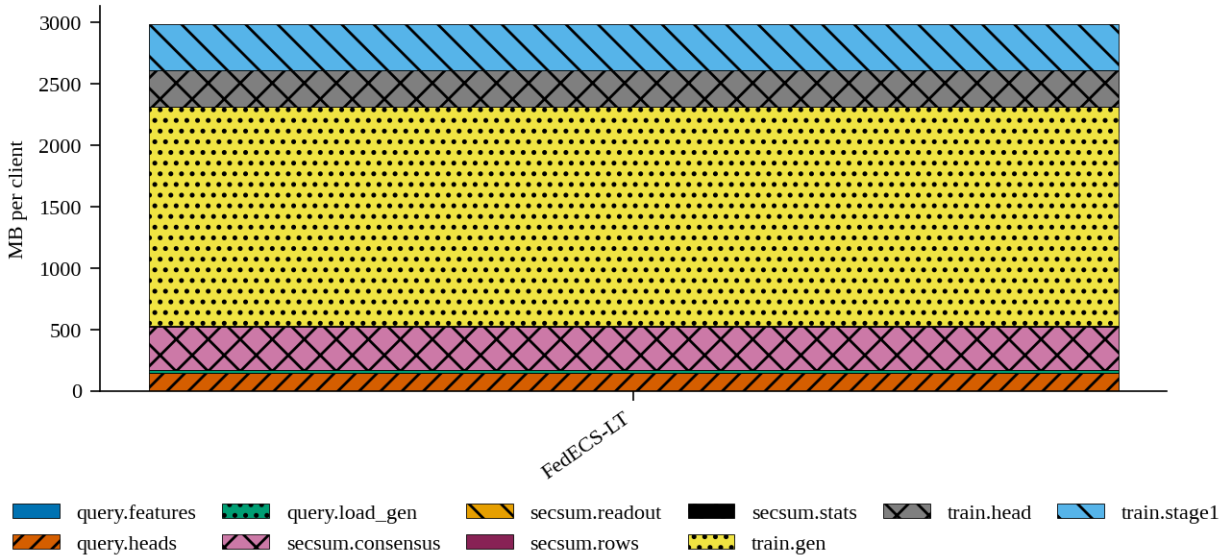


Fig. 94. Communication per client over the whole training, by message type (model weights, statistics, secure sums); first seed. Cached shared stages count once. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

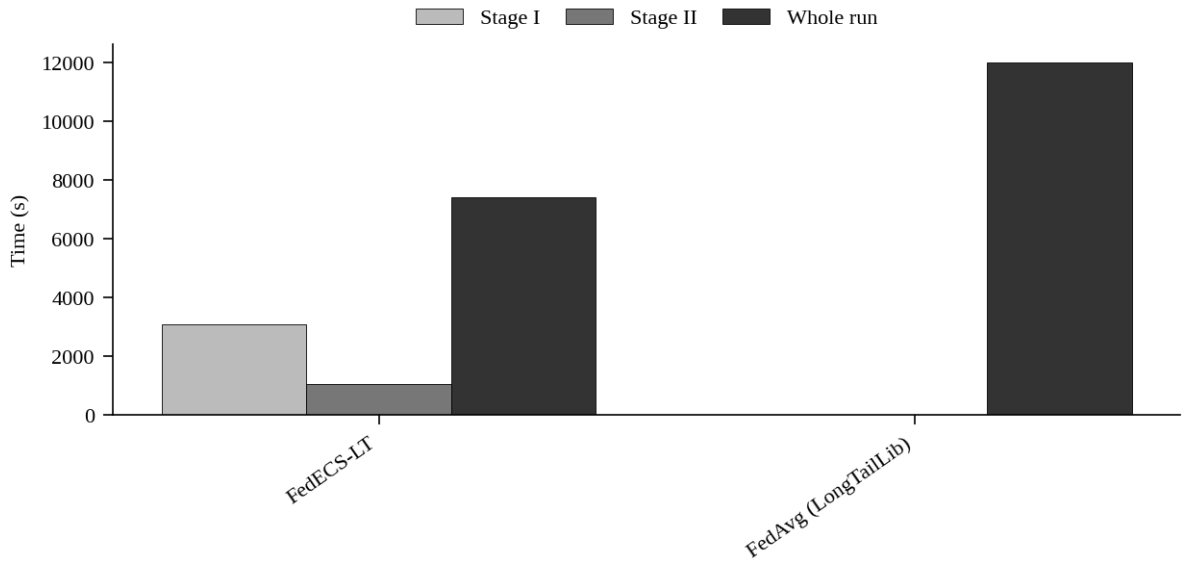


Fig. 95. Wall-clock time per stage on one GPU (simulation of all clients), mean over seeds; shared stages loaded from cache report the time of their first run. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

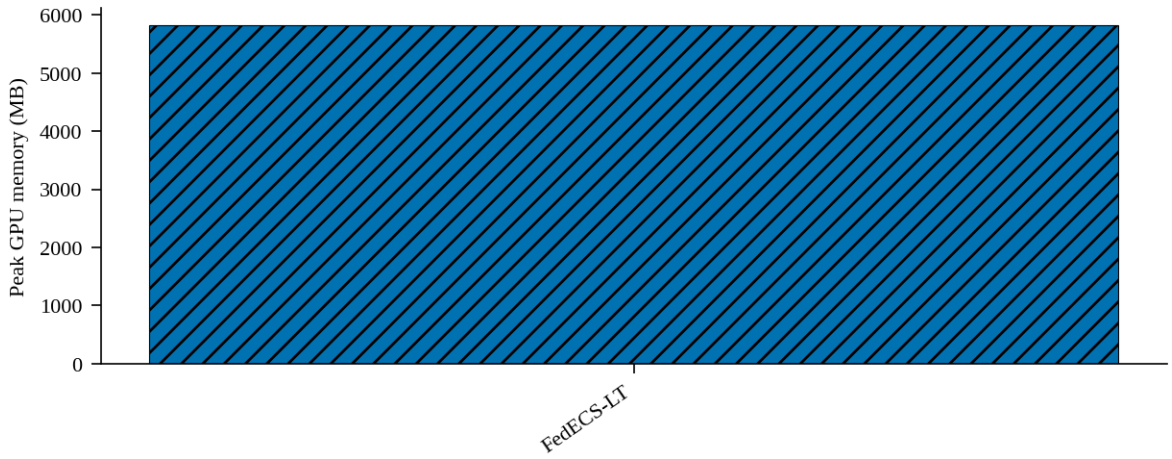


Fig. 96. Peak GPU memory of the server process during one run (mean over seeds). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

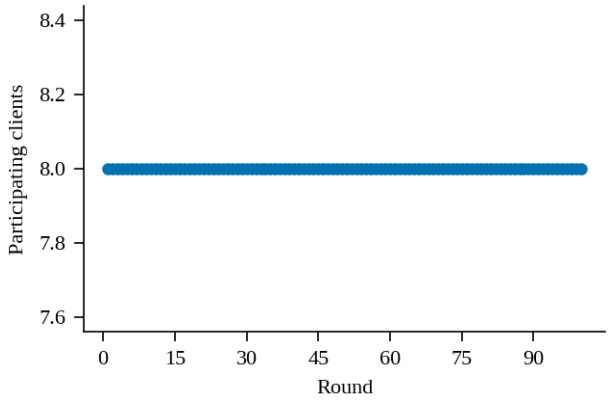


Fig. 97. Number of clients taking part in each round of Stage III (partial participation: the same seeded subset for every method). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

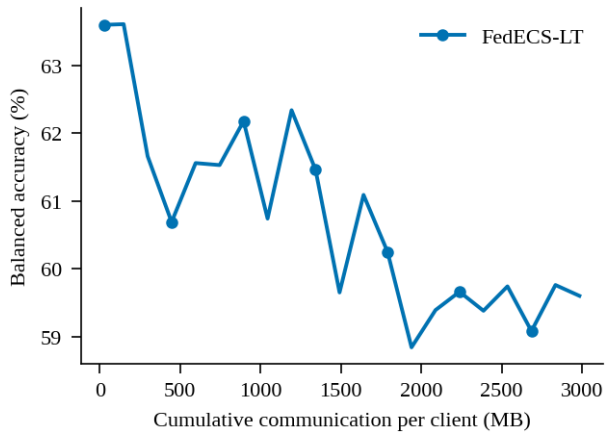


Fig. 98. Balanced accuracy of the aggregated heads against cumulative communication per client (Stage III rounds placed proportionally on the total). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

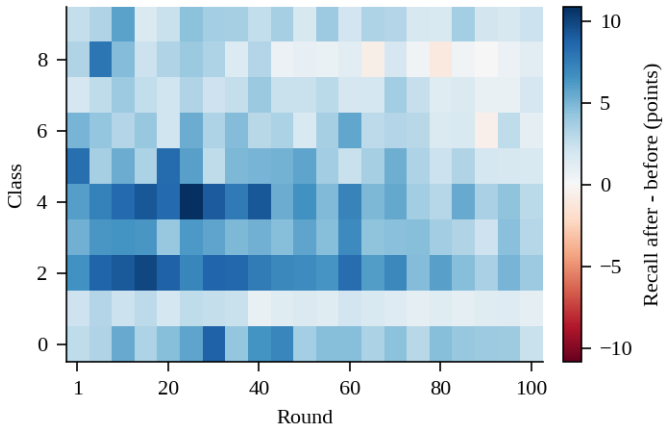


Fig. 99. Aggregation effect per class for FedECS-LT: recall of the aggregated heads minus the mean recall of the clients' local heads just before aggregation, at every evaluation round of Stage III (blue = aggregation helps); first seed. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

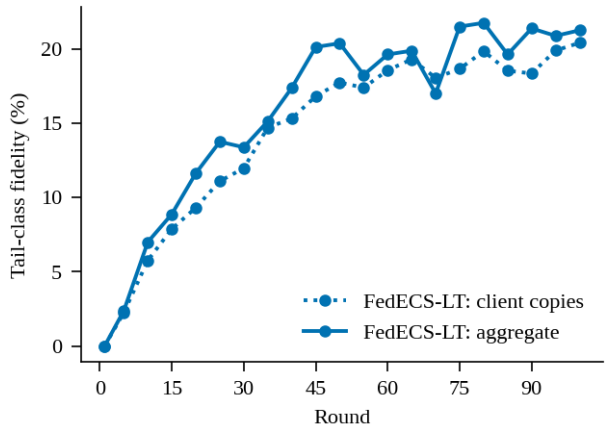


Fig. 100. Tail-class fidelity of the clients' generator copies just before aggregation (mean over clients) and of the aggregated generator, during Stage II. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

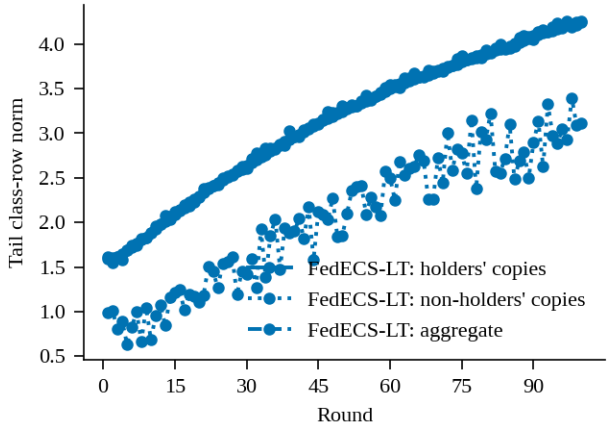


Fig. 101. Norm of the tail classes' conditioning rows in the copies of clients that hold the class, of clients that do not, and in the aggregate, during Stage II (class-row generators, first seed). CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

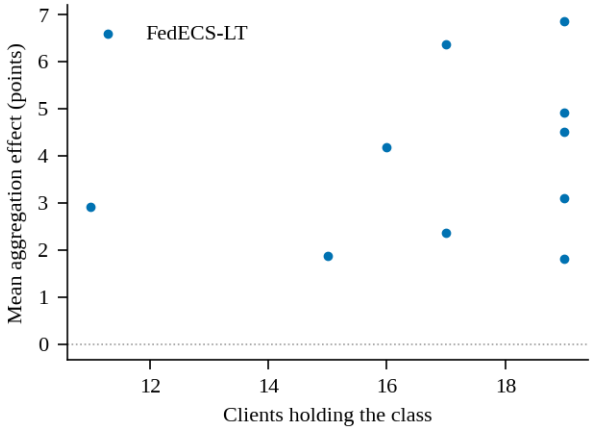


Fig. 102. Mean aggregation effect on a class's recall (aggregate minus local heads, over Stage III) against the number of clients holding the class; one point per class. CIFAR-10-LT (imbalance ratio 100), 20 clients, Dirichlet label split with alpha = 0.5, one ResNet-8 shared by all clients (standard CIFAR-LT class profile), 40% of the clients per round; seed 3.

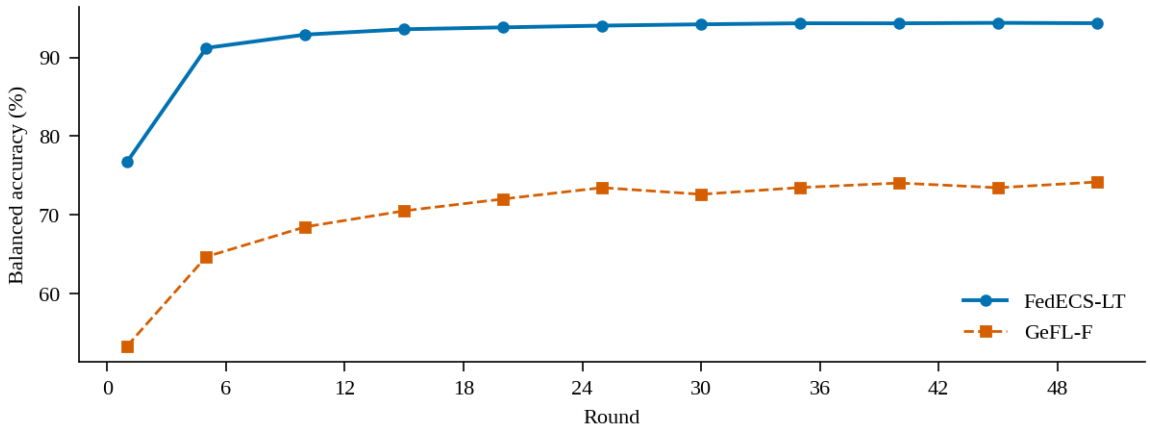


Fig. 103. Test-set balanced accuracy of the aggregated heads during head training (Stage III), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

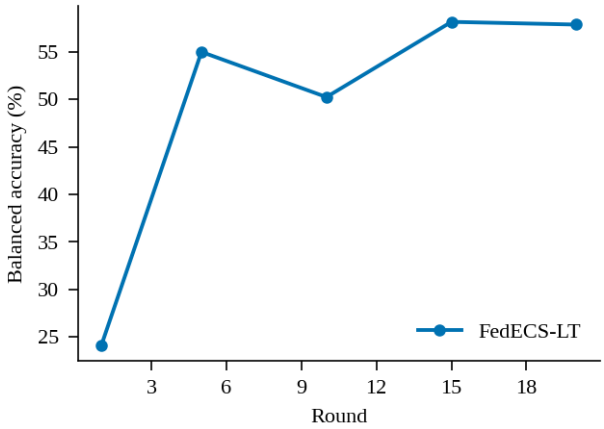


Fig. 104. Test-set balanced accuracy of the shared feature extractor with the architecture heads during Stage I, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

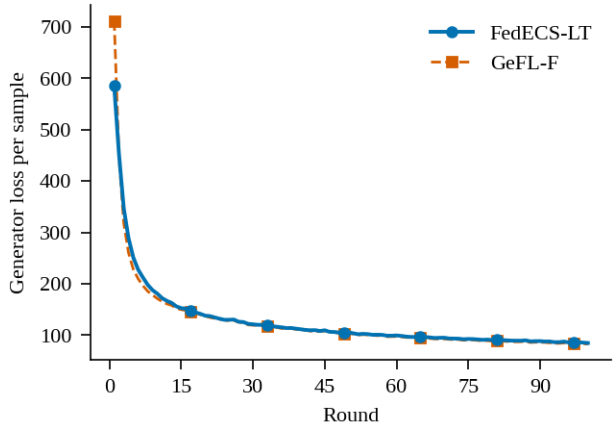


Fig. 105. Mean training loss of the clients' generator copies (reconstruction + KL) during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

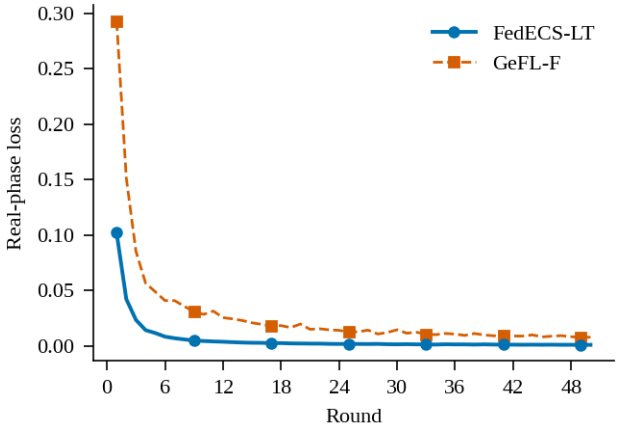


Fig. 106. Mean real-feature training loss of the clients' heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

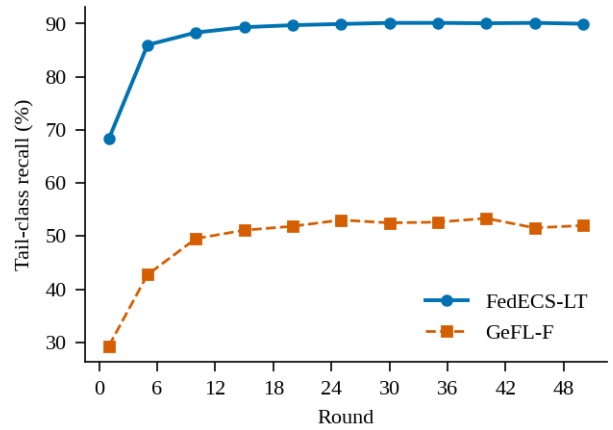


Fig. 107. Test-set mean recall of the tail classes (the rarest 40%) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

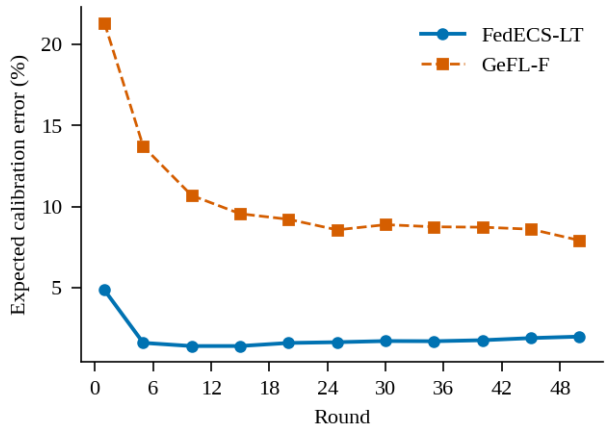


Fig. 108. Test-set expected calibration error (15 bins) of the aggregated heads during Stage III, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

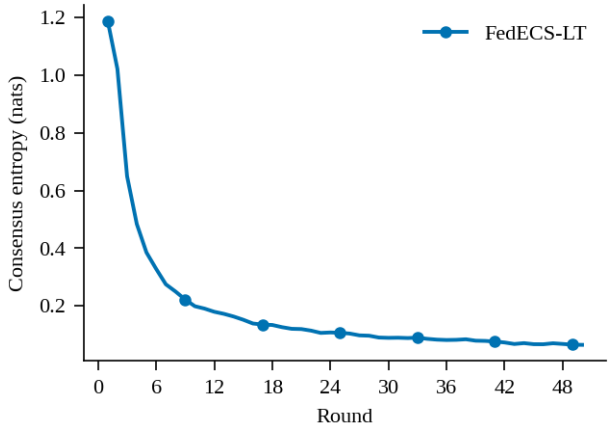


Fig. 109. Mean entropy of the consensus soft labels on the round's generated features, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

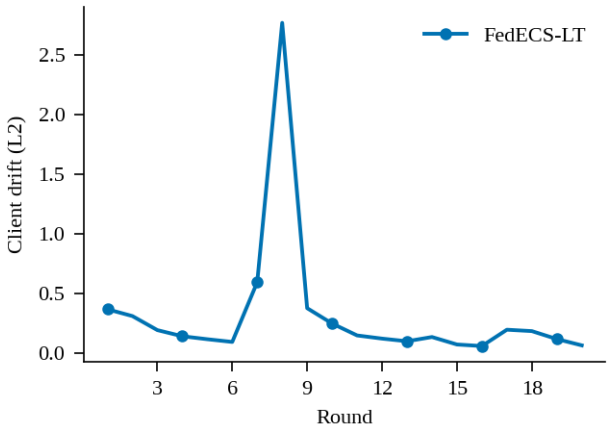


Fig. 110. Mean L2 distance of the clients' feature extractors from their average after each Stage I round (client drift), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

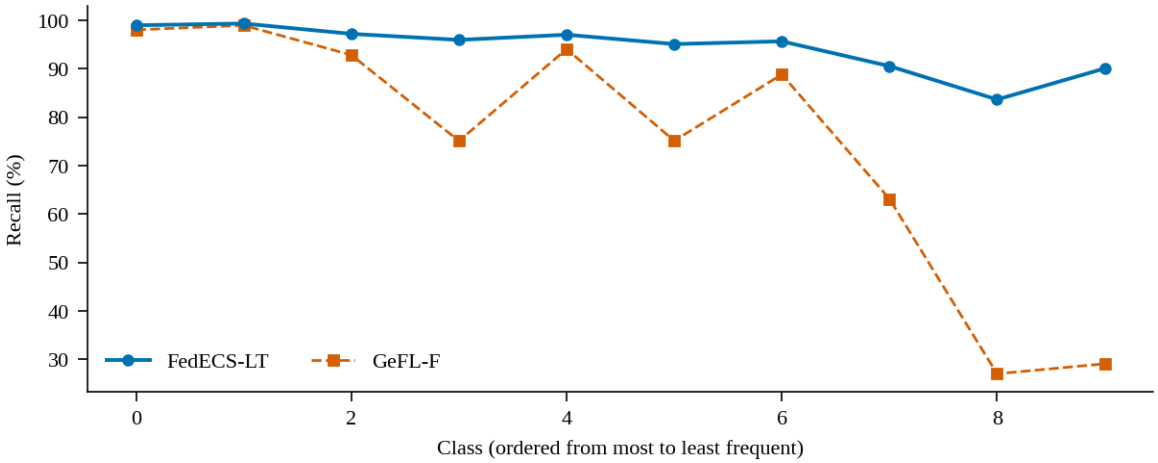


Fig. 111. Per-class recall on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

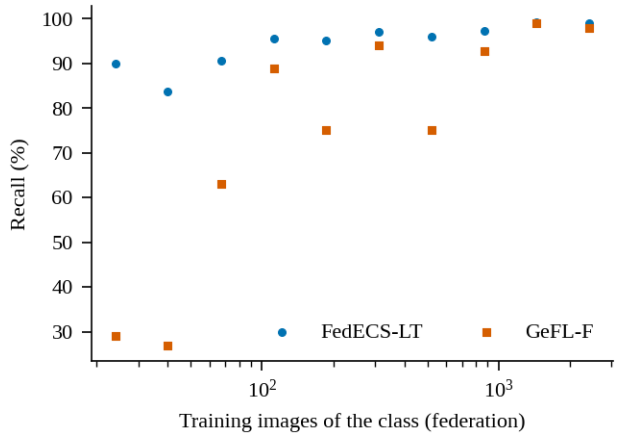


Fig. 112. Per-class recall against the number of training images of the class in the federation (logarithmic axis); each point is one class, mean over seeds. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

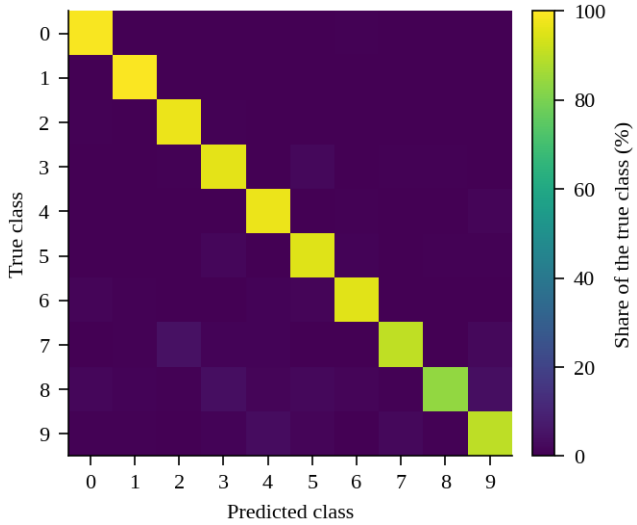


Fig. 113. Row-normalised confusion matrix of FedECS-LT on the test set (all client models, summed over seeds); classes ordered from most to least frequent. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

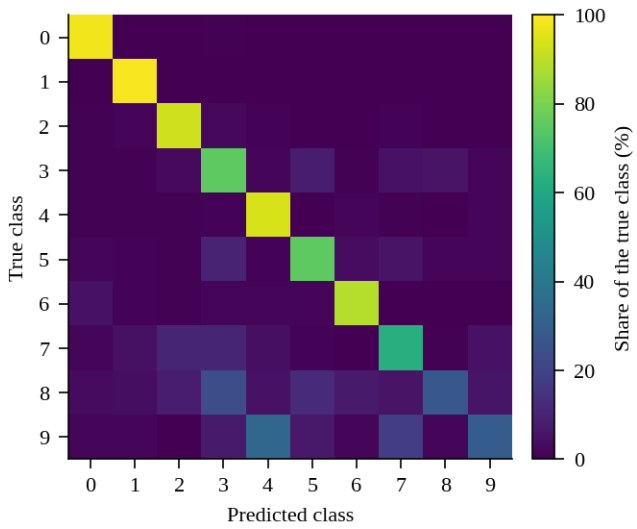


Fig. 114. Row-normalised confusion matrix of GeFL-F [Kang et al., IEEE TMC 2025] on the test set (all client models, summed over seeds); classes ordered from most to least frequent. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

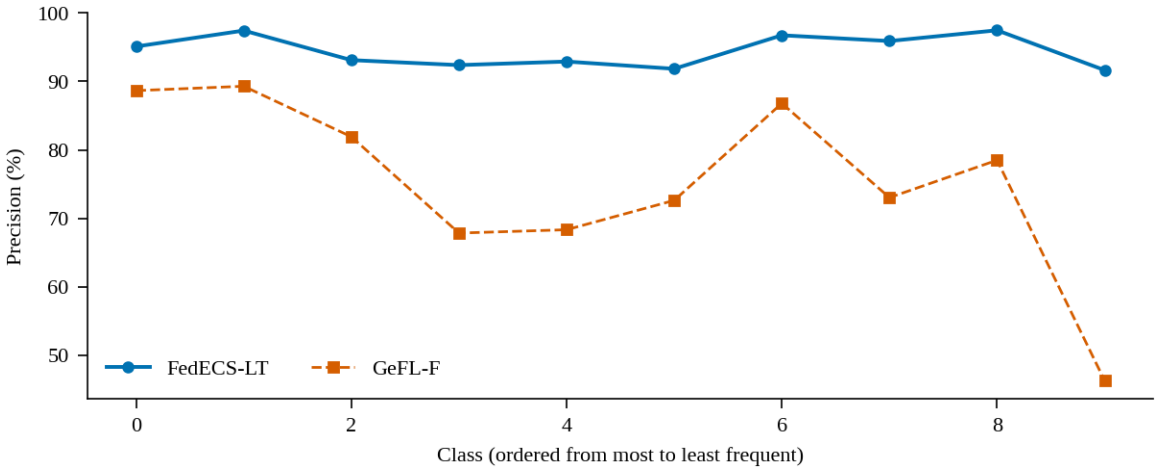


Fig. 115. Per-class precision on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

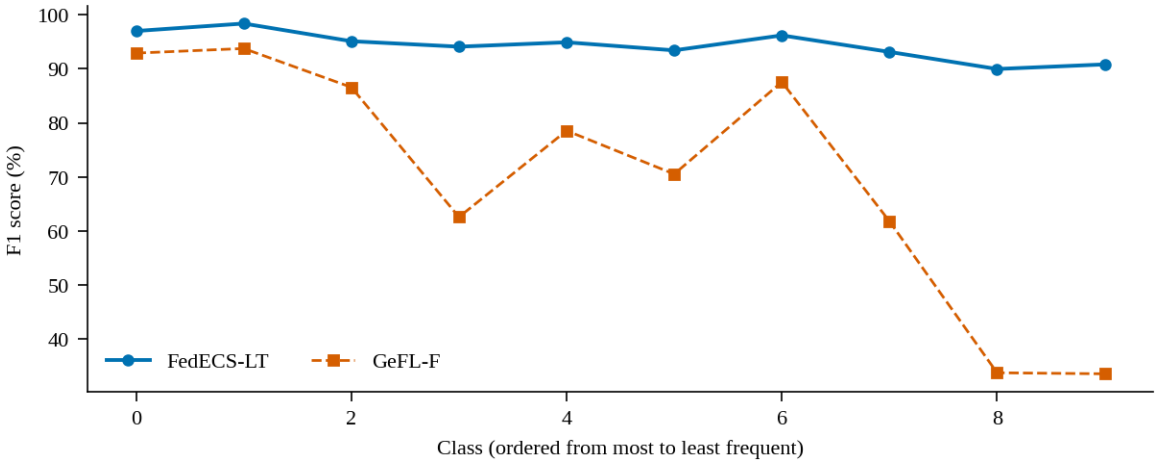


Fig. 116. Per-class F1 score on the test set after training (the clients' models as trained, mean over them and over seeds); classes are ordered from the most to the least frequent in the federation's training data. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

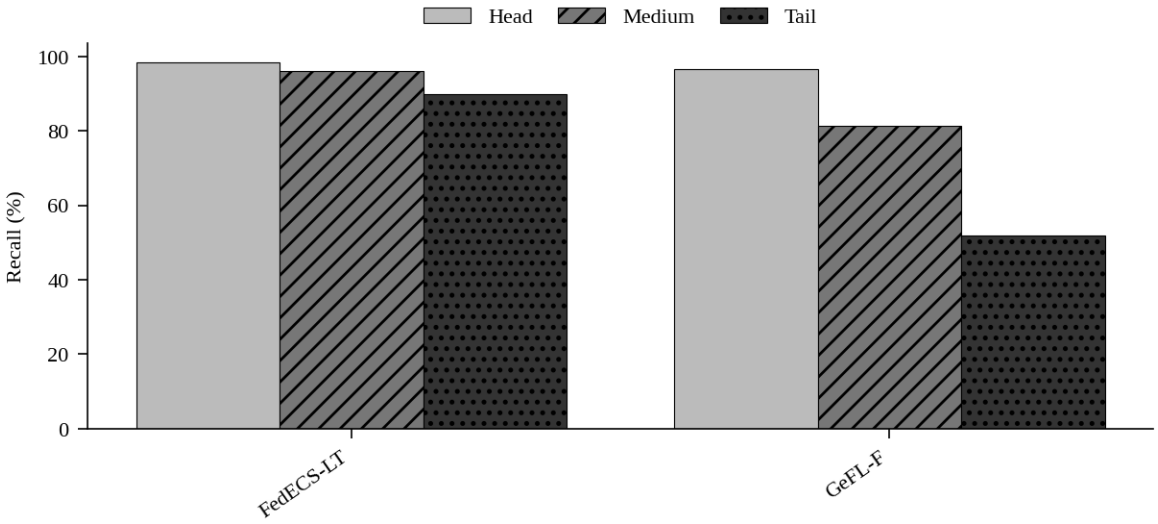


Fig. 117. Mean recall of the head (most frequent 30%), medium (next 30%) and tail (rarest 40%) classes; mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

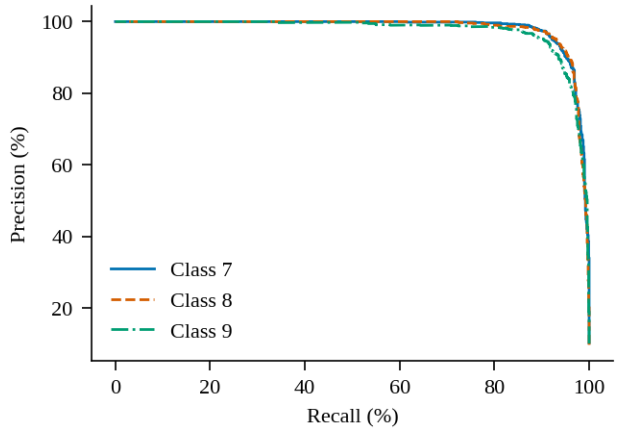


Fig. 118. Precision-recall curves of the three rarest classes for FedECS-LT (averaged softmax of all client models, first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

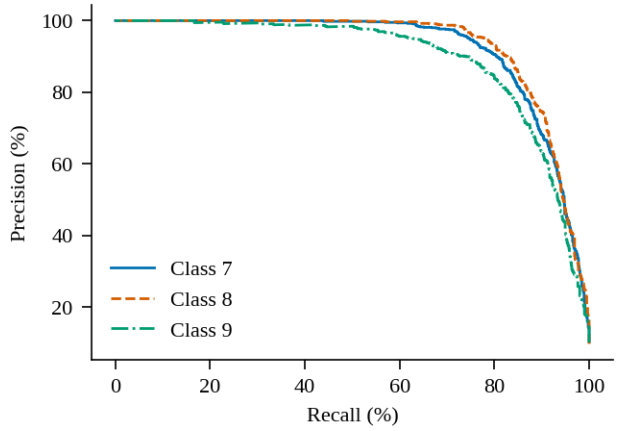


Fig. 119. Precision-recall curves of the three rarest classes for GeFL-F [Kang et al., IEEE TMC 2025] (averaged softmax of all client models, first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

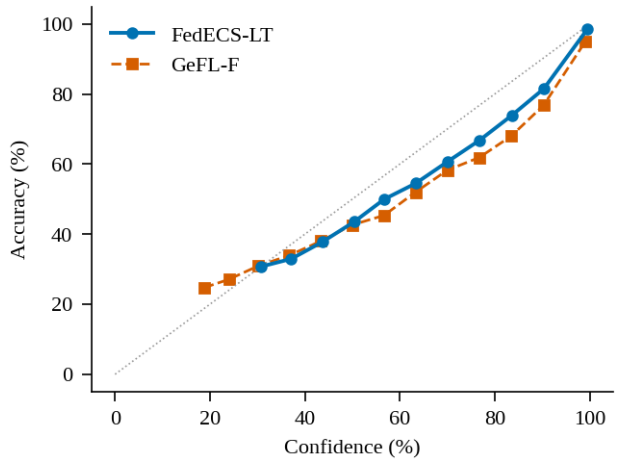


Fig. 120. Reliability diagram: accuracy against confidence in 15 equal-width bins (bins with more than 20 test images); the dotted line is perfect calibration. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

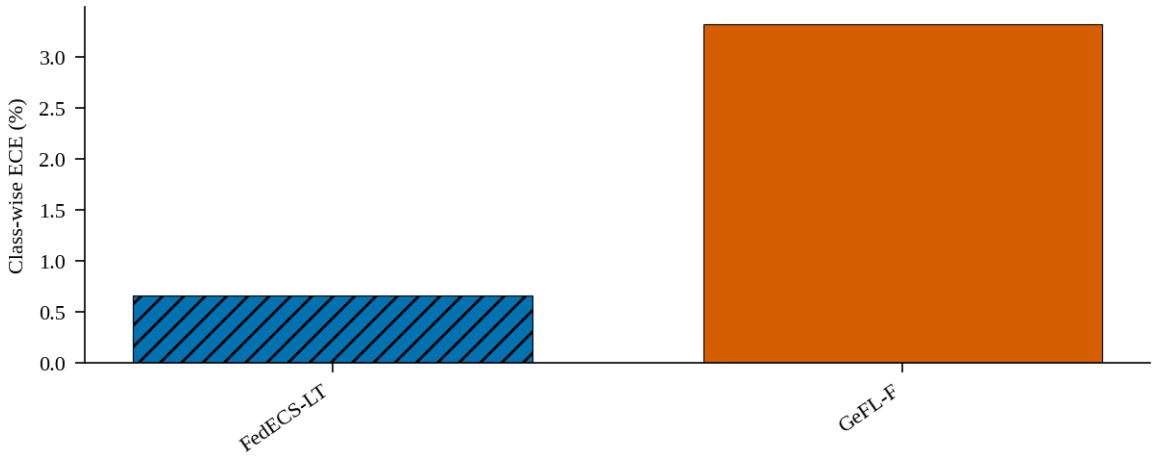


Fig. 121. Class-wise expected calibration error (mean over classes of the calibration error of p(c|x)); mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

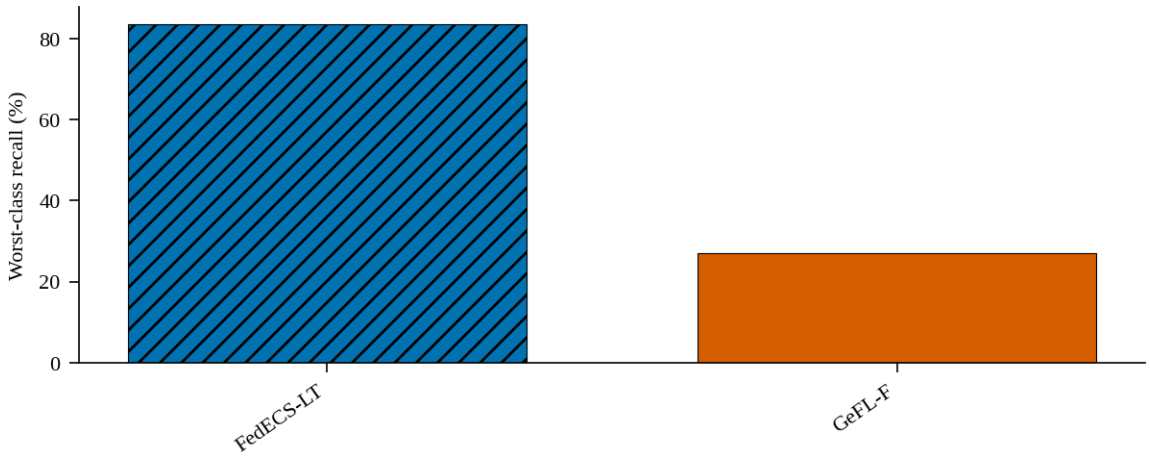


Fig. 122. Recall of the worst class; mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

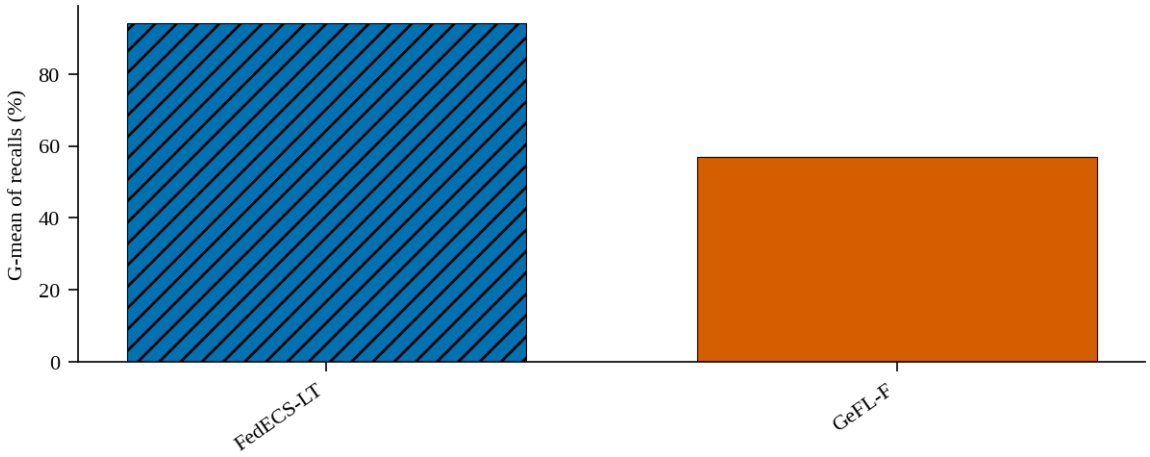


Fig. 123. Geometric mean of the class recalls (each floored at 1%); mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

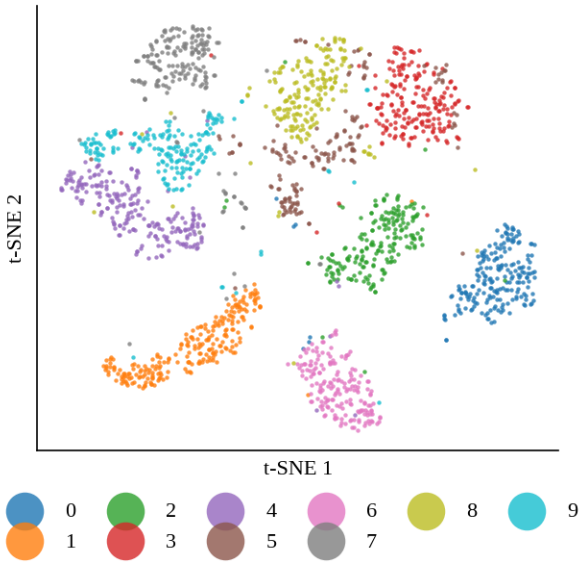


Fig. 124. t-SNE of the test-set embeddings of FedECS-LT's head CNN-1 (2,000 test images, first seed), coloured by true class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

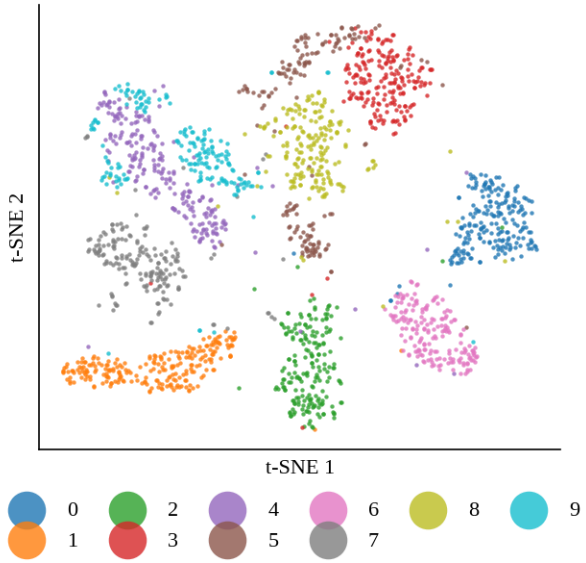
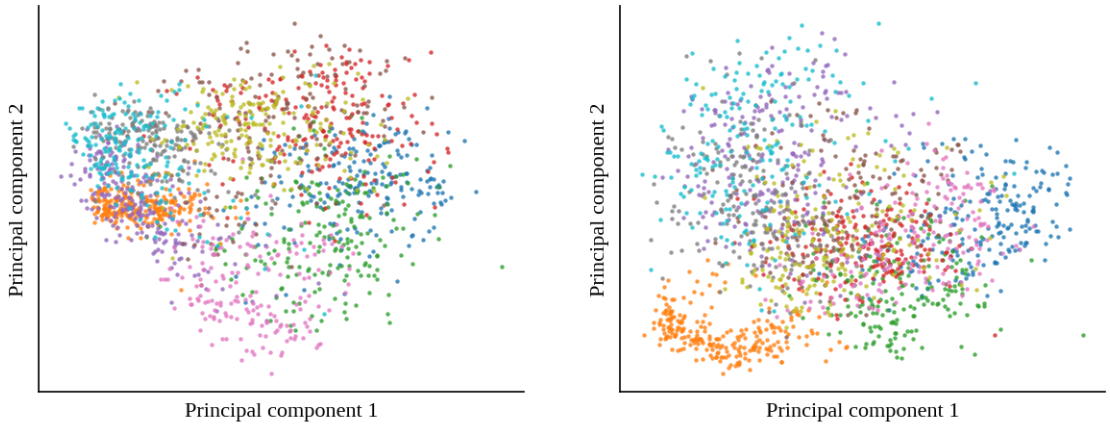


Fig. 125. t-SNE of the test-set embeddings of GeFL-F [Kang et al., IEEE TMC 2025]'s head CNN-1 (2,000 test images, first seed), coloured by true class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.



(a) (b)

Fig. 126. First two principal components of the test-set embeddings: (a) FedECS-LT (b) GeFL-F [Kang et al., IEEE TMC 2025]; coloured by true class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

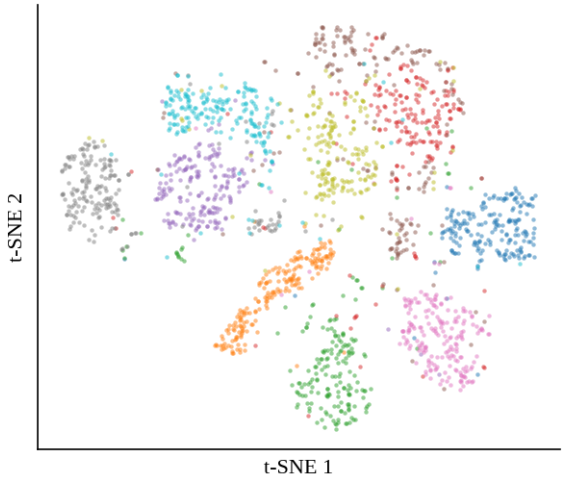


Fig. 127. t-SNE of real validation features (circles) and features generated by FedECS-LT's generator (crosses) in the frozen feature space, coloured by class (first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

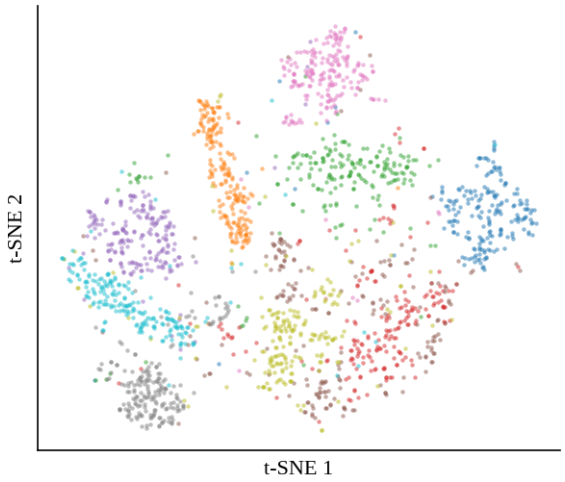


Fig. 128. t-SNE of real validation features (circles) and features generated by GeFL-F [Kang et al., IEEE TMC 2025]'s generator (crosses) in the frozen feature space, coloured by class (first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

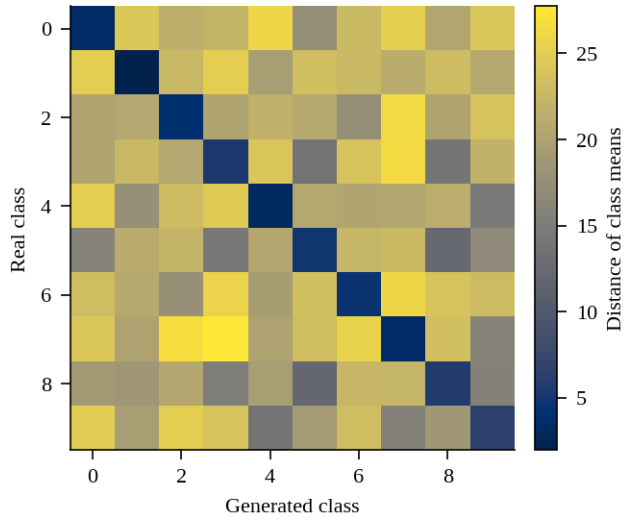


Fig. 129. Euclidean distance between the mean real feature of each class (validation set) and the mean generated feature of each class for FedECS-LT; a dark diagonal means generated classes sit at their real class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

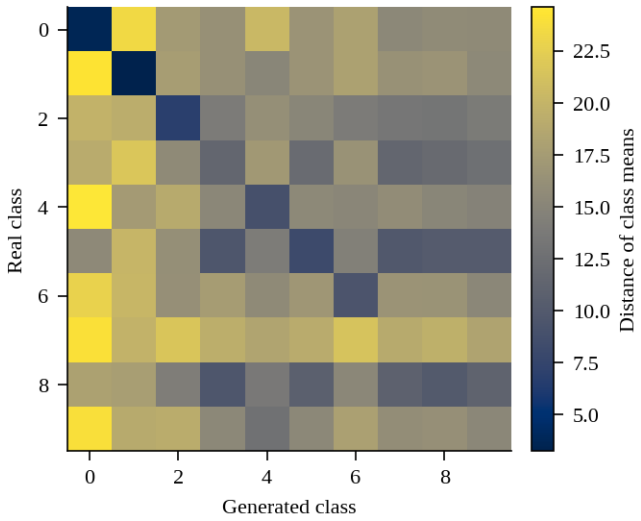


Fig. 130. Euclidean distance between the mean real feature of each class (validation set) and the mean generated feature of each class for GeFL-F [Kang et al., IEEE TMC 2025]; a dark diagonal means generated classes sit at their real class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

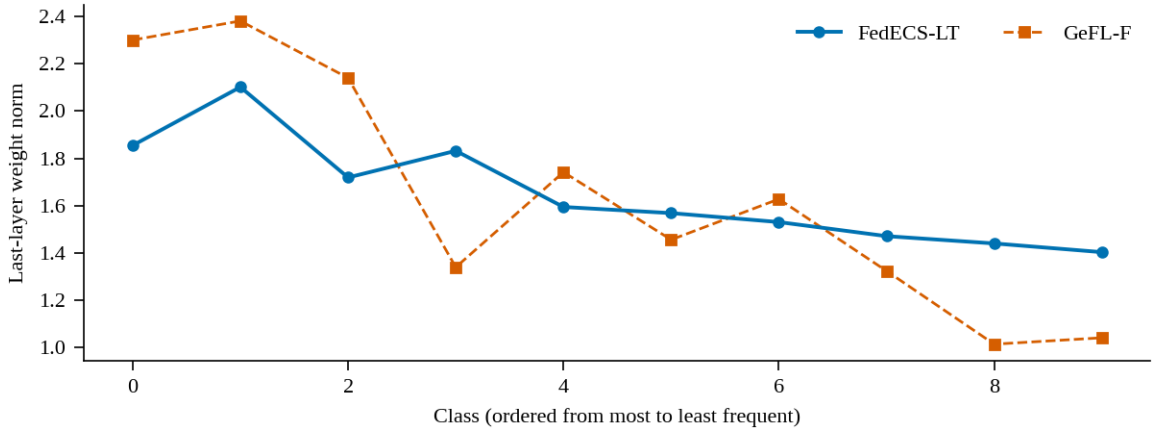


Fig. 131. L2 norm of each class's last-layer weight vector after Stage III, averaged over architectures and seeds; a falling line means the classifier favours frequent classes. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

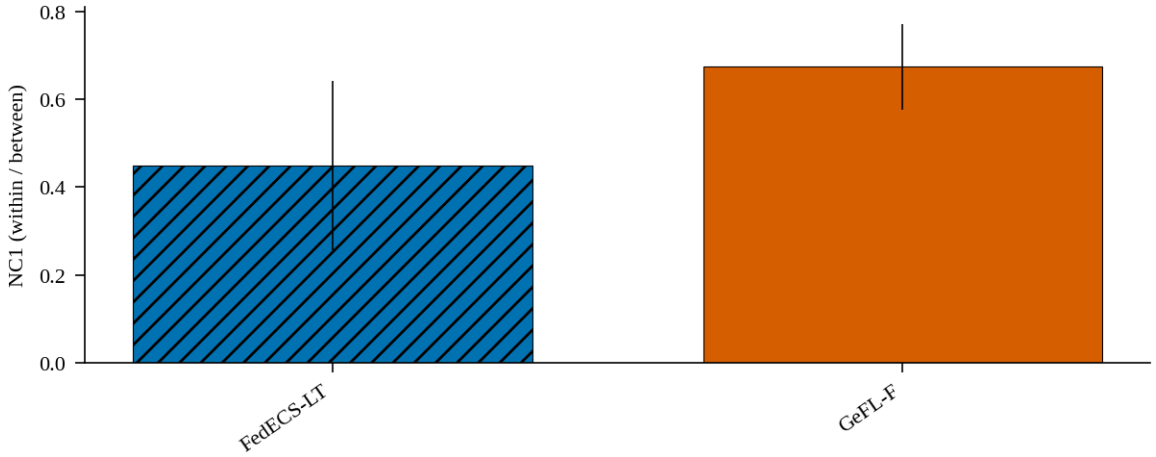


Fig. 132. Neural-collapse statistic NC1 = tr(S_W S_B^+)/C of the test-set embeddings (lower = tighter, better separated classes), mean over architectures and seeds. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

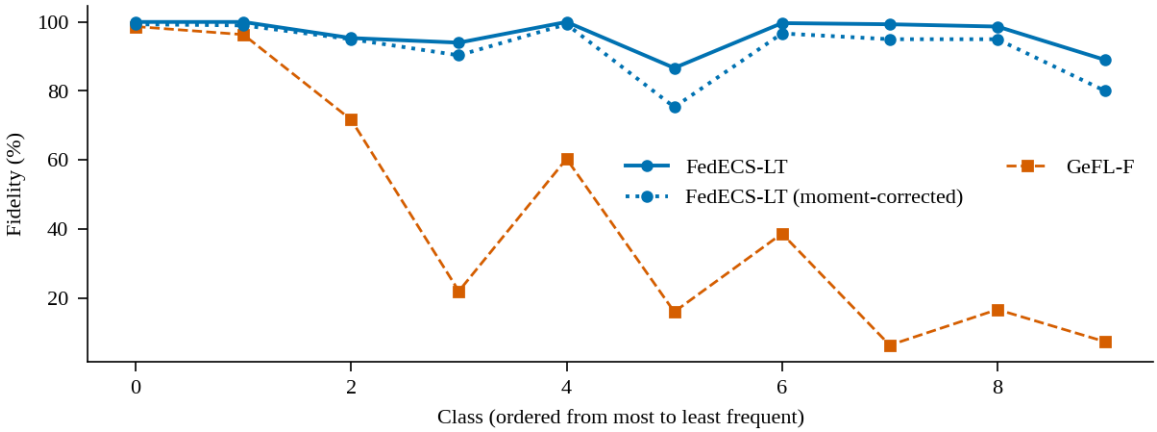


Fig. 133. Per-class fidelity of the trained generator: share of 300 generated class-c features that a referee classifier trained on the server's validation features assigns to class c; mean over seeds. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

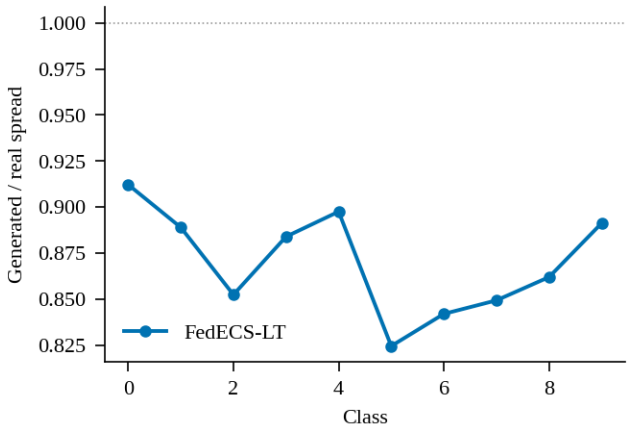


Fig. 134. Within-class spread of the generated features over the exact real spread (1 = realistic diversity, 0 = copies of the class mean), per class. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

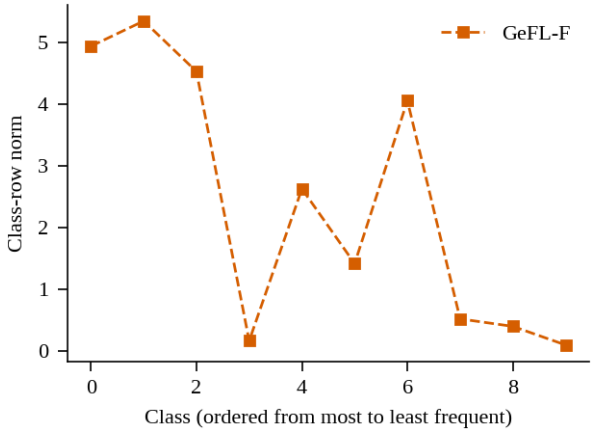


Fig. 135. L2 norm of each class's conditioning row of the generator after Stage II (class-row generators only); mean over seeds. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

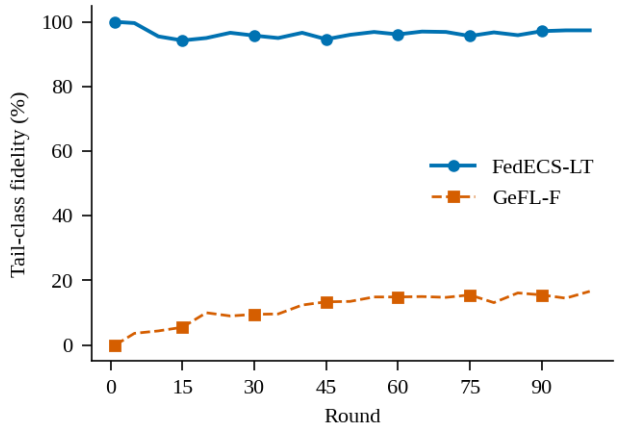


Fig. 136. Fidelity of the aggregated generator on the tail classes during Stage II (share of generated features the server's referee assigns to their class), per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

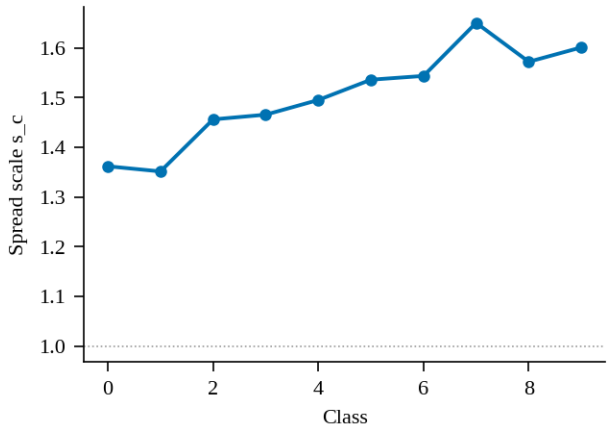


Fig. 137. Moment-correction scale s_c = sqrt(V_c / V~_c) per class: the factor by which the generator's within-class spread is stretched to the exact spread. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

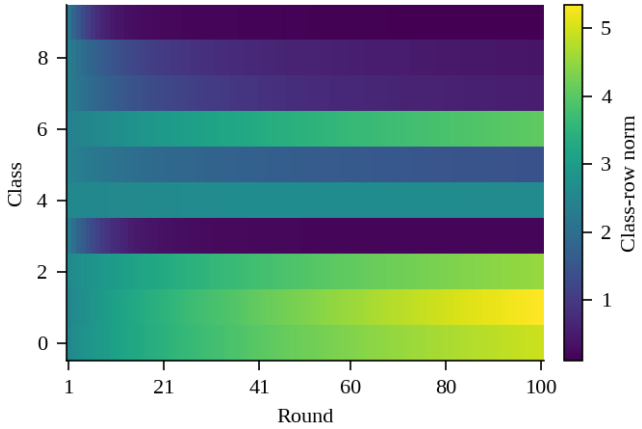


Fig. 138. Norm of each class's conditioning row of the aggregated generator over the rounds of Stage II for GeFL-F [Kang et al., IEEE TMC 2025] (first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

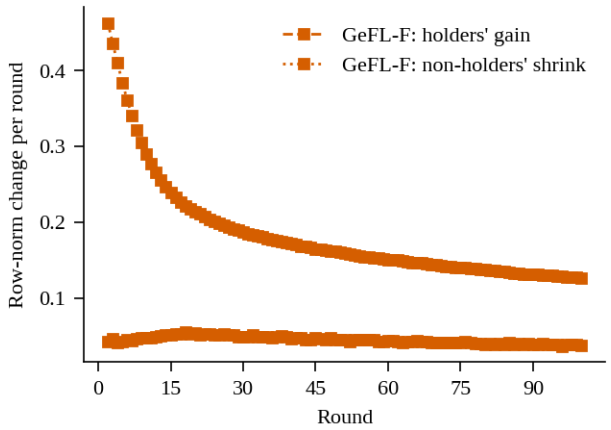


Fig. 139. Mean change of a class row's norm in one round of local training: growth on clients that hold the class against shrinkage on clients that do not (class-row generators, first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

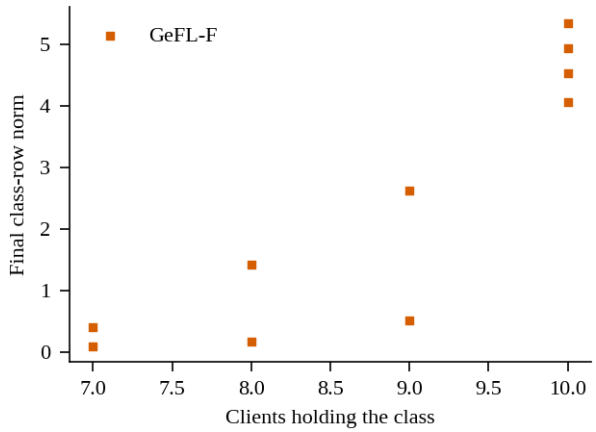


Fig. 140. Final norm of each class's conditioning row against the number of clients that hold the class; one point per class and seed. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

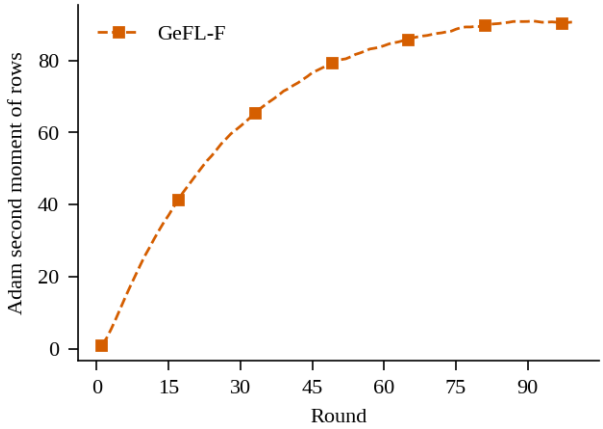


Fig. 141. Mean Adam second-moment estimate of the class rows over clients during Stage II, per evaluation round (every 5 rounds), mean over seeds; the shaded band is one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

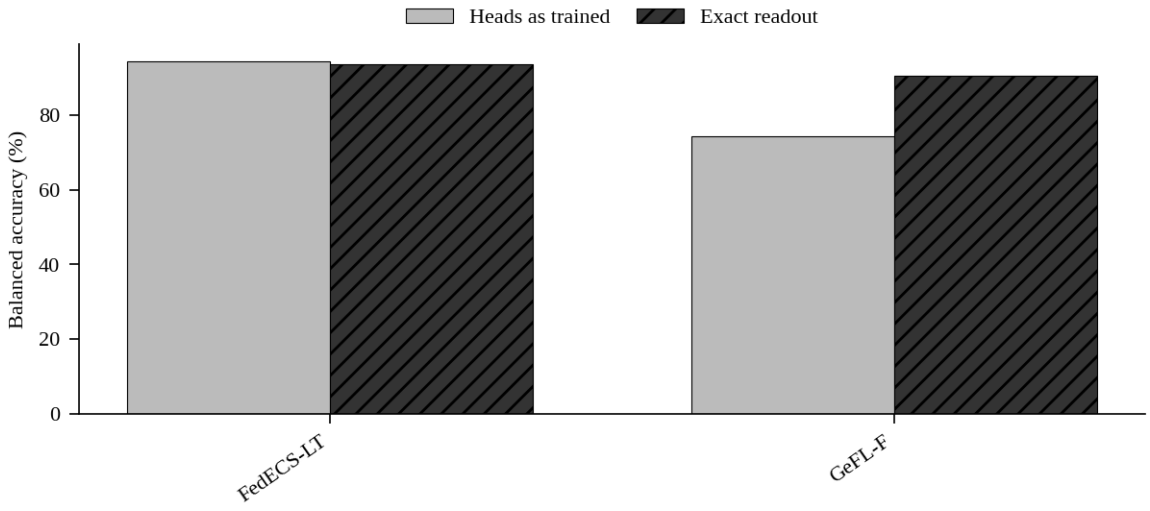


Fig. 142. Balanced accuracy of the heads as trained and after the exact class-balanced readout (last layer recomputed from secure sums of the clients' embeddings); mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

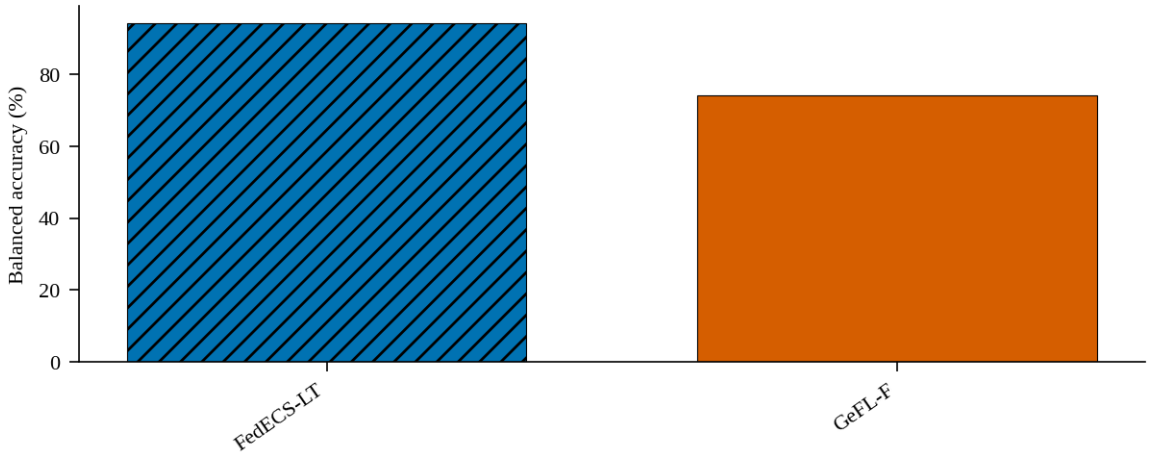


Fig. 143. Balanced accuracy on the test set (mean of the class recalls) of every method; mean over seeds, error bars one standard deviation. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

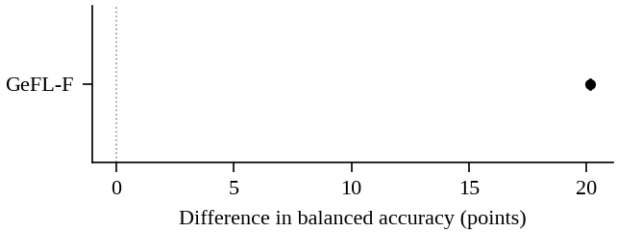


Fig. 144. Paired difference in balanced accuracy, FedECS-LT minus each baseline, over the same seeds and splits; points are means, bars 95% t confidence intervals; right of the dotted line FedECS-LT is better. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

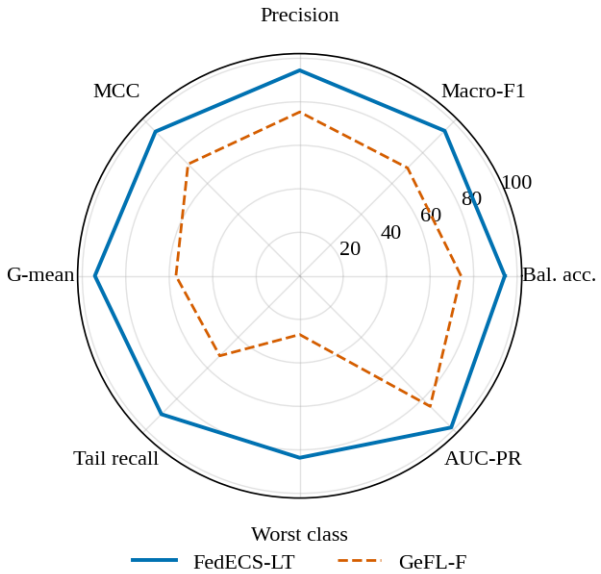


Fig. 145. Main metrics (%, test set, mean over seeds) of FedECS-LT and the five other methods with the highest balanced accuracy; larger is better on every axis. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

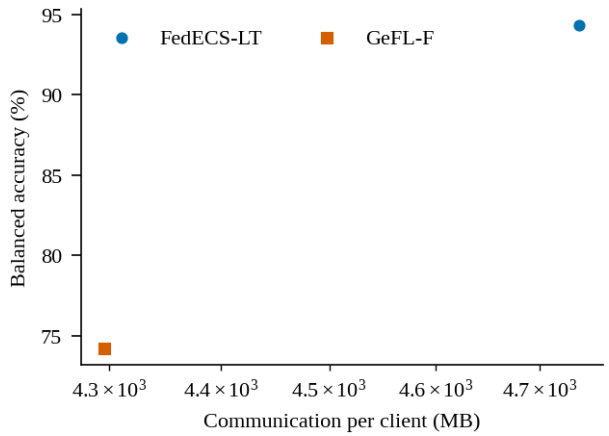


Fig. 146. Balanced accuracy against communication per client over the whole training (logarithmic axis); mean over seeds. Up and left is better. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

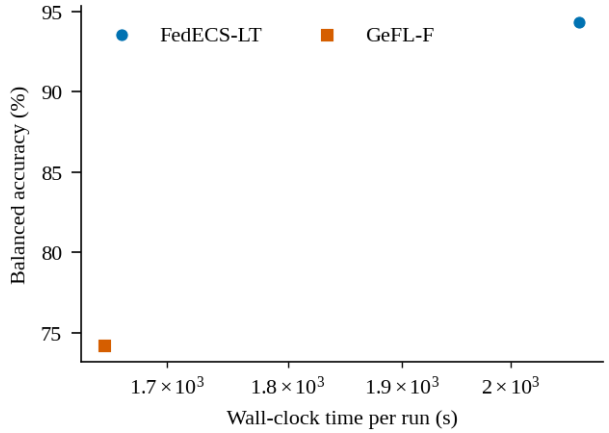


Fig. 147. Balanced accuracy against wall-clock time of one run on one GPU (logarithmic axis); mean over seeds. Up and left is better. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

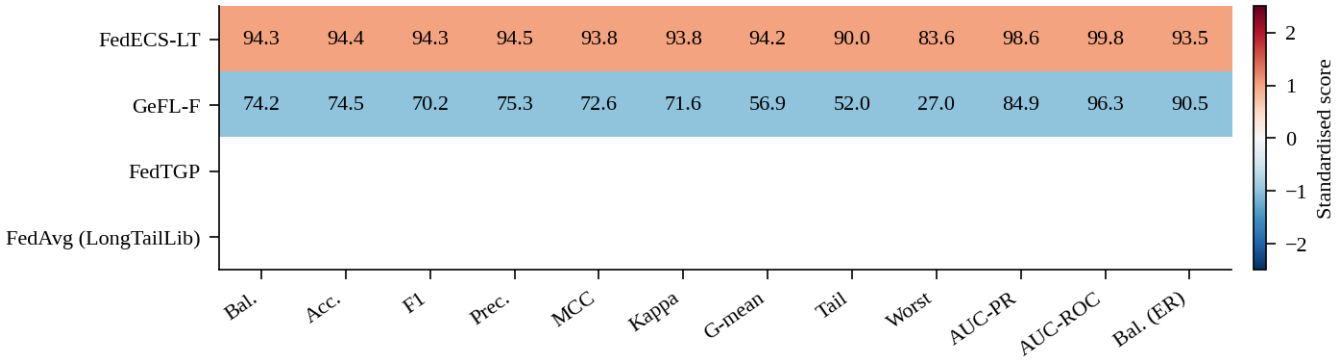


Fig. 148. Every main metric (%, test set, mean over seeds) of every method; colours are standardised per metric (red = better than the column mean). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

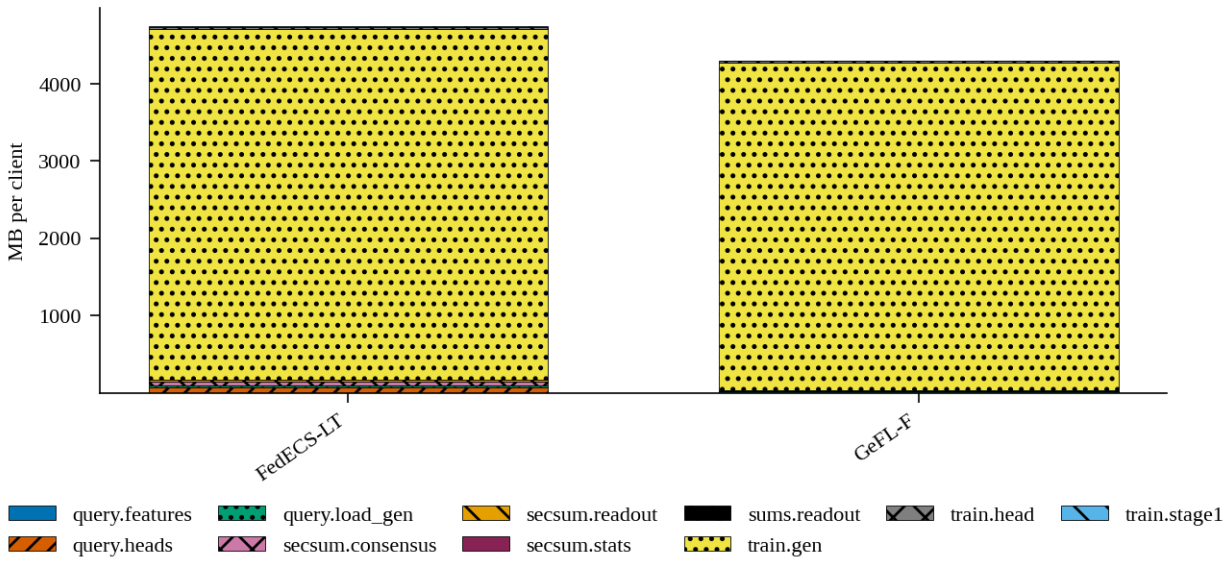


Fig. 149. Communication per client over the whole training, by message type (model weights, statistics, secure sums); first seed. Cached shared stages count once. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

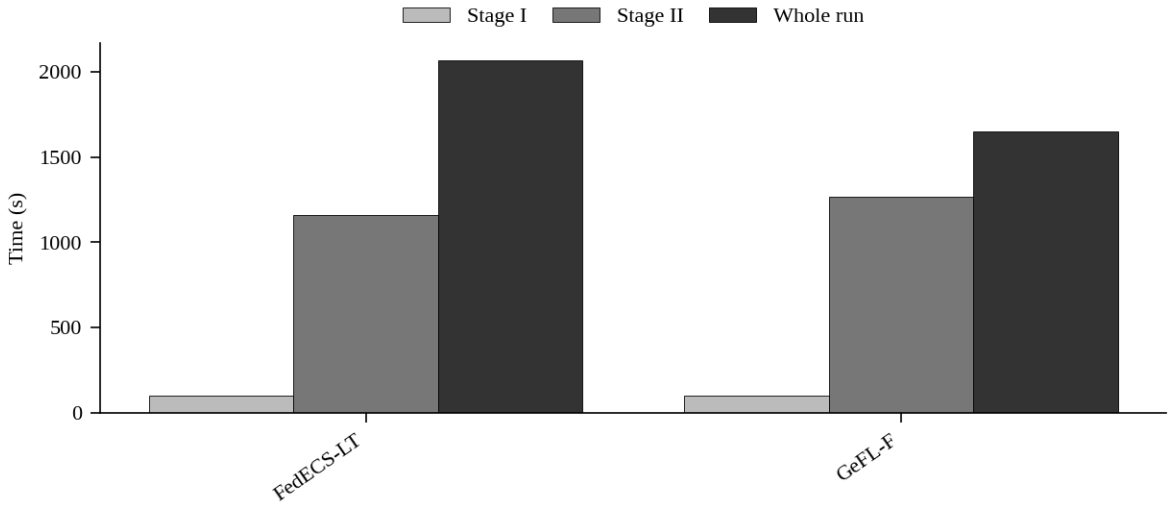


Fig. 150. Wall-clock time per stage on one GPU (simulation of all clients), mean over seeds; shared stages loaded from cache report the time of their first run. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

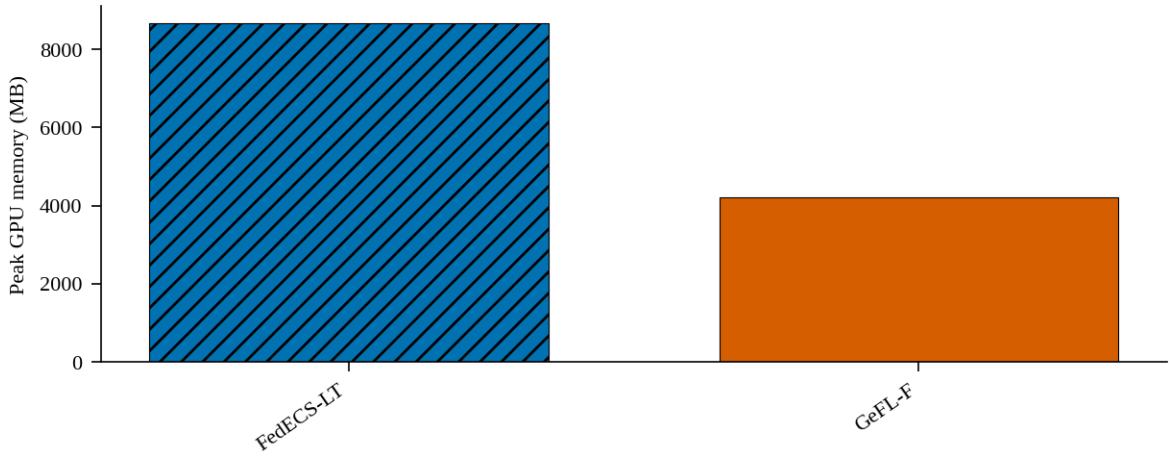


Fig. 151. Peak GPU memory of the server process during one run (mean over seeds). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

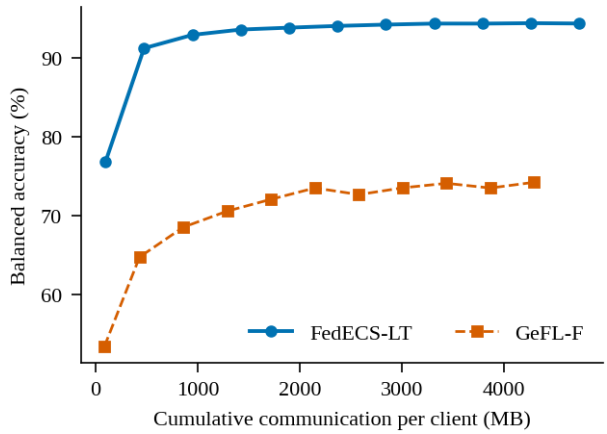


Fig. 152. Balanced accuracy of the aggregated heads against cumulative communication per client (Stage III rounds placed proportionally on the total). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

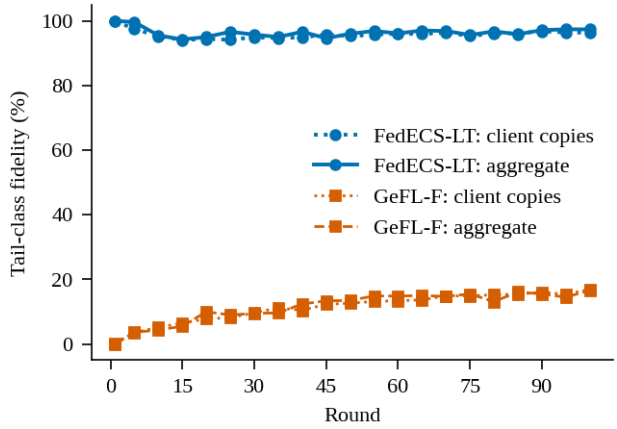


Fig. 153. Tail-class fidelity of the clients' generator copies just before aggregation (mean over clients) and of the aggregated generator, during Stage II. MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

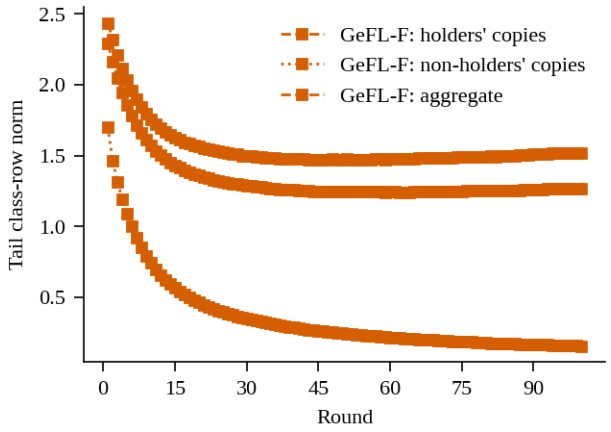


Fig. 154. Norm of the tail classes' conditioning rows in the copies of clients that hold the class, of clients that do not, and in the aggregate, during Stage II (class-row generators, first seed). MNIST-LT (imbalance ratio 100), 10 clients, Dirichlet label split with alpha = 0.5; seed 3.

Appendix A. Figures not applicable to this experiment

Catalogue no.	Group	Reason	Setting
26	C	made from the saved models (models.zip) by the repository script, not in the notebook	cifar10_IF100_a0.5_K10
40	F	the data this figure needs is not produced by this experiment	cifar10_IF100_a0.5_K10
45	F	ablation figure: made from the ablation experiment	cifar10_IF100_a0.5_K10
46	F	sensitivity figure: made from the sweep experiments	cifar10_IF100_a0.5_K10
50	G	all clients take part in every round	cifar10_IF100_a0.5_K10
52	H	privacy figure: made from the differential-privacy experiment	cifar10_IF100_a0.5_K10
53	H	privacy figure: made from the membership-inference experiment	cifar10_IF100_a0.5_K10
54	H	privacy figure: made from the differential-privacy experiment	cifar10_IF100_a0.5_K10
55	I	the pipeline diagram is drawn once for the paper	cifar10_IF100_a0.5_K10
56	17b	every head architecture has a single client here (10 clients, 10 architectures), so aggregating a head leaves it unchanged; the figure is made for the 50- and 100-client federations (or FedECS-LT and GeFL-F are not part of this experiment)	cifar10_IF100_a0.5_K10
59	17b	every head architecture has a single client here (10 clients, 10 architectures), so aggregating a head leaves it unchanged; the figure is made for the 50- and 100-client federations (or FedECS-LT and GeFL-F are not part of this experiment)	cifar10_IF100_a0.5_K10
60	17b	local-only training is not part of this experiment	cifar10_IF100_a0.5_K10
20	C	GeFL-F is not part of this experiment	cifar10_t2_IF100_a0.5_K20_p0.4
23	C	no generator in this experiment	cifar10_t2_IF100_a0.5_K20_p0.4
26	C	made from the saved models (models.zip) by the repository script, not in the notebook	cifar10_t2_IF100_a0.5_K20_p0.4
40	F	the data this figure needs is not produced by this experiment	cifar10_t2_IF100_a0.5_K20_p0.4
45	F	ablation figure: made from the ablation experiment	cifar10_t2_IF100_a0.5_K20_p0.4
46	F	sensitivity figure: made from the sweep experiments	cifar10_t2_IF100_a0.5_K20_p0.4
52	H	privacy figure: made from the differential-privacy experiment	cifar10_t2_IF100_a0.5_K20_p0.4
53	H	privacy figure: made from the membership-inference experiment	cifar10_t2_IF100_a0.5_K20_p0.4
54	H	privacy figure: made from the differential-privacy experiment	cifar10_t2_IF100_a0.5_K20_p0.4
55	I	the pipeline diagram is drawn once for the paper	cifar10_t2_IF100_a0.5_K20_p0.4
60	17b	local-only training is not part of this experiment	cifar10_t2_IF100_a0.5_K20_p0.4
26	C	made from the saved models (models.zip) by the repository script, not in the notebook	mnist_IF100_a0.5_K10
40	F	the data this figure needs is not produced by this experiment	mnist_IF100_a0.5_K10
45	F	ablation figure: made from the ablation experiment	mnist_IF100_a0.5_K10
46	F	sensitivity figure: made from the sweep experiments	mnist_IF100_a0.5_K10
50	G	all clients take part in every round	mnist_IF100_a0.5_K10
52	H	privacy figure: made from the differential-privacy experiment	mnist_IF100_a0.5_K10
53	H	privacy figure: made from the membership-inference experiment	mnist_IF100_a0.5_K10
54	H	privacy figure: made from the differential-privacy experiment	mnist_IF100_a0.5_K10
55	I	the pipeline diagram is drawn once for the paper	mnist_IF100_a0.5_K10
56	17b	every head architecture has a single client here (10 clients, 10 architectures), so aggregating a head leaves it unchanged; the figure is made for the 50- and 100-client federations (or FedECS-LT and GeFL-F are not part of this experiment)	mnist_IF100_a0.5_K10
59	17b	every head architecture has a single client here (10 clients, 10 architectures), so aggregating a head leaves it unchanged; the figure is made for the 50- and 100-client federations (or FedECS-LT and GeFL-F are not part of this experiment)	mnist_IF100_a0.5_K10
60	17b	local-only training is not part of this experiment	mnist_IF100_a0.5_K10

Appendix B. Figure sources

Fig.	Catalogue no.	File
1	1	figures/cifar10_IF100_a0.5_K10/fig_01_balanced_accuracy_rounds.pdf
2	2	figures/cifar10_IF100_a0.5_K10/fig_02_stage1_rounds.pdf
3	3	figures/cifar10_IF100_a0.5_K10/fig_03_generator_loss.pdf
4	4	figures/cifar10_IF100_a0.5_K10/fig_04_head_real_loss.pdf
5	5	figures/cifar10_IF100_a0.5_K10/fig_05_tail_recall_rounds.pdf
6	6	figures/cifar10_IF100_a0.5_K10/fig_06_ece_rounds.pdf
7	7	figures/cifar10_IF100_a0.5_K10/fig_07_consensus_entropy.pdf
8	8	figures/cifar10_IF100_a0.5_K10/fig_08_client_drift.pdf
9	9	figures/cifar10_IF100_a0.5_K10/fig_09_per_class_recall.pdf
10	10	figures/cifar10_IF100_a0.5_K10/fig_10_recall_vs_class_size.pdf
11	11	figures/cifar10_IF100_a0.5_K10/fig_11_confusion_ours.pdf
12	11	figures/cifar10_IF100_a0.5_K10/fig_11_confusion_geflf.pdf
13	12	figures/cifar10_IF100_a0.5_K10/fig_12_per_class_precision.pdf
14	13	figures/cifar10_IF100_a0.5_K10/fig_13_per_class_f1.pdf
15	14	figures/cifar10_IF100_a0.5_K10/fig_14_head_medium_tail.pdf
16	15	figures/cifar10_IF100_a0.5_K10/fig_15_pr_curves_ours.pdf
17	15	figures/cifar10_IF100_a0.5_K10/fig_15_pr_curves_geflf.pdf
18	16	figures/cifar10_IF100_a0.5_K10/fig_16_reliability.pdf
19	17	figures/cifar10_IF100_a0.5_K10/fig_17_classwise_ece.pdf
20	18	figures/cifar10_IF100_a0.5_K10/fig_18_worst_class.pdf

Fig.	Catalogue no.	File
21	18	figures/cifar10_IF100_a0.5_K10/fig_18_gmean.pdf
22	19	figures/cifar10_IF100_a0.5_K10/fig_19_tsne_ours.pdf
23	20	figures/cifar10_IF100_a0.5_K10/fig_20_tsne_geflf.pdf
24	21	figures/cifar10_IF100_a0.5_K10/fig_21_pca_embeddings.pdf
25	22	figures/cifar10_IF100_a0.5_K10/fig_22_tsne_real_vs_generated_ours.pdf
26	23	figures/cifar10_IF100_a0.5_K10/fig_23_tsne_real_vs_generated_geflf.pdf
27	24	figures/cifar10_IF100_a0.5_K10/fig_24_class_mean_distances_ours.pdf
28	24	figures/cifar10_IF100_a0.5_K10/fig_24_class_mean_distances_geflf.pdf
29	25	figures/cifar10_IF100_a0.5_K10/fig_25_classifier_norms.pdf
30	27	figures/cifar10_IF100_a0.5_K10/fig_27_neural_collapse_nc1.pdf
31	28	figures/cifar10_IF100_a0.5_K10/fig_28_generator_fidelity.pdf
32	29	figures/cifar10_IF100_a0.5_K10/fig_29_spread_ratio.pdf
33	30	figures/cifar10_IF100_a0.5_K10/fig_30_class_row_norms.pdf
34	31	figures/cifar10_IF100_a0.5_K10/fig_31_fidelity_rounds.pdf
35	32	figures/cifar10_IF100_a0.5_K10/fig_32_mc_scales.pdf
36	33	figures/cifar10_IF100_a0.5_K10/fig_33_row_norm_rounds_geflf.pdf
37	34	figures/cifar10_IF100_a0.5_K10/fig_34_holder_gain_vs_shrink.pdf
38	35	figures/cifar10_IF100_a0.5_K10/fig_35_row_norm_vs_holders.pdf
39	36	figures/cifar10_IF100_a0.5_K10/fig_36_adam_row_v.pdf
40	37	figures/cifar10_IF100_a0.5_K10/fig_37_readout_gap.pdf
41	38	figures/cifar10_IF100_a0.5_K10/fig_38_balanced_accuracy.pdf
42	39	figures/cifar10_IF100_a0.5_K10/fig_39_forest.pdf
43	41	figures/cifar10_IF100_a0.5_K10/fig_41_radar.pdf
44	42	figures/cifar10_IF100_a0.5_K10/fig_42_accuracy_vs_communication.pdf
45	43	figures/cifar10_IF100_a0.5_K10/fig_43_accuracy_vs_time.pdf
46	44	figures/cifar10_IF100_a0.5_K10/fig_44_metric_heatmap.pdf
47	47	figures/cifar10_IF100_a0.5_K10/fig_47_communication.pdf
48	48	figures/cifar10_IF100_a0.5_K10/fig_48_time_per_stage.pdf
49	49	figures/cifar10_IF100_a0.5_K10/fig_49_gpu_memory.pdf
50	51	figures/cifar10_IF100_a0.5_K10/fig_51_accuracy_vs_cumulative_communication.pdf
51	57	figures/cifar10_IF100_a0.5_K10/fig_57_fidelity_pre_post.pdf
52	58	figures/cifar10_IF100_a0.5_K10/fig_58_row_norm_holders_nonholders.pdf
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54	2	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_02_stage1_rounds.pdf
55	3	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_03_generator_loss.pdf
56	4	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_04_head_real_loss.pdf
57	5	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_05_tail_recall_rounds.pdf
58	6	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_06_ece_rounds.pdf
59	7	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_07_consensus_entropy.pdf
60	8	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_08_client_drift.pdf
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62	10	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_10_recall_vs_class_size.pdf
63	11	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_11_confusion_ours.pdf
64	12	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_12_per_class_precision.pdf
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66	14	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_14_head_medium_tail.pdf
67	15	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_15_pr_curves_ours.pdf
68	16	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_16_reliability.pdf
69	17	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_17_classwise_ece.pdf
70	18	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_18_worst_class.pdf
71	18	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_18_gmean.pdf
72	19	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_19_tsne_ours.pdf
73	21	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_21_pca_embeddings.pdf
74	22	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_22_tsne_real_vs_generated_ours.pdf
75	24	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_24_class_mean_distances_ours.pdf
76	25	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_25_classifier_norms.pdf
77	27	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_27_neural_collapse_nc1.pdf
78	28	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_28_generator_fidelity.pdf
79	29	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_29_spread_ratio.pdf
80	30	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_30_class_row_norms.pdf
81	31	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_31_fidelity_rounds.pdf
82	32	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_32_mc_scales.pdf
83	33	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_33_row_norm_rounds_fedecslt.pdf
84	34	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_34_holder_gain_vs_shrink.pdf
85	35	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_35_row_norm_vs_holders.pdf
86	36	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_36_adam_row_v.pdf
87	37	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_37_readout_gap.pdf
88	38	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_38_balanced_accuracy.pdf
89	39	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_39_forest.pdf

Fig.	Catalogue no.	File
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91	42	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_42_accuracy_vs_communication.pdf
92	43	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_43_accuracy_vs_time.pdf
93	44	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_44_metric_heatmap.pdf
94	47	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_47_communication.pdf
95	48	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_48_time_per_stage.pdf
96	49	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_49_gpu_memory.pdf
97	50	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_50_participation.pdf
98	51	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_51_accuracy_vs_cumulative_communication.pdf
99	56	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_56_aggregation_effect_ours.pdf
100	57	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_57_fidelity_pre_post.pdf
101	58	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_58_row_norm_holders_nonholders.pdf
102	59	figures/cifar10_t2_IF100_a0.5_K20_p0.4/fig_59_effect_vs_holders.pdf
103	1	figures/mnist_IF100_a0.5_K10/fig_01_balanced_accuracy_rounds.pdf
104	2	figures/mnist_IF100_a0.5_K10/fig_02_stage1_rounds.pdf
105	3	figures/mnist_IF100_a0.5_K10/fig_03_generator_loss.pdf
106	4	figures/mnist_IF100_a0.5_K10/fig_04_head_real_loss.pdf
107	5	figures/mnist_IF100_a0.5_K10/fig_05_tail_recall_rounds.pdf
108	6	figures/mnist_IF100_a0.5_K10/fig_06_ece_rounds.pdf
109	7	figures/mnist_IF100_a0.5_K10/fig_07_consensus_entropy.pdf
110	8	figures/mnist_IF100_a0.5_K10/fig_08_client_drift.pdf
111	9	figures/mnist_IF100_a0.5_K10/fig_09_per_class_recall.pdf
112	10	figures/mnist_IF100_a0.5_K10/fig_10_recall_vs_class_size.pdf
113	11	figures/mnist_IF100_a0.5_K10/fig_11_confusion_ours.pdf
114	11	figures/mnist_IF100_a0.5_K10/fig_11_confusion_geflf.pdf
115	12	figures/mnist_IF100_a0.5_K10/fig_12_per_class_precision.pdf
116	13	figures/mnist_IF100_a0.5_K10/fig_13_per_class_f1.pdf
117	14	figures/mnist_IF100_a0.5_K10/fig_14_head_medium_tail.pdf
118	15	figures/mnist_IF100_a0.5_K10/fig_15_pr_curves_ours.pdf
119	15	figures/mnist_IF100_a0.5_K10/fig_15_pr_curves_geflf.pdf
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127	22	figures/mnist_IF100_a0.5_K10/fig_22_tsne_real_vs_generated_ours.pdf
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137	32	figures/mnist_IF100_a0.5_K10/fig_32_mc_scales.pdf
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141	36	figures/mnist_IF100_a0.5_K10/fig_36_adam_row_v.pdf
142	37	figures/mnist_IF100_a0.5_K10/fig_37_readout_gap.pdf
143	38	figures/mnist_IF100_a0.5_K10/fig_38_balanced_accuracy.pdf
144	39	figures/mnist_IF100_a0.5_K10/fig_39_forest.pdf
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147	43	figures/mnist_IF100_a0.5_K10/fig_43_accuracy_vs_time.pdf
148	44	figures/mnist_IF100_a0.5_K10/fig_44_metric_heatmap.pdf
149	47	figures/mnist_IF100_a0.5_K10/fig_47_communication.pdf
150	48	figures/mnist_IF100_a0.5_K10/fig_48_time_per_stage.pdf
151	49	figures/mnist_IF100_a0.5_K10/fig_49_gpu_memory.pdf
152	51	figures/mnist_IF100_a0.5_K10/fig_51_accuracy_vs_cumulative_communication.pdf
153	57	figures/mnist_IF100_a0.5_K10/fig_57_fidelity_pre_post.pdf
154	58	figures/mnist_IF100_a0.5_K10/fig_58_row_norm_holders_nonholders.pdf

Appendix C. All logged values of every run

GeFL-F|mnist|IF0.01|a0.5|K10|s3

Quantity	Value
IF	0.01
K	10
alpha	0.5
comm_mb_per_client	4295.68
er_acc	0.90635
er_auc_pr	0.951306
er_auc_roc	0.993204
er_bal	0.904858
er_brier	0.17393
er_cw_ece	0.0172033
er_ece	0.0831203
er_ens_acc	0.9425
er_ens_bal	0.941669
er_f1	0.904855
er_gmean	0.901429
er_head	0.985335
er_kappa	0.895895
er_local	0.973198
er_mcc	0.896444
er_medium	0.917912
er_nll	0.898556
er_precision	0.909666
er_rec_std	0.0718137
er_tail	0.83471
er_top5	0.98413
er_worst	0.806145
er_worst_arch	0.855761
fe_time_s	98.9679
final_acc	0.74487
final_auc_pr	0.848753
final_auc_roc	0.963137
final_bal	0.74196
final_brier	0.357544
final_cw_ece	0.0332551
final_ece	0.0791932
final_ens_acc	0.8429
final_ens_bal	0.841745
final_f1	0.70154
final_gmean	0.569173
final_head	0.965943
final_kappa	0.716362
final_local	0.975724
final_mcc	0.726306
final_medium	0.814081
final_nll	0.840611
final_precision	0.753179
final_rec_std	0.255465
final_tail	0.519882
final_top5	0.97857
final_worst	0.269815
final_worst_arch	0.697324
gen_time_s	1266.75
generator_params	5546659
gpu_peak_mb	4203
head_time_s	20.2428
min_client	147
n_train	5996
pfrac	1
seed	3
time_s	1650.67
val_er_acc	0.89015
val_er_auc_pr	0.940805
val_er_auc_roc	0.991523
val_er_bal	0.89015
val_er_brier	0.203044
val_er_cw_ece	0.0207022
val_er_ece	0.09813

Quantity	Value
val_er_ens_acc	0.9345
val_er_ens_bal	0.9345
val_er_f1	0.890008
val_er_gmean	0.886072
val_er_head	0.975833
val_er_kappa	0.877944
val_er_local	0.963095
val_er_mcc	0.87862
val_er_medium	0.8995
val_er_nll	1.04405
val_er_precision	0.895329
val_er_rec_std	0.0767594
val_er_tail	0.818875
val_er_top5	0.98305
val_er_worst	0.779
val_er_worst_arch	0.8385
val_final_acc	0.7342
val_final_auc_pr	0.839565
val_final_auc_roc	0.959706
val_final_bal	0.7342
val_final_brier	0.372656
val_final_cw_ece	0.0346044
val_final_ece	0.0827947
val_final_ens_acc	0.833
val_final_ens_bal	0.833
val_final_f1	0.694352
val_final_gmean	0.565132
val_final_head	0.958667
val_final_kappa	0.704667
val_final_local	0.97085
val_final_mcc	0.71466
val_final_medium	0.802333
val_final_nll	0.880546
val_final_precision	0.743268
val_final_rec_std	0.255696
val_final_tail	0.51475
val_final_top5	0.9769
val_final_worst	0.263
val_final_worst_arch	0.685

Vector-valued fields (in runs.jsonl): bytes, client_counts, cond_norms_end, er_ap_c, er_confusion, er_f1_c, er_precision_c, er_recall_c, fidelity, final_ap_c, final_confusion, final_f1_c, final_per_arch, final_precision_c, final_recall_c, method, rounds, stage1_final, stage_time_s, val_er_ap_c, val_er_confusion, val_er_f1_c, val_er_precision_c, val_er_recall_c, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

FedECS-LT|mnist|IF0.01|a0.5|K10|s3

Quantity	Value
IF	0.01
K	10
alpha	0.5
class_sep_F	0.324849
comm_mb_per_client	4738.75
er_acc	0.93592
er_auc_pr	0.969931
er_auc_roc	0.996357
er_bal	0.934889
er_brier	0.120995
er_cw_ece	0.012095
er_ece	0.0585926
er_ens_acc	0.9483
er_ens_bal	0.947481
er_f1	0.935003
er_gmean	0.933347
er_head	0.990305
er_kappa	0.928762
er_local	0.981753
er_mcc	0.929104
er_medium	0.94852
er_nll	0.661212

Quantity	Value
er_precision	0.938232
er_rec_std	0.0519122
er_tail	0.883104
er_top5	0.98899
er_worst	0.856879
er_worst_arch	0.922933
fe_time_s	98.9679
final_acc	0.94398
final_auc_pr	0.98567
final_auc_roc	0.997575
final_bal	0.943342
final_brier	0.0861158
final_cw_ece	0.00661025
final_ece	0.019835
final_ens_acc	0.9547
final_ens_bal	0.954234
final_f1	0.942941
final_gmean	0.942016
final_head	0.984902
final_kappa	0.93773
final_local	0.983341
final_mcc	0.93794
final_medium	0.960084
final_nll	0.202523
final_precision	0.944619
final_rec_std	0.0462441
final_tail	0.899617
final_top5	0.99798
final_worst	0.836345
final_worst_arch	0.932933
gen_time_s	1161.18
generator_params	5937315
gpu_peak_mb	8674
head_time_s	45.0254
min_client	147
n_train	5996
pfrac	1
seed	3
time_s	2065.77
val_er_acc	0.92395
val_er_auc_pr	0.961808
val_er_auc_roc	0.995339
val_er_bal	0.92395
val_er_brier	0.143499
val_er_cw_ece	0.0147531
val_er_ece	0.0696282
val_er_ens_acc	0.938
val_er_ens_bal	0.938
val_er_f1	0.923429
val_er_gmean	0.922209
val_er_head	0.984167
val_er_kappa	0.9155
val_er_local	0.973547
val_er_mcc	0.915863
val_er_medium	0.932
val_er_nll	0.780945
val_er_precision	0.926194
val_er_rec_std	0.0535595
val_er_tail	0.87275
val_er_top5	0.98745
val_er_worst	0.841
val_er_worst_arch	0.9145
val_final_acc	0.93355
val_final_auc_pr	0.979161
val_final_auc_roc	0.996419
val_final_bal	0.93355
val_final_brier	0.105603
val_final_cw_ece	0.00947013
val_final_ece	0.0284292

Quantity	Value
val_final_ens_acc	0.9445
val_final_ens_bal	0.9445
val_final_f1	0.932997
val_final_gmean	0.931944
val_final_head	0.979
val_final_kappa	0.926167
val_final_local	0.978629
val_final_mcc	0.926381
val_final_medium	0.946
val_final_nll	0.25646
val_final_precision	0.934517
val_final_rec_std	0.0503619
val_final_tail	0.890125
val_final_top5	0.99645
val_final_worst	0.817
val_final_worst_arch	0.922

Vector-valued fields (in runs.jsonl): bytes, client_counts, er_ap_c, er_confusion, er_f1_c, er_precision_c, er_recall_c, fidelity, fidelity_mc, final_ap_c, final_confusion, final_f1_c, final_per_arch, final_precision_c, final_recall_c, mc_scale, method, rounds, spread_ratio, stage1_final, stage_time_s, val_er_ap_c, val_er_confusion, val_er_f1_c, val_er_precision_c, val_er_recall_c, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

FedECS-LT|cifar10_t2|IF0.01|a0.5|K20|s3|p0.4

Quantity	Value
IF	0.01
K	20
alpha	0.5
class_sep_F	0.0416905
comm_mb_per_client	2984.21
er_acc	0.5692
er_auc_pr	0.67862
er_auc_roc	0.929562
er_bal	0.5692
er_brier	0.763907
er_cw_ece	0.0755181
er_ece	0.353001
er_f1	0.554662
er_gmean	0.522452
er_head	0.846667
er_kappa	0.521333
er_local	0.812322
er_mcc	0.532015
er_medium	0.566333
er_nll	3.70526
er_precision	0.664588
er_rec_std	0.221721
er_tail	0.36325
er_top5	0.8792
er_worst	0.231
er_worst_arch	0.5692
fe_time_s	3100.65
final_acc	0.596
final_auc_pr	0.67316
final_auc_roc	0.924735
final_bal	0.596
final_brier	0.588252
final_cw_ece	0.043366
final_ece	0.185267
final_f1	0.587748
final_gmean	0.572608
final_head	0.789333
final_kappa	0.551111
final_local	0.773904
final_mcc	0.55573
final_medium	0.557667
final_nll	1.4197
final_precision	0.626717
final_rec_std	0.169577
final_tail	0.47975

Quantity	Value
final_top5	0.9481
final_worst	0.337
final_worst_arch	0.596
gen_time_s	1070
generator_params	5855076
gpu_peak_mb	5826
head_time_s	41.8893
min_client	158
n_train	11909
pfrac	0.4
seed	3
time_s	7409.47
val_er_acc	0.5745
val_er_auc_pr	0.682908
val_er_auc_roc	0.931503
val_er_bal	0.5745
val_er_brier	0.760082
val_er_cw_ece	0.0755035
val_er_ece	0.349351
val_er_f1	0.559687
val_er_gmean	0.521017
val_er_head	0.85
val_er_kappa	0.527222
val_er_local	0.817774
val_er_mcc	0.538818
val_er_medium	0.553333
val_er_nll	3.66018
val_er_precision	0.677851
val_er_rec_std	0.231251
val_er_tail	0.38375
val_er_top5	0.888
val_er_worst	0.225
val_er_worst_arch	0.5745
val_final_acc	0.587
val_final_auc_pr	0.673674
val_final_auc_roc	0.925657
val_final_bal	0.587
val_final_brier	0.592832
val_final_cw_ece	0.0466208
val_final_ece	0.203315
val_final_f1	0.576065
val_final_gmean	0.556199
val_final_head	0.765
val_final_kappa	0.541111
val_final_local	0.759293
val_final_mcc	0.545785
val_final_medium	0.535
val_final_nll	1.43147
val_final_precision	0.61178
val_final_rec_std	0.18534
val_final_tail	0.4925
val_final_top5	0.949
val_final_worst	0.285
val_final_worst_arch	0.587

Vector-valued fields (in runs.jsonl): bytes, client_counts, cond_norms_end, er_ap_c, er_confusion, er_f1_c, er_precision_c, er_recall_c, fidelity, fidelity_mc, final_ap_c, final_confusion, final_f1_c, final_per_arch, final_precision_c, final_recall_c, mc_scale, method, rounds, spread_ratio, stage1_final, stage_time_s, val_er_ap_c, val_er_confusion, val_er_f1_c, val_er_precision_c, val_er_recall_c, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

FedTGP|cifar10|IF0.01|a0.5|K10|s3

Quantity	Value
IF	0.01
K	10
alpha	0.5
comm_mb_per_client	18.8069
fe_time_s	432.925
final_acc	0.2332

Quantity	Value
final_auc_pr	0.267314
final_auc_roc	0.742581
final_bal	0.2332
final_brier	0.871818
final_cw_ece	0.0225376
final_ece	0.0887851
final_f1	0.161381
final_gmean	0.0776993
final_head	0.424367
final_kappa	0.148
final_mcc	0.174456
final_medium	0.250333
final_nll	2.17779
final_precision	0.250287
final_rec_std	0.170324
final_tail	0.076975
final_top5	0.746
final_worst	0.0107
final_worst_arch	0.1561
gpu_peak_mb	889
min_client	872
n_train	18485
pfrac	1
seed	3
time_s	7564.86
val_final_acc	0.22795
val_final_auc_pr	0.27255
val_final_auc_roc	0.741324
val_final_bal	0.22795
val_final_brier	0.87206
val_final_cw_ece	0.0227003
val_final_ece	0.0816618
val_final_f1	0.15655
val_final_gmean	0.074429
val_final_head	0.419333
val_final_kappa	0.142167
val_final_mcc	0.167704
val_final_medium	0.239667
val_final_nll	2.17914
val_final_precision	0.218892
val_final_rec_std	0.169326
val_final_tail	0.075625
val_final_top5	0.7438
val_final_worst	0.007
val_final_worst_arch	0.1425

Vector-valued fields (in runs.jsonl): bytes, client_counts, final_ap_c, final_confusion, final_f1_c, final_per_client, final_precision_c, final_recall_c, hyperparameters, method, rounds, stage_time_s, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

GeFL-F|cifar10|IF0.01|a0.5|K10|s3

Quantity	Value
IF	0.01
K	10
alpha	0.5
comm_mb_per_client	8659.06
er_acc	0.49278
er_auc_pr	0.51512
er_auc_roc	0.877496
er_bal	0.49278
er_brier	0.709262
er_cw_ece	0.0454033
er_ece	0.205957
er_ens_acc	0.5924
er_ens_bal	0.5924
er_f1	0.490107
er_gmean	0.480198
er_head	0.5915
er_kappa	0.436422

Quantity	Value
er_local	0.56354
er_mcc	0.437489
er_medium	0.430367
er_nll	1.79462
er_precision	0.500532
er_rec_std	0.103165
er_tail	0.46555
er_top5	0.91451
er_worst	0.3549
er_worst_arch	0.4369
fe_time_s	432.925
final_acc	0.32315
final_auc_pr	0.322858
final_auc_roc	0.742467
final_bal	0.32315
final_brier	1.13098
final_cw_ece	0.107789
final_ece	0.503143
final_ens_acc	0.438
final_ens_bal	0.438
final_f1	0.256224
final_gmean	0.147377
final_head	0.576433
final_kappa	0.247944
final_local	0.733317
final_mcc	0.267272
final_medium	0.3255
final_nll	4.31856
final_precision	0.359481
final_rec_std	0.201824
final_tail	0.131425
final_top5	0.81952
final_worst	0.0304
final_worst_arch	0.2463
gen_time_s	6160.67
generator_params	5613294
gpu_peak_mb	8679
head_time_s	20.4354
min_client	872
n_train	18485
pfrac	1
seed	3
time_s	6712.69
val_er_acc	0.49965
val_er_auc_pr	0.533662
val_er_auc_roc	0.882929
val_er_bal	0.49965
val_er_brier	0.692579
val_er_cw_ece	0.0457958
val_er_ece	0.199218
val_er_ens_acc	0.5965
val_er_ens_bal	0.5965
val_er_f1	0.496554
val_er_gmean	0.484241
val_er_head	0.598
val_er_kappa	0.444056
val_er_local	0.557253
val_er_mcc	0.445302
val_er_medium	0.417333
val_er_nll	1.72621
val_er_precision	0.508089
val_er_rec_std	0.115105
val_er_tail	0.487625
val_er_top5	0.9204
val_er_worst	0.3285
val_er_worst_arch	0.437
val_final_acc	0.32775
val_final_auc_pr	0.334689
val_final_auc_roc	0.746076

Quantity	Value
val_final_bal	0.32775
val_final_brier	1.11721
val_final_cw_ece	0.107219
val_final_ece	0.494816
val_final_ens_acc	0.4455
val_final_ens_bal	0.4455
val_final_f1	0.260944
val_final_gmean	0.149246
val_final_head	0.587167
val_final_kappa	0.253056
val_final_local	0.735893
val_final_mcc	0.272191
val_final_medium	0.319167
val_final_nll	4.24171
val_final_precision	0.358516
val_final_rec_std	0.203649
val_final_tail	0.139625
val_final_top5	0.8203
val_final_worst	0.0435
val_final_worst_arch	0.2535

Vector-valued fields (in runs.jsonl): bytes, client_counts, cond_norms_end, er_ap_c, er_confusion, er_f1_c, er_precision_c, er_recall_c, fidelity, final_ap_c, final_confusion, final_f1_c, final_per_arch, final_precision_c, final_recall_c, method, rounds, stage1_final, stage_time_s, val_er_ap_c, val_er_confusion, val_er_f1_c, val_er_precision_c, val_er_recall_c, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

FedAvg (LongTailLib)|cifar10_t2|IF0.01|a0.5|K20|s3|p0.4

Quantity	Value
IF	0.01
K	20
alpha	0.5
final_acc	0.6114
final_auc_pr	0.77024
final_auc_roc	0.953184
final_bal	0.6114
final_brier	0.644064
final_cw_ece	0.0638018
final_ece	0.285666
final_f1	0.592295
final_gmean	0.555774
final_head	0.909333
final_kappa	0.568222
final_mcc	0.578808
final_medium	0.539
final_nll	2.0954
final_precision	0.706081
final_rec_std	0.237713
final_tail	0.44225
final_top5	0.9639
final_worst	0.194
final_worst_arch	0.6114
library_accuracy	0.6114
pfrac	0.4
seed	3
time_s	12018
track	2

Vector-valued fields (in runs.jsonl): client_counts, final_ap_c, final_confusion, final_f1_c, final_precision_c, final_recall_c, history_library, method, rounds

FedECS-LT|cifar10|IF0.01|a0.5|K10|s3

Quantity	Value
IF	0.01
K	10
alpha	0.5
class_sep_F	0.0822147
comm_mb_per_client	11268.3
er_acc	0.51014

Quantity	Value
er_auc_pr	0.539242
er_auc_roc	0.88652
er_bal	0.51014
er_brier	0.695995
er_cw_ece	0.0470812
er_ece	0.215841
er_ens_acc	0.5873
er_ens_bal	0.5873
er_f1	0.507226
er_gmean	0.497353
er_head	0.619233
er_kappa	0.455711
er_local	0.579826
er_mcc	0.456848
er_medium	0.4364
er_nll	1.76425
er_precision	0.518503
er_rec_std	0.109312
er_tail	0.483625
er_top5	0.92099
er_worst	0.366
er_worst_arch	0.4613
fe_time_s	432.925
final_acc	0.407
final_auc_pr	0.427455
final_auc_roc	0.837789
final_bal	0.407
final_brier	0.936699
final_cw_ece	0.0834969
final_ece	0.389648
final_ens_acc	0.4993
final_ens_bal	0.4993
final_f1	0.384772
final_gmean	0.348605
final_head	0.6245
final_kappa	0.341111
final_local	0.685925
final_mcc	0.350547
final_medium	0.3461
final_nll	2.89932
final_precision	0.455624
final_rec_std	0.170475
final_tail	0.28955
final_top5	0.89484
final_worst	0.1772
final_worst_arch	0.3566
gen_time_s	4287.52
generator_params	6928622
gpu_peak_mb	5826
head_time_s	91.838
min_client	872
n_train	18485
pfrac	1
seed	3
time_s	6546.01
val_er_acc	0.5168
val_er_auc_pr	0.556089
val_er_auc_roc	0.890691
val_er_bal	0.5168
val_er_brier	0.682471
val_er_cw_ece	0.0478544
val_er_ece	0.211221
val_er_ens_acc	0.58
val_er_ens_bal	0.58
val_er_f1	0.513122
val_er_gmean	0.500248
val_er_head	0.624167
val_er_kappa	0.463111
val_er_local	0.576964

Quantity	Value
val_er_mcc	0.464414
val_er_medium	0.430667
val_er_nll	1.71584
val_er_precision	0.525959
val_er_rec_std	0.124389
val_er_tail	0.500875
val_er_top5	0.92345
val_er_worst	0.3365
val_er_worst_arch	0.4775
val_final_acc	0.415
val_final_auc_pr	0.444814
val_final_auc_roc	0.840825
val_final_bal	0.415
val_final_brier	0.923611
val_final_cw_ece	0.0842835
val_final_ece	0.384225
val_final_ens_acc	0.499
val_final_ens_bal	0.499
val_final_f1	0.393013
val_final_gmean	0.353349
val_final_head	0.641
val_final_kappa	0.35
val_final_local	0.690889
val_final_mcc	0.359599
val_final_medium	0.340333
val_final_nll	2.87094
val_final_precision	0.465818
val_final_rec_std	0.180995
val_final_tail	0.3015
val_final_top5	0.8943
val_final_worst	0.171
val_final_worst_arch	0.3605

Vector-valued fields (in runs.jsonl): bytes, client_counts, er_ap_c, er_confusion, er_f1_c, er_precision_c, er_recall_c, fidelity, fidelity_mc, final_ap_c, final_confusion, final_f1_c, final_per_arch, final_precision_c, final_recall_c, mc_scale, method, rounds, spread_ratio, stage1_final, stage_time_s, val_er_ap_c, val_er_confusion, val_er_f1_c, val_er_precision_c, val_er_recall_c, val_final_ap_c, val_final_confusion, val_final_f1_c, val_final_precision_c, val_final_recall_c

Appendix D. Failed runs

Run	Error
GeFL-F|cifar10|IF0.01|a0.5|K10|s3 (completed on retry)	OSError("Couldn't deserialize thrift: TProtocolException: Invalid data\nDeserializing page header failed.\n")
FedECS-LT|cifar10|IF0.01|a0.5|K10|s3 (completed on retry)	OSError("Error creating dataset. Could not read schema from '/tmp/fedecs/data/cifar10_hf/train.parquet'. Is this a 'parquet' file?: Could not open Parquet input source '/tmp/fedecs/data/cifar10_hf/train.parquet': Couldn't deserialize thrift: TProtocolException: Invalid data\n")